
User Manual

for MPC560XB MCU Driver

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Chapter 1

Revision History

Table 1-1. Revision History

Revision	Date	Author	Description
1.0	17/02/2012	Francesco Mazzearella	Initial Release
1.1	13/05/2013	Swarupa Chaudhury	RTM 1.0.0 Release
1.2	28/05/2014	Ronan Forest	RTM 1.0.1 Release



Chapter 2

Introduction

This User Manual describes Freescale Semiconductor AUTOSAR Micro Controller Unit (MCU) for MPC560XB .

AUTOSAR MCU driver configuration parameters and deviations from the specification are described in MCU Driver chapter of this document. AUTOSAR MCU driver requirements and APIs are described in the AUTOSAR MCU driver software specification document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of Freescale Semiconductor .

Table 2-1. MPC560XB Derivatives

Freescale Semiconductor	mpc5601dxlh_lqfp64 mpc5601dxll_lqfp100 mpc5602dxlh_lqfp64 mpc5602dxll_lqfp100 mpc5602bxlh_lqfp64 mpc5602bxll_lqfp100 mpc5602bxlq_lqfp144 mpc5602cxlh_lqfp64 mpc5602cxll_lqfp100 mpc5603bxlh_lqfp64 mpc5603bxll_lqfp100 mpc5603bxlq_lqfp144 mpc5603cxlh_lqfp64 mpc5603cxll_lqfp100 mpc5604bxlh_lqfp64 mpc5604bxll_lqfp100 mpc5604bxlq_lqfp144 mpc5604bxmg_mapbga208 mpc5604cxlh_lqfp64 mpc5604cxll_lqfp100 mpc5605bxll_lqfp100 mpc5605bxlq_lqfp144 mpc5605bxlu_lqfp176 mpc5606bkll_lqfp100
-------------------------	--

Table 2-1. MPC560XB Derivatives

	mpc5606bxlq_lqfp144 mpc5606bxlu_lqfp176 mpc5607bxlu_lqfp176 mpc5607bxmg_mapbga208
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All of the above microcontroller devices are collectively named as MPC560XB .

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About this Manual

This Technical Reference employs the following typographical conventions:

Boldface type: Bold is used for important terms, notes and warnings.

Italic font: Italic typeface is used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

2.4 Acronyms and Definitions

Table 2-2. Acronyms and Definitions

Term	Definition
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
ASM	Assembler
BSMI	Basic Software Make file Interface
CAN	Controller Area Network
DEM	Diagnostic Event Manager
DET	Development Error Tracer
C/CPP	C and C++ Source Code
VLE	Variable Length Encoding
N/A	Not Applicable
MCU	Micro Controller Unit

2.5 Reference List

Table 2-3. Reference List

#	Title	Version
1	AUTOSAR 4.0 Rev0003MCU Driver Software Specification Document.	3.2.0 Rev 3
2	MPC5607B Reference Manual	Rev. 7.2, May 2012
3	MPC5604B Reference Manual	Rev. 8.2, September 2013
4	5606B Reference Manual	Rev. 2, May 2014
5	5602D Reference Manual	Rev. 4.2, Dec 2013

Chapter 3 Driver

3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR 4.0 Rev0003MCU Driver Software Specification document (See Table [Reference List](#)).

3.2 Driver Design Summary

The MCU driver provides services for basic microcontroller initialization, power down functionality, reset and microcontroller specific functions required from other Microcontroller Abstraction Layer (MCAL) software modules.

Driver major features:

3.2.1 Initialization of MCU clock, PLL, MCU clock distribution

The following list summarizes the clock architecture implemented on the MPC560XB:

- **System clock** can be derived from the following sources:
 - progressive output clock of the main PLL called McuPLL in the plug-in
 - Internal RC oscillator (IRC) called McuIntOscRC16MHz in the plug-in
 - External oscillator (XOSC) called McuExtQuartzOsc in the plug-in

The system clock is adopted for generating the following clock sources:

- **SYS_CLK clock** used for the Core
- **DMA clock**

- **Peripheral set clock 1** is a reduced clock frequency for the peripherals that can run on a lower frequency without performance degradation: I2C, LINFLEX_n (n = 0..3..9). It is obtained from the system clock divided by a programmable clock divider ($\div 1, \div 2, \div 3, \div 4, \dots \div 16$)
- **Peripheral set clock 2** is a reduced clock frequency for the peripherals that can run on a lower frequency without performance degradation: DSPI_n (n = 0..2..5), FlexCAN_n (n=0..5..). It is obtained from the system clock divided by a programmable clock divider ($\div 1, \div 2, \div 3, \div 4, \dots \div 16$)
- **Peripheral set clock 3** is a reduced clock frequency for the peripherals that can run on a lower frequency without performance degradation: CTU, eMIOS_n (n = 0..1). It is obtained from the system clock divided by a programmable clock divider ($\div 1, \div 2, \div 3, \div 4, \dots \div 16$)

NOTE

The internal RC oscillator represents the default selection for the system clock after reset.

- **Output clock** can be derived from the following sources:
 - External oscillator (FXOSC)
 - non-progressive output clock of the main PLL (FMPLL_CLK) called PLL into the plug-in
 - Internal RC oscillator (IRC) called F16_MHz_RC_Internal into the plug-in
 - System Clock (only for certain devices)
 - RTC clock called RTC_clk into the plug-in (only for certain devices)

NOTE

The internal RC oscillator represents the default selection for the output clock after reset.

A programmable clock divider ($\div 1, \div 2, \div 4, \div 8$) can be adopted for the generation of a divided output clock.

- **FXOSC** represents the fast (4-16 MHz) external oscillator clock output
- **SXOSC** represents the slow (32 KHz) external oscillator clock output
- **Frequency Modulated Phase-Locked Loop (FMPLL):**
 - generates an high speed system clock from a common 4-16 Mhz input clock source. Its frequency is software configurable
 - supports software configurable frequency modulation

- is continuously monitored by the Lock detect circuitry which reports the lock status of the PLL
- if chosen as system clock can be continuously monitored by the primary control monitor unit (CMU_0) which is able to detect if the PLL leaves an upper or lower frequency boundary.
- **Fast Internal RC (FIRC)** oscillator, whose major features are:
 - nominal frequency of 16 Mhz
 - glitch free oscillation
 - back-up clock in the event of PLL or external oscillator clock failure
- **Slow Internal RC (SIRC)** oscillator, whose major features are:
 - nominal frequency of 128 KHz
 - Very low power consumption
 - Undivided, provides the clock for the Software Watchdog (SWT)
- **Real Time Clock/Autonomous Periodic Interrupt (RTC/API)** can be derived from the following sources:
 - Divided ($\div 1, \div 2, \div 3, \div 4, \dots, \div 32$) Slow External Oscillator (SXOSC)
 - Divided ($\div 1, \div 2, \div 3, \div 4, \dots, \div 32$) Slow Internal RC Oscillator (SIRC)
 - Undivided Fast Internal RC Oscillator (FIRC)

3.2.2 Initialization of RAM sections

3.2.3 Activation of MCU low power modes

The microcontroller offers several devices modes corresponding to different usage models of the device. Each mode is configurable and can define a policy for energy and processing power management to fit particular system requirements. An application can easily switch from one mode to another depending on the current needs of the system. In particular the following modes to reduce the activity of the uC for saving energy are supported:

- HALT mode is intended to be used by software to let the uC wait in a state of low power consumption at the cost of higher wakeup latency until the uC is required to do something and then to react quickly (i.e within a few system clock cycles of an interrupt event). When the uC enters into HALT mode the following actions programmable by software are performed:
 - the clocks to the core and the system memory are disabled
 - the clock to each peripheral may be disabled
 - the clock sources such as the PLLs, external and internal oscillators can be configured to be switched off. However a source can not be disabled if it has been chosen as the system clock to be used into the HALT mode.
 - Code flash memory may be switched off
- STOP mode is intended to be used by software to let the uC wait in a state of low power consumption until the uC is required to do something with no need to react quickly. When the uC enters into STOP mode the following actions programmable by software are performed: the core clock is frozen and most of the peripherals including oscillator may be configured to be switched off.
 - the clocks to the core and the system memory are disabled
 - the clock to each peripheral may be disabled
 - The external oscillator may be turned off.
 - Code flash memory may be switched off
 - The outputs of the pads may be forced to the high impedence state (I/O output power down)
- STANDBY mode is intended as an extreme low-power mode. It is used by software to wait until it is required to do something with no need to react quickly (for example allow for system power-up and system clock source to be re-started). The exit sequence of this mode is similar to the reset sequence, but, in addition to booting from the default location, the device can also be configured to boot from the backup RAM. In the case of booting from backup RAM, it is also possible to keep the flashes disabled. In this mode:
 - the core, the flashes, and almost all peripherals and memories are powered down.
 - all power domains except power domain #0 are configurable in order to reduce leakage consumption.

3.2.4 Activation of a MCU reset

3.2.5 Provides a service to get the reset reason from hardware

3.2.6 eDMA Control Register (*EDMA_CR*), eDMA Channel *n* Priority Register (*EDMA_CPRn*) and Channel configuration Registers (*CHCONFIGn*) are initialized in MCU (the DMA is available only for MPC5604B devices)

The enhanced direct memory access (eDMA) controller is able to perform movements of data block from a source to a destination with minimal intervention from the host processor. The source or destination can be a memory block or an I/O block capable of operation with the eDMA. The transfer can be requested by a source like a peripheral via one of the DMA channels.

The DMA Channel Mux (DMA_MUX) allows to assign a DMA source to a DMA channel. The multiplexer is needed because the sources are twenty seven against sixteen channels. Associating multiple DMA channels with the same source will result in unpredictable behaviour of the DMA module. A channel configuration register *CHCONFIG#n* relating to the DMA channel *n* offers a 6 bit *source* field that specifies which DMA source is routed to the given DMA channel *n*. For example if the source #1 (DSPI_0) has to be configured for use with DMA channel 2 then the field *source* of the *CHCONFIG2* register must be written with the value 0x5.

Multiple service requests from different sources Arbitration can be configured to use either fixed-priority or round-robin policy. In the former, the highest priority channel requesting service is selected to execute. In particular a unique priority is associated with each DMA channel and is expressed by a numeric value which belongs to the interval [0..15] where 0 means the lowest priority, 1 is the next higher priority and so on until 15 which is the highest priority. Channel preemption can be enabled to allow the executing channel's data transfers to be temporarily suspended in favor of starting a higher priority channel. After the preempting channel has completed, the preempted channel is restored and resumes execution. Nested preemption (attempting to preempt a preempting channel) is not supported. Preemption is only available when fixed arbitration is selected. In round-robin arbitration mode, the channel priorities are ignored and the channels are cycled through, from channel 15 down to channel 0. The eDMA control register (*DMA_CR*) can be configured to enable the round-robin or the fixed priority arbitration

mechanism. The eDMA channel n Priority Register (*DCHPRIn*) can be programmed by software to establish the preemption (on/off) and the arbitration priority (0..15) of the DMA channel n.

3.3 Driver Limitations / Restrictions

In the Driver there are following limitations and/or restrictions.

DMAMUX Configuration:Limitation for Bolero256k.

For Bolerosmall256K,there are 16 sources to route to 16 channels.But in actual only 15 sources are available to route.So all these 15 sources could map to 0 to 14 th any channel,channel 15th is not configurable.

Crystal clock monitor:Limitation for Bolero256k,512 and 1M

Functional FXOSC monitoring can only be guaranteed when the FXOSC frequency is greater than $(FIRC / 2RCDIV) + 0.5$ MHz.

FMPLL clock monitor:Limitation for Bolero256k,512 and 1M

Functional FMPLL monitoring can only be guaranteed when the FMPLL frequency is greater than $(FIRC / 4) + 0.5$ MHz.

3.4 Deviation from Requirements

The driver deviates from the AUTOSAR MCU Driver software specification in some places.

There are also some additional requirements (on top of requirements detailed in AUTOSAR MCU Driver software specification) which need to be satisfied for correct operation.

Table 3-1. Deviations Status Column Description

Term	Definition
N/A	Not available
N/T	Not testable
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, or out of scope for the driver.

Table 3-2. Driver Deviations Table

Requirement	Status	Description	Notes
MCU200	N/I	Each variable that shall be accessible by AUTOSAR Debugging, shall be defined as global variable.	Not implemented
MCU201	N/I	All type definitions of variables which shall be debugged, shall be accessible by the header file Mcu.h.	Not implemented
MCU202	N/I	The declaration of variables in the header file shall allow to calculate the size of the variables by C-"sizeof".	Not implemented
MCU203	N/I	Variables available for debugging shall be described in the respective Basic Software Module Description.	Not implemented
MCU215	N/F	The type definitions for Mcu_Lcfg.c and Mcu_PBcfg.c are located in the file Mcu.h.	Not implemented because the platform-dependent types are located in Mcu_Cfg.h
MCU216	N/A	Mcu_Lcfg.c shall include Mcu_Cbk.h for a link time configuration if the call back function is linked to the module via the ROM structure.	No callback functions in Mcu driver
MCU218	N/A	Mcu_PBcfg.c shall include Mcu_Cbk.h for post build time configuration if the call back function is linked to the module via the ROM structure.	No callback functions in Mcu driver
MCU219	N/A	Mcu.c shall include Mcu_Cbk.h for pre-compile time configuration.	No callback functions in Mcu driver

3.5 Runtime Errors

The driver generates the following DEM errors at runtime.

Table 3-3. Runtime Errors

Function	Error Code	Condition triggering the error
Mcu_Apply_Mode()	MCU_E_TIMEOUT_TRANSITION	TIME transition timeout expired.
Mcu_Check_Init()	MCU_E_TIMEOUT_TRANSITION	Timeout expired. Mode and system clock are not configured correctly.
Mcu_Report_ClkFail()	MCU_E_QUARTZ_FAILURE	Oscillator frequency lower than RC frequency event.
Mcu_Report_ClkFail()	MCU_E_CLOCK_FAILURE	1. FMPLL clock frequency lower than low reference event. 2. FMPLL clock frequency higher than high reference interrupt.

Table continues on the next page...

Table 3-3. Runtime Errors (continued)

Function	Error Code	Condition triggering the error
Mcu_Report_ClkFail()	MCU_E_CLOCK_FAILURE	A FMPLL fail event has occurred.
Mcu_Report_ClkFail()	MCU_E_CLOCK_FAILURE	Oscillator stabilization Timeout expired (for 32 KHz or 128 kHz oscillator).
Mcu_Report_ClkFail()	MCU_E_CLOCK_FAILURE	Oscillator stabilization Timeout expired (for 32 KHz or 128 kHz oscillator).
Mcu_Report_ClkFail()	MCU_E_LOCK_FAILURE	The FMPLL has lost the lock

3.6 Software specification

The following sections contains driver software specifications.

3.6.1 Define Reference

Constants supported by the MCU driver.

3.6.1.1 Define MCU_APPLY_LOWPOWER_ID

API service ID of Mcu_Apply_LowPower_Mode function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03018

**Table 3-4. Define MCU_APPLY_LOWPOWER_ID
Description**

Name	MCU_APPLY_LOWPOWER_ID
Initializer	((uint8)0x23U)

3.6.1.2 Define MCU_APPLY_MODE_ID

Service ID of Mcu_Apply_Mode function.

Details:

Parameter used when raising an error/exception

Implements:**Table 3-5. Define MCU_APPLY_MODE_ID Description**

Name	MCU_APPLY_MODE_ID
Initializer	((uint8)0x25U)

3.6.1.3 Define MCU_APPLY_RUNMODE_ID

API service ID of Mcu_Apply_LowPower_Mode function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03019**Table 3-6. Define MCU_APPLY_RUNMODE_ID Description**

Name	MCU_APPLY_RUNMODE_ID
Initializer	((uint8)0x24U)

3.6.1.4 Define MCU_CHANGE_LOWPOWER_ID

API service ID of Mcu_Change_LowPower_Mode function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03017**Table 3-7. Define MCU_CHANGE_LOWPOWER_ID Description**

Name	MCU_CHANGE_LOWPOWER_ID
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Table continues on the next page...

**Table 3-7. Define MCU_CHANGE_LOWPOWER_ID Description
(continued)**

Initializer	((uint8)0x22U)
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3.6.1.5 Define MCU_CHANGE_MODE_ID

API service ID of Mcu_Change_Mode function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03016

Table 3-8. Define MCU_CHANGE_MODE_ID Description

Name	MCU_CHANGE_MODE_ID
Initializer	((uint8)0x21U)

3.6.1.6 Define MCU_DISABLE_CMU_ID

API service ID of Mcu_Disable_CMU function.

Details:

Parameters used when raising an error/exception

Implements: PR-MCAL-3319

Table 3-9. Define MCU_DISABLE_CMU_ID Description

Name	MCU_DISABLE_CMU_ID
Initializer	((uint8)0x26U)

3.6.1.7 Define MCU_DISTRIBUTEPLLCLOCK_ID

API service ID of Mcu_DistributePllClock function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03006

Table 3-10. Define MCU_DISTRIBUTEPLLCLOCK_ID Description

Name	MCU_DISTRIBUTEPLLCLOCK_ID
Initializer	((uint8)0x03U)

3.6.1.8 Define MCU_E_CLOCK_UNINIT

Error: Mcu_InitClock() was not executed before Mcu_GetPllStatus() or Mcu_DistributePllClock().

Details:

Errors and exceptions that will be detected by the MCU driver

Implements: DMCU03027

Table 3-11. Define MCU_E_CLOCK_UNINIT Description

Name	MCU_E_CLOCK_UNINIT
Initializer	((uint8)0x11U)

3.6.1.9 Define MCU_E_MODE_NOT_ENABLED

Disabled mode transition error.

Details:

Mcu_Apply_Mode() tries to apply a disabled mode

Implements:

Table 3-12. Define MCU_E_MODE_NOT_ENABLED Description

Name	MCU_E_MODE_NOT_ENABLED
Initializer	((uint8)0x11U)

3.6.1.10 Define MCU_E_PARAM_CLOCK

ClockSetting shall be within the settings defined in the configuration.

Details:

Errors and exceptions that will be detected by the MCU driver

Implements: DMCU03021

Table 3-13. Define MCU_E_PARAM_CLOCK Description

Name	MCU_E_PARAM_CLOCK
Initializer	((uint8)0x0BU)

3.6.1.11 Define MCU_E_PARAM_CMU

CMU shall be within the sections defined in the configuration data.

Details:

Errors and exceptions that will be detected by the MCU driver

Implements: PR-MCAL-3319

Table 3-14. Define MCU_E_PARAM_CMU Description

Name	MCU_E_PARAM_CMU
Initializer	((uint8)0x12U)

3.6.1.12 Define MCU_E_PARAM_CONFIG

API service called with wrong parameter.

Details:

Errors and exceptions that will be detected by the MCU driver

Implements: DMCU03020

Table 3-15. Define MCU_E_PARAM_CONFIG Description

Name	MCU_E_PARAM_CONFIG
Initializer	((uint8)0x0AU)

3.6.1.13 Define MCU_E_PARAM_MODE

McuMode shall be within the modes defined in the configuration data.

Details:

Errors and exceptions that will be detected by the MCU driver

Implements: DMCU03022

Table 3-16. Define MCU_E_PARAM_MODE Description

Name	MCU_E_PARAM_MODE
Initializer	((uint8)0x0CU)

3.6.1.14 Define MCU_E_PARAM_POINTER

The error MCU_E_PARAM_POINTER shall be reported in case the value is a NULL pointer in Mcu_GetVersionInfo() API.

Details:

If development error detection is enabled, the parameter versioninfo shall be checked for being NULL.

Implements: DMCU03026

Table 3-17. Define MCU_E_PARAM_POINTER Description

Name	MCU_E_PARAM_POINTER
Initializer	((uint8)0x10U)

3.6.1.15 Define MCU_E_PARAM_RAMSECTION

RamSection shall be within the sections defined in the configuration data.

Details:

Errors and exceptions that will be detected by the MCU driver

Implements: DMCU03023

Table 3-18. Define MCU_E_PARAM_RAMSECTION Description

Name	MCU_E_PARAM_RAMSECTION
Initializer	((uint8)0x0DU)

3.6.1.16 Define MCU_E_PLL_NOT_LOCKED

A error shall be reported if the status of the PLL is detected as not locked.

Details:

Errors and exceptions that will be detected by the MCU driver

Implements: DMCU03024

Table 3-19. Define MCU_E_PLL_NOT_LOCKED Description

Name	MCU_E_PLL_NOT_LOCKED
Initializer	((uint8)0x0EU)

3.6.1.17 Define MCU_E_UNINIT

API service used without module initialization.

Details:

Errors and exceptions that will be detected by the MCU driver

Implements: DMCU03025

Table 3-20. Define MCU_E_UNINIT Description

Name	MCU_E_UNINIT
Initializer	((uint8)0x0FU)

3.6.1.18 Define MCU_E_WRONG_MODE

CER Error.

Details:

Wrong mode transition

Implements: DMCU03029

Table 3-21. Define MCU_E_WRONG_MODE Description

Name	MCU_E_WRONG_MODE
Initializer	((uint8)0x20U)

3.6.1.19 Define MCU_ENABLECLOCKOUTPUT_ID

API service ID of Mcu_Enable_ClockOutput function.

Details:

Parameters used when raising an error/exception

Implements:

Table 3-22. Define MCU_ENABLECLOCKOUTPUT_ID Description

Name	MCU_ENABLECLOCKOUTPUT_ID
Initializer	((uint8)0x20U)

3.6.1.20 Define MCU_GETPLLSTATUS_ID

API service ID of Mcu_GetPllStatus function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03007

Table 3-23. Define MCU_GETPLLSTATUS_ID Description

Name	MCU_GETPLLSTATUS_ID
Initializer	((uint8)0x04U)

3.6.1.21 Define MCU_GETTRAMSTATE_ID

API service ID of Mcu_GetVersionInfo function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03013

Table 3-24. Define MCU_GETTRAMSTATE_ID Description

Name	MCU_GETTRAMSTATE_ID
Initializer	((uint8)0x10U)

3.6.1.22 Define MCU_GETRESETRAWVALUE_ID

API service ID of Mcu_GetResetRawValue function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03009

**Table 3-25. Define MCU_GETRESETRAWVALUE_ID
Description**

Name	MCU_GETRESETRAWVALUE_ID
Initializer	((uint8)0x06U)

3.6.1.23 Define MCU_GETRESETREASON_ID

API service ID of Mcu_GetResetReason function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03008

**Table 3-26. Define MCU_GETRESETREASON_ID
Description**

Name	MCU_GETRESETREASON_ID
Initializer	((uint8)0x05U)

3.6.1.24 Define MCU_GETVERSIONINFO_ID

API service ID of Mcu_GetVersionInfo function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03012

Table 3-27. Define MCU_GETVERSIONINFO_ID Description

Name	MCU_GETVERSIONINFO_ID
Initializer	((uint8)0x09U)

3.6.1.25 Define MCU_INIT_ID

API service ID of Mcu_Init function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03003

Table 3-28. Define MCU_INIT_ID Description

Name	MCU_INIT_ID
Initializer	((uint8)0x00U)

3.6.1.26 Define MCU_INITCLOCK_ID

API service ID of Mcu_InitClock function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03005

Table 3-29. Define MCU_INITCLOCK_ID Description

Name	MCU_INITCLOCK_ID
Initializer	((uint8)0x02U)

3.6.1.27 Define MCU_INITOSC_ID

API service ID of Mcu_Init_Osc function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03015

Table 3-30. Define MCU_INITOSC_ID Description

Name	MCU_INITOSC_ID
Initializer	((uint8)0x20U)

3.6.1.28 Define MCU_INITRAMSECTION_ID

API service ID of Mcu_InitRamSection function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03004

Table 3-31. Define MCU_INITRAMSECTION_ID Description

Name	MCU_INITRAMSECTION_ID
Initializer	((uint8)0x01U)

3.6.1.29 Define MCU_INSTANCE_ID

MCU Instance ID.

Implements: DMCU03014

Table 3-32. Define MCU_INSTANCE_ID Description

Name	MCU_INSTANCE_ID
Initializer	((uint8)0x0U)

3.6.1.30 Define MCU_PERFORMRESET_ID

API service ID of Mcu_PerformReset function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03010

**Table 3-33. Define MCU_PERFORMRESET_ID
Description**

Name	MCU_PERFORMRESET_ID
Initializer	((uint8)0x07U)

3.6.1.31 Define MCU_RAW_RESET_DEFAULT

Raw Reset Value returned in a case when Mcu_GetResetRawValue() is called before Mcu_Init() is called.

Details:

Errors and exceptions that will be detected by the MCU driver

Implements: DMCU03028

**Table 3-34. Define MCU_RAW_RESET_DEFAULT
Description**

Name	MCU_RAW_RESET_DEFAULT
Initializer	((uint32)0xFFFFFFFFFUL)

3.6.1.32 Define MCU_SETMODE_ID

API service ID of Mcu_SetMode function.

Details:

Parameters used when raising an error/exception

Implements: DMCU03011

Table 3-35. Define MCU_SETMODE_ID Description

Name	MCU_SETMODE_ID
Initializer	((uint8)0x08U)

3.6.1.33 Define MCU_CONF_PC

Switches the Production Error Detection and Notification OFF.

Implements: DMCU06059

Table 3-36. Define MCU_CONF_PC Description

Name	MCU_CONF_PC
Initializer	extern CONST(Mcu_ConfigType, MCU_CONST) MCU_INIT_CONFIG_PC;

3.6.1.34 Define MCU_DEV_ERROR_DETECT

development error detection enabled/disabled (STD_ON/STD_OFF)

Implements: DMCU06001

Implements: DMCU04009

Table 3-37. Define MCU_DEV_ERROR_DETECT Description

Name	MCU_DEV_ERROR_DETECT
Initializer	([!IF "McuGeneralConfiguration/McuDevErrorDetect"!])STD_ON[! ELSE!]STD_OFF[!ENDIF!])

3.6.1.35 Define MCU_DMA_INIT

dma initialization is enabled/disabled from the MCU driver (TRUE/FALSE)

Table 3-38. Define MCU_DMA_INIT Description

Name	MCU_DMA_INIT
Initializer	([!IF "McuGeneralConfiguration/McuDmaInit"!])STD_ON[!ELSE!]STD_OFF[!ENDIF!])

3.6.1.36 Define MCU_EMIOS_CONFIGURE_GPREN_API

The Api function Mcu_eMios_Configure_GPREN is/isn't available (STD_ON/STD_OFF).

Table 3-39. Define MCU_EMIOS_CONFIGURE_GPREN_API Description

Name	MCU_EMIOS_CONFIGURE_GPREN_API
Initializer	([!IF "McuGeneralConfiguration/McucMiosConfigureGPRENApi"!])(STD_ON)[!ELSE!](STD_OFF)[!ENDIF!])

3.6.1.37 Define MCU_ENABLE_DISABLE_NOTIFICATION_API

The callout clock notification is/isn't available (STD_ON/STD_OFF).

Table 3-40. Define MCU_ENABLE_DISABLE_NOTIFICATION_API Description

Name	MCU_ENABLE_DISABLE_NOTIFICATION_API
Initializer	([!IF "McuGeneralConfiguration/McuEnableDisableNotificationApi"!])STD_ON[!ELSE!]STD_OFF[!ENDIF!])

3.6.1.38 Define MCU_ENTER_LOW_POWER_MODE

if enter low power mode transition enabled (STD_ON/STD_OFF)

Implements: DMCU06068

Table 3-41. Define MCU_ENTER_LOW_POWER_MODE
Description

Name	MCU_ENTER_LOW_POWER_MODE
Initializer	([!IF "McuGeneralConfiguration/McuEnterLowPowerMode = 'true'"!]STD_ON[!ELSE!]STD_OFF[!ENDIF!])

3.6.1.39 Define MCU_FLASH_CONFIGURATION_USED

the Api function Mcu_Flash_Configure is/isn't available (STD_ON/STD_OFF)

Table 3-42. Define MCU_FLASH_CONFIGURATION_USED
Description

Name	MCU_FLASH_CONFIGURATION_USED
Initializer	([!IF "McuGeneralConfiguration/McuConfigureFlash = 'true'"!]STD_ON[!ELSE!]STD_OFF[!ENDIF!])

3.6.1.40 Define MCU_GET_DEVICE_INFO_API

the Api function Mcu_GetDeviceInfo is/isn't available (STD_ON/STD_OFF)

Table 3-43. Define MCU_GET_DEVICE_INFO_API Description

Name	MCU_GET_DEVICE_INFO_API
Initializer	([!IF "NonAutosar/McuGetDeviceInfoApi"!]STD_ON[!ELSE!]STD_OFF[!ENDIF!])

3.6.1.41 Define MCU_GET_RAM_STATE_API

the Api function Mcu_GetRamState is/isn't available (STD_ON/STD_OFF)

Table 3-44. Define MCU_GET_RAM_STATE_API Description

Name	MCU_GET_RAM_STATE_API
Initializer	([!IF "McuGeneralConfiguration/McuGetRamStateApi"!]STD_ON[!ELSE!]STD_OFF[!ENDIF!])

3.6.1.42 Define MCU_INIT_CLOCK

clock initialization is enabled/disabled from the MCU driver (TRUE/FALSE)

Table 3-45. Define MCU_INIT_CLOCK Description

Name	MCU_INIT_CLOCK
Initializer	([!IF "McuGeneralConfiguration/McuInitClock"!] TRUE[!ELSE!]FALSE[! ENDIF!])

3.6.1.43 Define MCU_INIT_CLOCK_DISABLE_PERIPHERAL

enable/disable the fact to call Mcu_Disable_Peripheral that gate all peripherals during clock initialization done by MCU driver (TRUE/FALSE)

Table 3-46. Define MCU_INIT_CLOCK_DISABLE_PERIPHERAL Description

Name	MCU_INIT_CLOCK_DISABLE_PERIPHERAL
Initializer	([!IF "McuGeneralConfiguration/ McuDisablePeripheralDuringInitclock"!] TRUE[!ELSE!]FALSE[! ENDIF!])

3.6.1.44 Define MCU_LP_TRANSITION_CHECK

if low power mode transition check should be enabled (STD_ON/STD_OFF)

Table 3-47. Define MCU_LP_TRANSITION_CHECK Description

Name	MCU_LP_TRANSITION_CHECK
Initializer	([!IF "McuGeneralConfiguration/McuLPTransitionCheck = 'true'"!] STD_ON[! ELSE!]STD_OFF[! ENDIF!])

3.6.1.45 Define MCU_NO_PLL

MCU_DistributePllClock isn't/is available (TRUE/FALSE).

Table 3-48. Define MCU_NO_PLL Description

Name	MCU_NO_PLL
Initializer	([!IF "McuGeneralConfiguration/McuNoPll"!] TRUE [!ELSE!] FALSE [!ENDIF!])

3.6.1.46 Define MCU_PERFORM_RESET_API

the Api function Mcu_PerformReset is/isn't available (STD_ON/STD_OFF)

Table 3-49. Define MCU_PERFORM_RESET_API Description

Name	MCU_PERFORM_RESET_API
Initializer	([!IF "McuGeneralConfiguration/McuPerformResetApi"!] STD_ON [!ELSE!] STD_OFF [!ENDIF!])

3.6.1.47 Define MCU_SKIP_FUNCTION

The remaining settings are/aren't applied after a missed transition (STD_ON/STD_OFF).

Table 3-50. Define MCU_SKIP_FUNCTION Description

Name	MCU_SKIP_FUNCTION
Initializer	([!IF "McuGeneralConfiguration/McuSkipFunction"!] STD_ON [!ELSE!] STD_OFF [!ENDIF!])

3.6.1.48 Define MCU_SSCM_ERROR_CONFIG

SSCM ERROR configurations is enabled/disabled (STD_ON/STD_OFF).

Table 3-51. Define MCU_SSCM_ERROR_CONFIG Description

Name	MCU_SSCM_ERROR_CONFIG
Initializer	([!IF "(McuGeneralConfiguration/McuSscmRAE[.='true'] or McuGeneralConfiguration/McuSscmPAE[.='true'])"!] STD_ON [!ELSE!] STD_OFF [!ENDIF!])

3.6.1.49 Define MCU_SSCM_ERROR_CONFIG

SSCM ERROR configurations is enabled/disabled (STD_ON/STD_OFF).

Table 3-52. Define MCU_SSCM_ERROR_CONFIG
Description

Name	MCU_SSCM_ERROR_CONFIG
Initializer	([!IF "(McuGeneralConfiguration/McuSscmRAE[.='true'] or McuGeneralConfiguration/McuSscmPAE[.='true'])"!])STD_ON[! ELSE!]STD_OFF[!ENDIF!])

3.6.1.50 Define MCU_SSCM_ERROR_PAE_ENABLE

SSCM Error Peripheral Bus Abort is enabled/disabled (STD_ON/STD_OFF).

Table 3-53. Define MCU_SSCM_ERROR_PAE_ENABLE
Description

Name	MCU_SSCM_ERROR_PAE_ENABLE
Initializer	([!IF "McuGeneralConfiguration/McuSscmPAE[.='true']"!])STD_ON[! ELSE!]STD_OFF[!ENDIF!])

3.6.1.51 Define MCU_SSCM_ERROR_RAE_ENABLE

SSCM Error Register Bus Abort is enabled/disabled (STD_ON/STD_OFF).

Table 3-54. Define MCU_SSCM_ERROR_RAE_ENABLE
Description

Name	MCU_SSCM_ERROR_RAE_ENABLE
Initializer	([!IF "McuGeneralConfiguration/McuSscmRAE[.='true']"!])STD_ON[! ELSE!]STD_OFF[!ENDIF!])

3.6.1.52 Define MCU_SSCM_GET_MEMCONFIG_API

the Api function Mcu_SscmGetMemConfig is/isn't available (STD_ON/STD_OFF)

Table 3-55. Define MCU_SSCM_GET_MEMCONFIG_API
Description

Name	MCU_SSCM_GET_MEMCONFIG_API
Initializer	((!!IF "NonAutosar/McuSscmGetMemConfig"!)STD_ON[!ELSE!]STD_OFF[! ENDIF!])

3.6.1.53 Define MCU_SSCM_GET_STATUS_API

the Api function Mcu_SscmGetStatus is/isn't available (STD_ON/STD_OFF)

Table 3-56. Define MCU_SSCM_GET_STATUS_API
Description

Name	MCU_SSCM_GET_STATUS_API
Initializer	((!!IF "NonAutosar/McuSscmGetSystemStatus"!)STD_ON[!ELSE!]STD_OFF[! ENDIF!])

3.6.1.54 Define MCU_SSCM_GET_UOPS_API

the Api function Mcu_SscmGetUops is/isn't available (STD_ON/STD_OFF)

Table 3-57. Define MCU_SSCM_GET_UOPS_API
Description

Name	MCU_SSCM_GET_UOPS_API
Initializer	(STD_OFF)

3.6.1.55 Define MCU_VERSION_INFO_API

version info API enabled/disabled (STD_ON/STD_OFF)

Implements: DMCU04010

Note

BSW00411 satisfied by MCU103, MCU104

Table 3-58. Define MCU_VERSION_INFO_API Description

Name	MCU_VERSION_INFO_API
Initializer	([!IF "McuGeneralConfiguration/McuVersionInfoApi"!] STD_ON[! ELSE!]STD_OFF[!ENDIF!])

3.6.2 Macros Reference

Macros supported by the MCU driver.

3.6.2.1 Macro Dma_ack_interrupt

Acknowledge a DMA interrupt.

Details:

Clears the interrupt request bit for a specific DMA channel

Prototype: `Dma_ack_interrupt(dma_channel)`

Table 3-59. Dma_ack_interrupt Arguments

Name	Direction	Description
dma_channel	input	Dma channel for which to acknowledge the interrupt.

3.6.2.2 Macro Dma_configure_channel

Configure a DMA transfer.

Details:

Configures the DMA Transfer Control Descriptor

Prototype:

```
Dma_configure_channel(dma_channel,src_addr,src_transfer_size,dest_transfer_size,src_next_offset,n_bytes_to_transfer,src_last_adj,dest_addr,crt_iter_cnt,dest_next_offset,dlast_sg_addr,begin_iter_cnt,tcd_flags)
```

Table 3-60. Dma_configure_channel Arguments

Name	Direction	Description
dma_channel	input	Dma channel for which to disable the interrupt.
src_addr	input	Source address for DMA transfer.
src_transfer_size	input	Source access transfer size.
dest_transfer_size	input	Destination access transfer size.
src_next_offset	input	Offset to be applied to compute next src addr.
n_bytes_to_transfer	input	Number of bytes to transfer.
src_last_adj	input	Adjustment to be applied to src addr at the completion of the outer major iteration count.
dest_addr	input	Destination address for DMA transfer.
crt_iter_cnt	input	Current major iteration count.
dest_next_offset	input	Offset to be applied to compute next destination address.
dlast_sg_addr	input	Adjustment to be applied to destination address at the completion of the outer major iteration count (if E_SG bit is 0), or scatter/gather address (if E_SG bit is 1).
begin_iter_cnt	input	Beginning major iteration count.
tcd_flags	input	Mask to configure E_SG(bit4)/D_REQ(bit3)/INT_HALF(bit2)/INT_MAJ(bit1) bits of TCD.

3.6.2.3 Macro Dma_start_channel

Start the transfer on a specific DMA channel.

Details:

Start a DMA transfer

Prototype: `Dma_start_channel(dma_channel)`

Table 3-61. Dma_start_channel Arguments

Name	Direction	Description
dma_channel	input	Dma channel for which to disable the interrupt.

3.6.2.4 Macro MCU_CLOCK_NOTIFICATION_CALLOUT

Prototype: `MCU_CLOCK_NOTIFICATION_CALLOUT(cmuIndex)`

3.6.2.5 Macro MCU_EMIOS_GPREN_CLEAR

Prototype: `MCU_EMIOS_GPREN_CLEAR(emios_idx)`

3.6.2.6 Macro MCU_EMIOS_GPREN_Conf

Prototype: `MCU_EMIOS_GPREN_Conf(emios_idx,value)`

3.6.2.7 Macro MCU_EMIOS_GPREN_SET

Prototype: `MCU_EMIOS_GPREN_SET(emios_idx)`

3.6.2.8 Macro CLEAR_DMA_INTC

Prototype: `CLEAR_DMA_INTC(dma_channel)`

3.6.2.9 Macro DMA_CDNE

Prototype: `DMA_CDNE(mIdx)`

3.6.2.10 Macro DMA_CEEI

Prototype: `DMA_CEEI(mIdx)`

3.6.2.11 Macro DMA_CERQ

Prototype: DMA_CERQ (mIdx)

3.6.2.12 Macro DMA_CERR

Prototype: DMA_CERR (mIdx)

3.6.2.13 Macro DMA_CINT

Prototype: DMA_CINT (mIdx)

3.6.2.14 Macro DMA_CPR

Prototype: DMA_CPR (Channel)

3.6.2.15 Macro DMA_CPR0

Prototype: DMA_CPR0 (mIdx)

3.6.2.16 Macro DMA_CPR1

Prototype: DMA_CPR1 (mIdx)

3.6.2.17 Macro DMA_CPR10

Prototype: DMA_CPR10 (mIdx)

3.6.2.18 Macro DMA_CPR11

Prototype: DMA_CPR11 (mIdx)

3.6.2.19 Macro DMA_CPR12

Prototype: DMA_CPR12 (mIdx)

3.6.2.20 Macro DMA_CPR13

Prototype: DMA_CPR13 (mIdx)

3.6.2.21 Macro DMA_CPR14

Prototype: DMA_CPR14 (mIdx)

3.6.2.22 Macro DMA_CPR15

Prototype: DMA_CPR15 (mIdx)

3.6.2.23 Macro DMA_CPR16

Prototype: DMA_CPR16 (mIdx)

3.6.2.24 Macro DMA_CPR17

Prototype: DMA_CPR17 (mIdx)

3.6.2.25 Macro DMA_CPR18

Prototype: DMA_CPR18 (mIdx)

3.6.2.26 Macro DMA_CPR19

Prototype: DMA_CPR19 (mIdx)

3.6.2.27 Macro DMA_CPR2

Prototype: DMA_CPR2 (mIdx)

3.6.2.28 Macro DMA_CPR20

Prototype: DMA_CPR20 (mIdx)

3.6.2.29 Macro DMA_CPR21

Prototype: DMA_CPR21 (mIdx)

3.6.2.30 Macro DMA_CPR22

Prototype: DMA_CPR22 (mIdx)

3.6.2.31 Macro DMA_CPR23

Prototype: DMA_CPR23 (mIdx)

3.6.2.32 Macro DMA_CPR24

Prototype: DMA_CPR24 (mIdx)

3.6.2.33 Macro DMA_CPR25

Prototype: DMA_CPR25 (mIdx)

3.6.2.34 Macro DMA_CPR26

Prototype: DMA_CPR26 (mIdx)

3.6.2.35 Macro DMA_CPR27

Prototype: DMA_CPR27 (mIdx)

3.6.2.36 Macro DMA_CPR28

Prototype: DMA_CPR28 (mIdx)

3.6.2.37 Macro DMA_CPR29

Prototype: DMA_CPR29 (mIdx)

3.6.2.38 Macro DMA_CPR3

Prototype: DMA_CPR3 (mIdx)

3.6.2.39 Macro DMA_CPR30

Prototype: DMA_CPR30 (mIdx)

3.6.2.40 Macro DMA_CPR31

Prototype: DMA_CPR31 (mIdx)

3.6.2.41 Macro DMA_CPR32

Prototype: DMA_CPR32 (mIdx)

3.6.2.42 Macro DMA_CPR33

Prototype: DMA_CPR33 (mIdx)

3.6.2.43 Macro DMA_CPR34

Prototype: DMA_CPR34 (mIdx)

3.6.2.44 Macro DMA_CPR35

Prototype: DMA_CPR35 (mIdx)

3.6.2.45 Macro DMA_CPR36

Prototype: DMA_CPR36 (mIdx)

3.6.2.46 Macro DMA_CPR37

Prototype: DMA_CPR37 (mIdx)

3.6.2.47 Macro DMA_CPR38

Prototype: DMA_CPR38 (mIdx)

3.6.2.48 Macro DMA_CPR39

Prototype: DMA_CPR39 (mIdx)

3.6.2.49 Macro DMA_CPR4

Prototype: DMA_CPR4 (mIdx)

3.6.2.50 Macro DMA_CPR40

Prototype: DMA_CPR40 (mIdx)

3.6.2.51 Macro DMA_CPR41

Prototype: DMA_CPR41 (mIdx)

3.6.2.52 Macro DMA_CPR42

Prototype: DMA_CPR42 (mIdx)

3.6.2.53 Macro DMA_CPR43

Prototype: DMA_CPR43 (mIdx)

3.6.2.54 Macro DMA_CPR44

Prototype: DMA_CPR44 (mIdx)

3.6.2.55 Macro DMA_CPR45

Prototype: DMA_CPR45 (mIdx)

3.6.2.56 Macro DMA_CPR46

Prototype: DMA_CPR46 (mIdx)

3.6.2.57 Macro DMA_CPR47

Prototype: DMA_CPR47 (mIdx)

3.6.2.58 Macro DMA_CPR48

Prototype: DMA_CPR48 (mIdx)

3.6.2.59 Macro DMA_CPR49

Prototype: DMA_CPR49 (mIdx)

3.6.2.60 Macro DMA_CPR5

Prototype: DMA_CPR5 (mIdx)

3.6.2.61 Macro DMA_CPR50

Prototype: DMA_CPR50 (mIdx)

3.6.2.62 Macro DMA_CPR51

Prototype: DMA_CPR51 (mIdx)

3.6.2.63 Macro DMA_CPR52

Prototype: DMA_CPR52 (mIdx)

3.6.2.64 Macro DMA_CPR53

Prototype: DMA_CPR53 (mIdx)

3.6.2.65 Macro DMA_CPR54

Prototype: DMA_CPR54 (mIdx)

3.6.2.66 Macro DMA_CPR55

Prototype: DMA_CPR55 (mIdx)

3.6.2.67 Macro DMA_CPR56

Prototype: DMA_CPR56 (mIdx)

3.6.2.68 Macro DMA_CPR57

Prototype: DMA_CPR57 (mIdx)

3.6.2.69 Macro DMA_CPR58

Prototype: DMA_CPR58 (mIdx)

3.6.2.70 Macro DMA_CPR59

Prototype: DMA_CPR59 (mIdx)

3.6.2.71 Macro DMA_CPR6

Prototype: DMA_CPR6 (mIdx)

3.6.2.72 Macro DMA_CPR60

Prototype: DMA_CPR60 (mIdx)

3.6.2.73 Macro DMA_CPR61

Prototype: DMA_CPR61 (mIdx)

3.6.2.74 Macro DMA_CPR62

Prototype: DMA_CPR62 (mIdx)

3.6.2.75 Macro DMA_CPR63

Prototype: DMA_CPR63 (mIdx)

3.6.2.76 Macro DMA_CPR7

Prototype: DMA_CPR7 (mIdx)

3.6.2.77 Macro DMA_CPR8

Prototype: DMA_CPR8 (mIdx)

3.6.2.78 Macro DMA_CPR9

Prototype: DMA_CPR9 (mIdx)

3.6.2.79 Macro DMA_CR

Prototype: DMA_CR (mIdx)

3.6.2.80 Macro DMA_EEIH

Prototype: DMA_EEIH (mIdx)

3.6.2.81 Macro DMA_EEIL

Prototype: DMA_EEIL (mIdx)

3.6.2.82 Macro DMA_EERH

Prototype: DMA_EERH (mIdx)

3.6.2.83 Macro DMA_EERL

Prototype: DMA_EERL (mIdx)

3.6.2.84 Macro DMA_ERQH

Prototype: DMA_ERQH (mIdx)

3.6.2.85 Macro DMA_ERQL

Prototype: DMA_ERQL (mIdx)

3.6.2.86 Macro DMA_ES

Prototype: DMA_ES (mIdx)

3.6.2.87 Macro DMA_GPOR

Prototype: DMA_GPOR (mIdx)

3.6.2.88 Macro DMA_HRSH

Prototype: DMA_HRSH (mIdx)

3.6.2.89 Macro DMA_HRSL

Prototype: DMA_HRSL (mIdx)

3.6.2.90 Macro DMA_INTH

Prototype: DMA_INTH (mIdx)

3.6.2.91 Macro DMA_INTL

Prototype: DMA_INTL (mIdx)

3.6.2.92 Macro DMA_MOD_BASE_ADDR

macro to get the DMA Module address by module index

Prototype: DMA_MOD_BASE_ADDR (mIdx)

3.6.2.93 Macro DMA_MOD_BASE_ADDR_CH

macro to get the DMA Module base address by channel

Prototype: DMA_MOD_BASE_ADDR_CH (channel)

3.6.2.94 Macro DMA_MOD_IDX

macro to get the DMA Module index by channel

Prototype: DMA_MOD_IDX (channel)

3.6.2.95 Macro DMA_SEEI

Prototype: DMA_SEEI (mIdx)

3.6.2.96 Macro DMA_SERQ

Prototype: DMA_SERQ (mIdx)

3.6.2.97 Macro DMA_SSRT

Prototype: DMA_SSRT (mIdx)

3.6.2.98 Macro DMA_TCD

Prototype: DMA_TCD (Channel)

3.6.2.99 Macro DMA_TCD0

Prototype: DMA_TCD0 (mIdx)

3.6.2.100 Macro DMA_TCD1

Prototype: DMA_TCD1 (mIdx)

3.6.2.101 Macro DMA_TCD10

Prototype: DMA_TCD10 (mIdx)

3.6.2.102 Macro DMA_TCD11

Prototype: DMA_TCD11 (mIdx)

3.6.2.103 Macro DMA_TCD12

Prototype: DMA_TCD12 (mIdx)

3.6.2.104 Macro DMA_TCD13

Prototype: DMA_TCD13 (mIdx)

3.6.2.105 Macro DMA_TCD14

Prototype: DMA_TCD14 (mIdx)

3.6.2.106 Macro DMA_TCD15

Prototype: DMA_TCD15 (mIdx)

3.6.2.107 Macro DMA_TCD16

Prototype: DMA_TCD16 (mIdx)

3.6.2.108 Macro DMA_TCD17

Prototype: DMA_TCD17 (mIdx)

3.6.2.109 Macro DMA_TCD18

Prototype: DMA_TCD18 (mIdx)

3.6.2.110 Macro DMA_TCD19

Prototype: DMA_TCD19 (mIdx)

3.6.2.111 Macro DMA_TCD2

Prototype: DMA_TCD2 (mIdx)

3.6.2.112 Macro DMA_TCD20

Prototype: DMA_TCD20 (mIdx)

3.6.2.113 Macro DMA_TCD21

Prototype: DMA_TCD21 (mIdx)

3.6.2.114 Macro DMA_TCD22

Prototype: DMA_TCD22 (mIdx)

3.6.2.115 Macro DMA_TCD23

Prototype: DMA_TCD23 (mIdx)

3.6.2.116 Macro DMA_TCD24

Prototype: DMA_TCD24 (mIdx)

3.6.2.117 Macro DMA_TCD25

Prototype: DMA_TCD25 (mIdx)

3.6.2.118 Macro DMA_TCD26

Prototype: DMA_TCD26 (mIdx)

3.6.2.119 Macro DMA_TCD27

Prototype: DMA_TCD27 (mIdx)

3.6.2.120 Macro DMA_TCD28

Prototype: DMA_TCD28 (mIdx)

3.6.2.121 Macro DMA_TCD29

Prototype: DMA_TCD29 (mIdx)

3.6.2.122 Macro DMA_TCD3

Prototype: DMA_TCD3 (mIdx)

3.6.2.123 Macro DMA_TCD30

Prototype: DMA_TCD30 (mIdx)

3.6.2.124 Macro DMA_TCD31

Prototype: DMA_TCD31 (mIdx)

3.6.2.125 Macro DMA_TCD32

Prototype: DMA_TCD32 (mIdx)

3.6.2.126 Macro DMA_TCD33

Prototype: DMA_TCD33 (mIdx)

3.6.2.127 Macro DMA_TCD34

Prototype: DMA_TCD34 (mIdx)

3.6.2.128 Macro DMA_TCD35

Prototype: DMA_TCD35 (mIdx)

3.6.2.129 Macro DMA_TCD36

Prototype: DMA_TCD36 (mIdx)

3.6.2.130 Macro DMA_TCD37

Prototype: DMA_TCD37 (mIdx)

3.6.2.131 Macro DMA_TCD38

Prototype: DMA_TCD38 (mIdx)

3.6.2.132 Macro DMA_TCD39

Prototype: DMA_TCD39 (mIdx)

3.6.2.133 Macro DMA_TCD4

Prototype: DMA_TCD4 (mIdx)

3.6.2.134 Macro DMA_TCD40

Prototype: DMA_TCD40 (mIdx)

3.6.2.135 Macro DMA_TCD41

Prototype: DMA_TCD41 (mIdx)

3.6.2.136 Macro DMA_TCD42

Prototype: DMA_TCD42 (mIdx)

3.6.2.137 Macro DMA_TCD43

Prototype: DMA_TCD43 (mIdx)

3.6.2.138 Macro DMA_TCD44

Prototype: DMA_TCD44 (mIdx)

3.6.2.139 Macro DMA_TCD45

Prototype: DMA_TCD45 (mIdx)

3.6.2.140 Macro DMA_TCD46

Prototype: DMA_TCD46 (mIdx)

3.6.2.141 Macro DMA_TCD47

Prototype: DMA_TCD47 (mIdx)

3.6.2.142 Macro DMA_TCD48

Prototype: DMA_TCD48 (mIdx)

3.6.2.143 Macro DMA_TCD49

Prototype: DMA_TCD49 (mIdx)

3.6.2.144 Macro DMA_TCD5

Prototype: DMA_TCD5 (mIdx)

3.6.2.145 Macro DMA_TCD50

Prototype: DMA_TCD50 (mIdx)

3.6.2.146 Macro DMA_TCD51

Prototype: DMA_TCD51 (mIdx)

3.6.2.147 Macro DMA_TCD52

Prototype: DMA_TCD52 (mIdx)

3.6.2.148 Macro DMA_TCD53

Prototype: DMA_TCD53 (mIdx)

3.6.2.149 Macro DMA_TCD54

Prototype: DMA_TCD54 (mIdx)

3.6.2.150 Macro DMA_TCD55

Prototype: DMA_TCD55 (mIdx)

3.6.2.151 Macro DMA_TCD56

Prototype: DMA_TCD56 (mIdx)

3.6.2.152 Macro DMA_TCD57

Prototype: DMA_TCD57 (mIdx)

3.6.2.153 Macro DMA_TCD58

Prototype: DMA_TCD58 (mIdx)

3.6.2.154 Macro DMA_TCD59

Prototype: DMA_TCD59 (mIdx)

3.6.2.155 Macro DMA_TCD5_CITER

Prototype: DMA_TCD5_CITER (Channel)

3.6.2.156 Macro DMA_TCD6

Prototype: DMA_TCD6 (mIdx)

3.6.2.157 Macro DMA_TCD60

Prototype: DMA_TCD60 (mIdx)

3.6.2.158 Macro DMA_TCD61

Prototype: DMA_TCD61 (mIdx)

3.6.2.159 Macro DMA_TCD62

Prototype: DMA_TCD62 (mIdx)

3.6.2.160 Macro DMA_TCD63

Prototype: DMA_TCD63 (mIdx)

3.6.2.161 Macro DMA_TCD7

Prototype: DMA_TCD7 (mIdx)

3.6.2.162 Macro DMA_TCD7_BITER

Prototype: DMA_TCD7_BITER (Channel)

3.6.2.163 Macro DMA_TCD8

Prototype: DMA_TCD8 (mIdx)

3.6.2.164 Macro DMA_TCD9

Prototype: DMA_TCD9 (mIdx)

3.6.2.165 Macro DMA_TCD_CHANNEL

Prototype: DMA_TCD_CHANNEL (Channel, Offset)

3.6.2.166 Macro DMA_TCD_SR

Prototype: DMA_TCD_SR (Channel)

3.6.2.167 Macro LLM_Clear_DMA_TCDADDR_INT

Prototype: LLM_Clear_DMA_TCDADDR_INT (TcdAddr)

3.6.2.168 Macro LLM_Get_DMA_CITER

Prototype: LLM_Get_DMA_CITER (Channel)

3.6.2.169 Macro LLM_Get_DMA_INTERRUPT_ENABLE

Prototype: LLM_Get_DMA_INTERRUPT_ENABLE (dma_channel)

3.6.2.170 Macro LLM_Get_DMA_INTERRUPT_STATUS

Prototype: LLM_Get_DMA_INTERRUPT_STATUS (dma_channel)

3.6.2.171 Macro LLM_Get_DMA_Status

Prototype: `LLM_Get_DMA_Status(Channel)`

3.6.2.172 Macro LLM_Rd_DMA_INTH

Prototype: `LLM_Rd_DMA_INTH(mIdx)`

3.6.2.173 Macro LLM_Rd_DMA_INTL

Prototype: `LLM_Rd_DMA_INTL(mIdx)`

3.6.2.174 Macro LLM_Rd_DMA_TCDADDR_FLAGS

Prototype: `LLM_Rd_DMA_TCDADDR_FLAGS(TcdAddr)`

3.6.2.175 Macro LLM_Set_DMA_TCDADDR_INT

Prototype: `LLM_Set_DMA_TCDADDR_INT(TcdAddr)`

3.6.2.176 Macro LLM_Wr_DMA_CDNE

Prototype: `LLM_Wr_DMA_CDNE(channel)`

3.6.2.177 Macro LLM_Wr_DMA_CDNE_IDX

Prototype: `LLM_Wr_DMA_CDNE_IDX(mIdx,value)`

3.6.2.178 Macro LLM_Wr_DMA_CINT

Prototype: `LLM_Wr_DMA_CINT(channel)`

3.6.2.179 Macro LLM_Wr_DMA_CINT_IDX

Prototype: `LLM_Wr_DMA_CINT_IDX(mIdx,value)`

3.6.2.180 Macro LLM_Wr_DMA_SERQ

Prototype: `LLM_Wr_DMA_SERQ(channel)`

3.6.2.181 Macro LLM_Wr_DMA_SERQ_IDX

Prototype: `LLM_Wr_DMA_SERQ_IDX(mIdx,value)`

3.6.2.182 Macro LLM_Wr_DMA_SSRT

Prototype: `LLM_Wr_DMA_SSRT(channel)`

3.6.2.183 Macro LLM_Wr_DMA_SSRT_IDX

Prototype: `LLM_Wr_DMA_SSRT_IDX(mIdx,value)`

3.6.2.184 Macro LLM_Wr_DMA_TCD_DADDR

Prototype: `LLM_Wr_DMA_TCD_DADDR(Channel,dest_addr)`

3.6.2.185 Macro LLM_Wr_DMA_TCD_SADDR

Prototype: `LLM_Wr_DMA_TCD_SADDR(Channel,src_addr)`

3.6.2.186 Macro LLM_Wr_DMA_TCDADDR_BITER

Prototype: `LLM_Wr_DMA_TCDADDR_BITER(TcdAddr,link_iter)`

3.6.2.187 Macro LLM_Wr_DMA_TCDADDR_CITER

Prototype: `LLM_Wr_DMA_TCDADDR_CITER(TcdAddr,link_iter)`

3.6.2.188 Macro LLM_Wr_DMA_TCDADDR_CITER_DOFF**Prototype:** `LLM_Wr_DMA_TCDADDR_CITER_DOFF(TcdAddr, link_iter_doff)`**3.6.2.189 Macro LLM_Wr_DMA_TCDADDR_DADDR****Prototype:** `LLM_Wr_DMA_TCDADDR_DADDR(TcdAddr, dest_addr)`**3.6.2.190 Macro LLM_Wr_DMA_TCDADDR_DLAST_SGA****Prototype:** `LLM_Wr_DMA_TCDADDR_DLAST_SGA(TcdAddr, dlast_sg_addr)`**3.6.2.191 Macro LLM_Wr_DMA_TCDADDR_FLAGS****Prototype:** `LLM_Wr_DMA_TCDADDR_FLAGS(TcdAddr, flags)`**3.6.2.192 Macro LLM_Wr_DMA_TCDADDR_SADDR****Prototype:** `LLM_Wr_DMA_TCDADDR_SADDR(TcdAddr, src_addr)`**3.6.2.193 Macro LLM_Wr_DMA_TCDADDR_SOFF****Prototype:** `LLM_Wr_DMA_TCDADDR_SOFF(TcdAddr, src_off)`**3.6.2.194 Macro DMAMUX_CHANNEL_INDEX**

Macro to extract the channel index in a controller for a given channel.

Violates: Function-like macro defined because of code complexity**Prototype:** `DMAMUX_CHANNEL_INDEX(channel)`**3.6.2.195 Macro DMAMUX_CHCONF**

Macro to extract the address of CHCONF register based on a given channel.

Violates: Function-like macro defined because of code complexity

Prototype: `DMAMUX_CHCONF(channel)`

3.6.2.196 Macro **DMAMUX_CTRL_INDEX**

Macro to extract the controller index for a given channel.

Violates: Function-like macro defined because of code complexity

Violates: In the definition of a function-like macro each instance of a parameter shall be enclosed in parentheses

Prototype: `DMAMUX_CTRL_INDEX(channel)`

3.6.2.197 Macro **EMIOS_MCR**

EMIOS Module Registers - global registers, mIdx : emios module index 0 for emiosA, 1 for emiosB.

Prototype: `EMIOS_MCR(mIdx)`

3.6.2.198 Macro **EMIOS_GFR**

EMIOS Module Registers - global registers, mIdx : emios module index 0 for emiosA, 1 for emiosB.

Prototype: `EMIOS_GFR(mIdx)`

3.6.2.199 Macro **EMIOS_OUDR**

EMIOS Module Registers - global registers, mIdx : emios module index 0 for emiosA, 1 for emiosB.

Prototype: `EMIOS_OUDR(mIdx)`

3.6.2.200 Macro EMIOS_CADR

EMIOS Module Registers - Unified Channel specific registers, mIdx : emios module index 0 for emiosA, 1 for emiosB, chIdx : unified channel index.

Prototype: EMIOS_CADR (mIdx, chIdx)

3.6.2.201 Macro EMIOS_CBDR

EMIOS Module Registers - Unified Channel specific registers, mIdx : emios module index 0 for emiosA, 1 for emiosB, chIdx : unified channel index.

Prototype: EMIOS_CBDR (mIdx, chIdx)

3.6.2.202 Macro EMIOS_CCNTR

EMIOS Module Registers - Unified Channel specific registers, mIdx : emios module index 0 for emiosA, 1 for emiosB, chIdx : unified channel index.

Prototype: EMIOS_CCNTR (mIdx, chIdx)

3.6.2.203 Macro EMIOS_CCR

EMIOS Module Registers - Unified Channel specific registers, mIdx : emios module index 0 for emiosA, 1 for emiosB, chIdx : unified channel index.

Prototype: EMIOS_CCR (mIdx, chIdx)

3.6.2.204 Macro EMIOS_CSR

EMIOS Module Registers - Unified Channel specific registers, mIdx : emios module index 0 for emiosA, 1 for emiosB, chIdx : unified channel index.

Prototype: EMIOS_CSR (mIdx, chIdx)

3.6.2.205 Macro EMIOS_ALTA

EMIOS Module Registers - Unified Channel specific registers, mIdx : emios module index 0 for emiosA, 1 for emiosB, chIdx : unified channel index.

Prototype: EMIOS_ALTA (mIdx, chIdx)

3.6.2.206 Macro AUX_CLK_0

Prototype: AUX_CLK_0(x)

3.6.2.207 Macro AUX_CLK_00_DIV

Prototype: AUX_CLK_00_DIV(x)

3.6.2.208 Macro AUX_CLK_01_DIV

Prototype: AUX_CLK_01_DIV(x)

3.6.2.209 Macro AUX_CLK_1

Prototype: AUX_CLK_1(x)

3.6.2.210 Macro AUX_CLK_1_DIV

Prototype: AUX_CLK_1_DIV(x)

3.6.2.211 Macro AUX_CLK_2

Prototype: AUX_CLK_2(x)

3.6.2.212 Macro AUX_CLK_2_DIV

Prototype: AUX_CLK_2_DIV(x)

3.6.2.213 Macro AUX_CLK_3

Prototype: AUX_CLK_3(x)

3.6.2.214 Macro AUX_CLK_3_DIV

Prototype: AUX_CLK_3_DIV(x)

3.6.2.215 Macro AUX_CLK_4

Prototype: AUX_CLK_4(x)

3.6.2.216 Macro CLOCK_OUT_DIV

Prototype: CLOCK_OUT_DIV(x)

3.6.2.217 Macro CLOCK_OUT_SOURCE

Prototype: CLOCK_OUT_SOURCE(x)

3.6.2.218 Macro CMU0_HFREF

Prototype: CMU0_HFREF(x)

3.6.2.219 Macro CMU0_LFREF

Prototype: CMU0_LFREF(x)

3.6.2.220 Macro CMU0_MDR

Prototype: CMU0_MDR(x)

3.6.2.221 Macro CMU1_HFREF

Prototype: CMU1_HFREF(x)

3.6.2.222 Macro CMU1_LFREF

Prototype: CMU1_LFREF(x)

3.6.2.223 Macro EOCV_OSC_VALUE

Prototype: EOCV_OSC_VALUE(x)

3.6.2.224 Macro FMPLL_CR_NDIV

Prototype: FMPLL_CR_NDIV(x)

3.6.2.225 Macro FMPLL_INC_STEP

Prototype: FMPLL_INC_STEP(x)

3.6.2.226 Macro FMPLL_MOD_PERIOD

Prototype: FMPLL_MOD_PERIOD(x)

3.6.2.227 Macro ME_MCTL_TARGET_MODE

Prototype: ME_MCTL_TARGET_MODE(x)

3.6.2.228 Macro ME_MODE_CFAON

Prototype: ME_MODE_CFAON(x)

3.6.2.229 Macro ME_MODE_DFAON

Prototype: ME_MODE_DFAON(x)

3.6.2.230 Macro ME_MODE_SYSCCLK

Prototype: ME_MODE_SYSCCLK(x)

3.6.2.231 Macro ME_PCTLn

Prototype: ME_PCTLn(run, lp)

3.6.2.232 Macro OSC_DIV

Prototype: OSC_DIV(x)

3.6.2.233 Macro PUT_VALUE_ON_REG

Prototype: PUT_VALUE_ON_REG(type, value, shift)

3.6.2.234 Macro RC_TRIM

Prototype: RC_TRIM(x)

3.6.2.235 Macro STATUS_BITS_READ

Prototype: STATUS_BITS_READ(type, value, shift)

3.6.2.236 Macro SIUL_GPDI

Prototype: SIUL_GPDI(PIN)

3.6.2.237 Macro SIUL_GPDO

Prototype: SIUL_GPDO(PIN)

3.6.2.238 Macro SIUL_IFMC

Prototype: SIUL_IFMC(IFMC_OFFSET)

3.6.2.239 Macro SIUL_MPGPDO

Prototype: SIUL_MPGPDO(MPGPDO_OFFSET)

3.6.2.240 Macro SIUL_PCR

Prototype: `SIUL_PCR (PIN)`

3.6.2.241 Macro SIUL_PGPDI

Prototype: `SIUL_PGPDI (PGPDI_OFFSET)`

3.6.2.242 Macro SIUL_PGPDO

Prototype: `SIUL_PGPDO (PGPDO_OFFSET)`

3.6.2.243 Macro SIUL_PSMI

Prototype: `SIUL_PSMI (PSMI_OFFSET)`

3.6.3 Enum Reference

Enumeration of all constants supported by the MCU driver.

3.6.4 Function Reference

Functions supported by the MCU driver.

3.6.4.1 Function Mcu_EMIOS_Init

Initializes the EMIOS MCR register for EMIOS_A and EMIOS_B.



Figure 3-1. Inverse references of fuction Mcu_EMIOS_Init

Details:

function to initialize the Emios MCR register value configured

Implements: DMCU05420

Prototype: void Mcu_EMIOS_Init(const eMIOS_ConfigType *ClockPtr);

Table 3-62. Mcu_EMIOS_Init Arguments

Type	Name	Direction	Description
ConfigLLDPtr	input		
None	output		
None	input, output		

Table 3-63. Mcu_EMIOS_Init Return Values

Name	Description
None	

3.6.4.2 Function Mcu_Disable_ClockOutput

Clock Output Disable function.

Details:

This function disables the clock output.

Implements: DMCU06056

Prototype: void Mcu_Disable_ClockOutput(void);

3.6.4.3 Function Mcu_Disable_CMU

CMU Disable function.

Details:

This function disables the CMU.

Implements: DMCU06052

Prototype: void Mcu_Disable_CMU(uint8 cmuIndex);

Table 3-64. Mcu_Disable_CMU Arguments

Type	Name	Direction	Description
uint8	cmuIndex	input	0 --> Select CMU0 1 --> Select CMU1 2 --> Select CMU2 3 --> Select CMU3 4 --> Select CMU4 5 --> Select CMU5 6 --> Select CMU6 7 --> Select CMU7 8 --> Select CMU8 9 --> Select CMU9 10 --> Select CMU10 11 --> Select CMU11 12 --> Select CMU12.

3.6.4.4 Function Mcu_eMios_Configure_GPREN

eMios Global Prescaler Enable.

Details:

This function enable/disable the GPREN bit of the EMIOSMCR register of an addressed eMIOS instance.

Pre: Function requires an execution of Mcu_Init() before it can be used, otherwise it reports error to DET.

Return: void.

Implements: DMCU06051

Violates: Required Rule 8.10, API could not be made static.

Prototype: void Mcu_eMios_Configure_GPREN(uint8 eMios_module, uint8 value);

Table 3-65. Mcu_eMios_Configure_GPREN Arguments

Type	Name	Direction	Description
uint8	eMios_module	input	MCU_EMIOS_MODULE_0 --> Select eMios 0 MCU_EMIOS_MODULE_1 --> Select eMios 1.
uint8	value	input	MCU_EMIOS_GPREN_BIT_ENABLE --> Global Prescaler Enabled MCU_EMIOS_GPREN_BIT_DISABLE --> Global Prescaler Disabled (no clock).

3.6.4.5 Function Mcu_Enable_ClockOutput

Clock Output Enable function.

Details:

This function enables the clock output. The clock output setting is provided from the configuration structure.

Pre: Function requires an execution of `Mcu_Init()` before it can be used, otherwise it reports error to DET.

Implements: DMCU06056

Prototype: `void Mcu_Enable_ClockOutput(Mcu_ClockType ClockSetting);`

Table 3-66. Mcu_Enable_ClockOutput Arguments

Type	Name	Direction	Description
ConfigPtr	input	Pointer to MCU configuration structure.	

3.6.4.6 Function `Mcu_GetDeviceInfo`

This function extract MCU identifier information.

Details:

This function retrieve MCU information from MIDR1 and MIDR2 registers of SIUL IP component.

Return: `Mcu_DeviceInfoType`.

Pre: .

Implements: DMCU06050

Prototype: `void Mcu_GetDeviceInfo(Mcu_DeviceInfoType *deviceinfo);`

Table 3-67. Mcu_GetDeviceInfo Arguments

Type	Name	Direction	Description
<code>Mcu_DeviceInfoType</code>	output	Mcu Information.	

Table 3-68. Mcu_GetDeviceInfo Return Values

Name	Description
MCU_RAMSTATE_INVALID	Ram content is not valid or unknown (default).
MCU_RAMSTATE_VALID	Ram content is valid.

3.6.5 Structs Reference

Data structures supported by the MCU driver.

3.6.6 Types Reference

Types structures supported by the MCU driver.

3.6.7 Variables Reference

Variables supported by the MCU driver.

3.7 Mcu Clock tree configuration

This chapter provides further details about the clock tree configuration in the MCU plug-in.

MCU module plug-in defines a container `McuClockSettingConf` (multiplicity 1..*) each instance of which represents a configuration of the clock sources (internal and external oscillators, PLL etc..) in terms of frequency and prescalers values and specific source features (e.g FM modulation parameters) that may be applied to the microcontroller by the MCU driver. The `McuClockSettingConf` includes a subcontainer `McuClockReferencePoint` (multiplicity 1..*) providing the following parameters:

1. `McuClockFrequencySelect` defines the clock source that may be used by other modules (e.g. `CanDrv`, `LinDrv`, `PWM`).
2. `McuClockReferencePointFrequency` defines the frequency value (in Hz) corresponding to the selected clock source.

Other modules should define in their own configuration which container `McuClockReferencePoint` to be referred to, that allows them to select the frequency used as an input value.

In particular the module plug-in may provide the following ten clock source choices:

- SystemClockFrequency for the Core and all the peripherals that derive the clock from this source
- Motor control clock frequency called MotorControlClockFrequency in the plugin
- Fast (4-40 MHz) External Oscillator (XOSC) called ExternalOscillatorClockFrequency in the plugin
- Internal RC Oscillator (IRC) (16 MHz) called RC16MHzFrequency in the plugin
- Safety Port clock frequency called SafetyPortClockFrequency in the plugin
- FlexRay Clock Frequency called FlexRayClockFrequency in the plugin

At least one McuClockReferencePoint instance should exist in the MCU plug-in containing those clock sources:

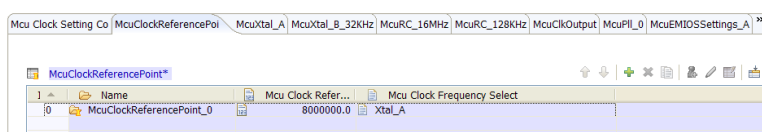


Figure 3-2. Tressos Plugin snapshot for Mcu Clock Setting Config form.

with the corresponding frequencies calculated automatically:

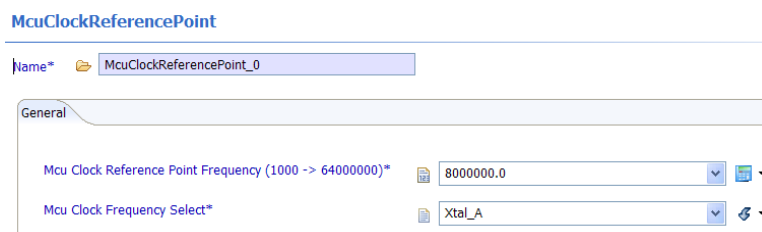


Figure 3-3. Tressos Plugin snapshot for Mcu Clock Reference Point form.

3.8 Frequency-Modulated Phase-Locked Loop (FMPLL) parameters configuration

This chapter provides further details about the configuration of the parameters of the FMPLL module implemented in the device.

The FMPLL enables the generation of high speed system clocks from a common 4-40 MHz input clock. Further, the FMPLL supports programmable frequency modulation of the system clock. The FMPLL multiplication factor, input clock divider and output clock divider ratio are all software configurable.

The parameters can be configured in the plugin as shown in the figure:

The screenshot shows the 'Mcu PLL 0' configuration window. The 'PLL0 Mode' is set to 'FM'. The 'McuPLL0_Parameter' section includes:

- PLL0 Frequency (16000000 -> 64000000): 64000000
- Output Divisor Factor (ODF) (2 -> 16): 4
- Input Divisor Factor (IDF) (1 -> 15): 3
- Multiplier Factor (NDIV) (32 -> 96): 96
- Progressive Clock Switching: ☐
- Fail Interrupt: ☐

 The 'Mcu Fm PLL 0 Parameter' section includes:

- Modulation Frequency (0 -> 64000000): 10000

Figure 3-4. Tresos Plugin snapshot of FMPLL configuration settings form

When FMPLL is in lock state, the FMPLL output clock is derived by the reference clock through the following relation:

$$\phi = \frac{clkin \cdot NDIV}{IDF \cdot ODF}$$

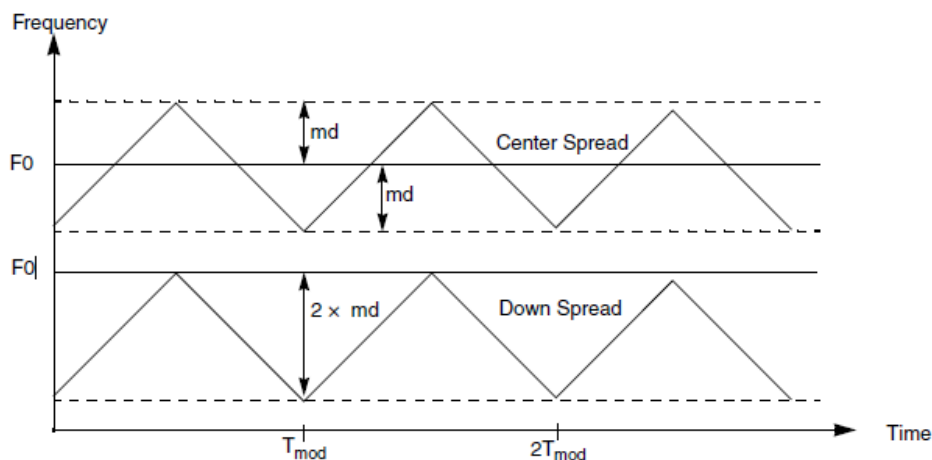
where ϕ is the PLL clock frequency, $clkin$ is the PLL reference clock frequency, $NDIV$ is McuMultiplicationFactor, IDF is McuInputDivisionFactor and ODF is McuOutputDivisionFactor. **NOTE:** the inner VCO (Voltage Controlled Oscillator) output frequency, whose value is given by $\phi \times ODF$, should reside in the range 256 MHz to 512 MHz.

The table below shows some examples of parameter configuration, according to different values of reference clock frequency:

PLL Reference Frequency (MHz)	PLL0 Frequency (MHz)	ODF (2, 4, 8, 16)	IDF (1..15)	NDIV (32..96)	VCO frequency (MHz)
8	32	8	1	32	256
8	64	8	1	64	512
16	32	8	2	32	256
16	64	8	2	64	512
40	32	8	10	64	256
40	64	4	10	64	256

The parameter *progressive clock switching* may be activated when the PLL is expected to be selected as system clock. When it occurs the PLL locks at a divided frequency, then gradually decreases the division factor until it is divided by 1. The effect is to gradually increase current consumption instead of a single large increase.

Moreover the user may enable the frequency modulation of the PLL by choosing the *FM* as value of the parameter *PLL0 mode*. When the frequency modulation is switched on, the carrier wave represented by the PLL output, varies its instantaneous frequency according to the modulation waveform which is always a triangle wave whose parameters are programmable. In particular the desired level of modulation is achieved through three parameters: the *modulation frequency*, the *modulation depth* and the *Spread selection*. The *modulation frequency* and *depth* represent respectively the frequency and the peak modulation depth in percentage of the triangle waveform. The *spread selection* specifies if the frequency of the carrier spreads around the original unmodulated PLL frequency F_0 (*Center spread*) or keeps below F_0 (*down spread*) that becomes the maximum value (see figure below). In the first case the *modulation depth* corresponds to the peak value of the triangle wave measured with respect to the mean value that is equal to F_0 . Instead with reference to the *Down spread*, the *modulation depth* represents half the peak-to-peak amplitude of the modulation signal whose maximum value becomes F_0 .



3.9 Symbolic Names DISCLAIMER

All containers having the symbolic name tag set as true in the Autosar schema will generate defines like:

```
#define <Container_Short_Name> <Container_ID>
```

For this reason it is forbidden to duplicate the name of such containers across the MCAL configuration, or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

3.10 Mcu Run and Low Power modes Configuration

This chapter provides further details about the API function that can be used for generating a mode transition request.

The microcontroller offers several devices modes corresponding to different usage models of the device. Each mode is configurable and can define a policy for energy and processing power management to fit particular system requirements. An application can easily switch from one mode to another depending on the current needs of the system. The operating modes are divided into system and user modes. The system modes are modes such as **RESET**, **DRUN**, **SAFE** and **TEST**. These modes aim to ease the configuration and monitoring of the system. The user modes are modes such as **RUN0..3**, **HALT** and **STOP** which can be configured to meet the application requirements in terms of energy management and available processing power. The modes **DRUN**, **SAFE**, **TEST** and **RUN0...3** are also called device software running modes.

In order to switch from one mode to another, the application can use the two API functions *Mcu_InitClock* and *Mcu_SetMode* that are described in the following sections.

3.10.1 Mcu_InitClock function

The *Mcu_InitClock(x)* function applies to the microcontroller the clock configuration specified by its argument *x*. The configuration can be set through the TRESOS plug-in interface that typically appears as shown below:

The screenshot shows the 'General Clock Settings' tab in the TRESOS Plugin. It contains several configuration options:

- System Clock Select:** A dropdown menu set to 'McuExtQuartzOsc'.
- RUN Mode:** A dropdown menu set to 'RUN0'.
- XOSC:** A checkbox that is checked.
- Voltage Regulator:** A checkbox that is checked.
- Code Flash Status:** A dropdown menu set to 'NormalMode'.
- Data Flash Status:** A dropdown menu set to 'NormalMode'.
- RC 16MHz:** A checkbox that is checked.
- Output Power Down Control:** A checkbox that is unchecked.

Figure 3-5. TRESOS Plugin snapshot for Mcu Clock Configuration form.

Mcu_InitClock(x) can request only a device software running mode transition (**DRUN**, **SAFE**, **TEST** or **RUN0...3**). The mode is chosen through the Listbox control “*RUN Mode*”. It also loads into the ME_XXX_MC register corresponding to the selected mode, the target mode configuration information provided by the user by setting the controls contained into the GeneralClockSettings Tab shown above. However if the user requests the PLL to be used as system clock, the *Mcu_InitClock(x)* prudentially sets the internal RC oscillator as system clock and turns off the PLLs. Then it demands to the *Mcu_DistributePllClock* function the task of changing the system clock to the PLL and turning on the PLL. The *Mcu_DistributePllClock()* must be called after the *Mcu_InitClock(x)* and when the PLL is locked. The *Mcu_InitClock(x)* also configures the reference clocks for the following SoC blocks:

- External oscillator (XOSC)

The screenshot shows the 'Mcu Xtal A' configuration form. It contains the following settings:

- Name:** A text field containing 'McuXtal_A'.
- ByPass XTAL Circuits:** A checkbox that is unchecked.
- Mcu Counter Oscillator Value (0 -> 255):** A numeric input field set to '100'.
- Xtal Mask Interrupt Status:** A checkbox that is checked.
- Xtal Divisor (1 -> 32):** A numeric input field set to '1'.

Figure 3-6. TRESOS Plugin snapshot for XOSC configuration parameters form

- FMPLL_0 if the FMPLL_0 has to be turned on when entering the mode chosen by the user

The screenshot shows the 'Mcu Fm Pll 0 Parameter' configuration form. It contains the following settings:

- Modulation Frequency (0 -> 64000000):** A numeric input field set to '10000'.
- Modulation Depth (%) (0.25 -> 4):** A numeric input field set to '0.5'.
- Spread Selection:** A dropdown menu set to 'Down_Spread'.

Figure 3-7. TRESOS Plugin snapshot for FMPLL_0 configuration parameters form

- Motor control clock (Auxiliary clock selector 0 and divider 0)
- Clock out selector and divider

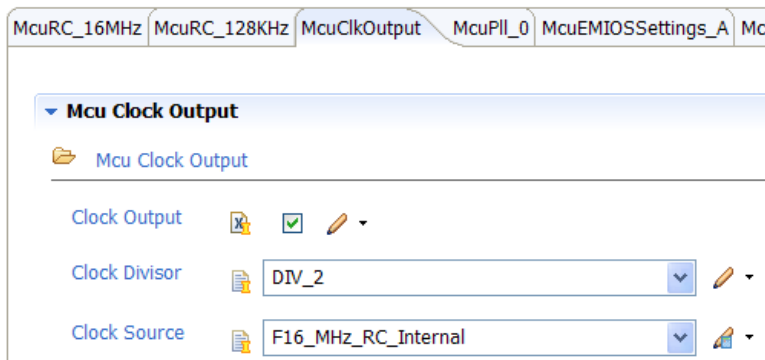


Figure 3-8. TRESOS Plugin snapshot for Clock out clock selector configuration parameters form

3.10.2 Mcu_SetMode function

The *Mcu_SetMode(x)* function applies to the microcontroller the mode configuration specified by its argument x. The configuration can be set through the TRESOS plug-in interface that typically appears as shown below:

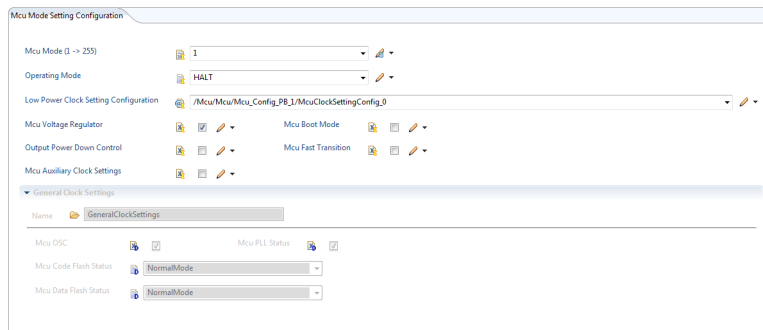


Figure 3-9. TRESOS Plugin snapshot for Mcu Mode Configuration form.

Mcu_SetMode(x) can request a microcontroller transition towards a low power mode (HALT, STOP). The function *Mcu_SetMode(x)* is designed to only apply the target Mode without doing any change about the clock configuration. If the PLL is selected as the source clock for the system clock in the target mode, then the function *Mcu_DistributePllClock* shall be used. It can be also adopted to pass into a device software running mode (**DRUN**, **SAFE**, **TEST** or **RUN0...3**). The mode is chosen through the Listbox control "*Operating Mode*". It also loads into the **ME_XXX_MC** register corresponding to the selected mode, the target mode configuration information provided by the user by setting the controls contained into the "*GeneralClockSettings*" Tab of the clock configuration specified by the "*LowPowerClockSettingConfig*" field

shown above. Then the *Mcu_SetMode(x)* also configures the following reference clocks with the parameters contained in the clock configuration specified by the *"LowPowerClockSettingConfig"* field:

- External oscillator (XOSC)
- Aux clock selector 3 for FlexRay clock source selection
- FMPLL_0 if the FMPLL_0 has to be turned on when entering the mode chosen by the user.
- FMPLL_1 if the FMPLL_1 has to be turned on when entering the mode chosen by the user (its availability depends on the subderivative).
- Motor control clock (Auxiliary clock selector 0 and divider 0)
- FlexRay clock (Auxiliary clock selector 1 and divider 0)
- Safety Port clock (Auxiliary clock selector 2 and divider 0)
- Clock out selector and divider

However if the fast transition is enabled through the *"McuFastTransition"* checkbox in the plug-in GUI, the mode transition is performed without applying any change to the reference clocks for the SoC blocks. A further option is represented by the *"McuAuxiliaryClockSettings"* field. It enables the settings into the *GeneralClockSettings* tab of the mode configuration container. These settings are used for updating the target configuration (ME_XXX_MC) corresponding to the current mode (for example RUN0) before entering into a low power mode. Then when the device exits from a low power mode, the microcontroller will come back to the previous mode (for example RUN0) whose target configuration has been updated. An example can clarify the purpose of the *McuAuxiliaryClockSettings*.

Problem: Let's say the microcontroller is in RUN0 mode with XOSC disabled and has to enter into the STOP mode. The XOSC interrupt has to be used for waking up the microcontroller and after exiting from STOP and coming back to RUN0, the XOSC has to be turned on.

Solution: The *Mcu_SetMode(x)* function can be used with the argument x relating to the mode configuration shown below.

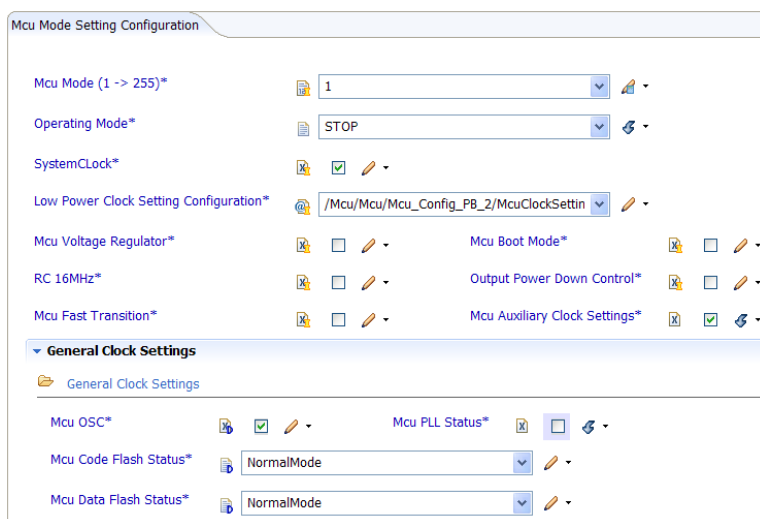


Figure 3-10. TRESOS Plugin snapshot for mode configuration form

3.10.3 Mcu_InitClock function

The function `Mcu_Init` initializes the MCU module by applying to the microcontroller the user configuration settings for:

1. specifying the behavior of the peripherals during run and low-power modes
2. determining the behavior of the device during the reset events
3. direct memory access (DMA)
4. RAM and FLASH

With reference to the first point during all device modes, each peripheral can be associated with a particular clock gating policy determined by two groups of peripheral configuration registers. The run peripheral configuration registers `ME_RUN_PC0...7` and the Low-power configuration registers `ME_LP_PC0...7`. All configurations are programmable by software according to the needs of the application. In particular a `ME_RUN_PCx` ($x=0...7$) register contains a bit for each software running mode (DRUN, TEST, SAFE and RUN0...3) which determines whether or not a peripheral has to be frozen with clock gated during the running mode.

The user settings in the `McuRunConfig` form of the TRESOS plug-in provides the configuration information to fill the `ME_RUN_PC0...7` registers.

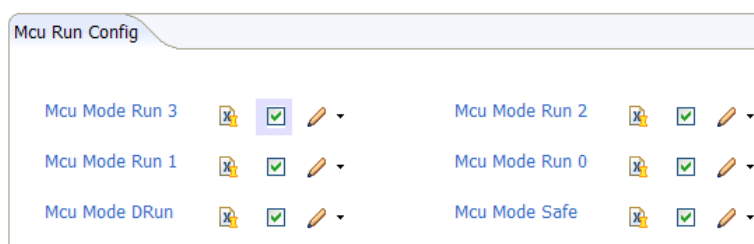


Figure 3-11. TRESOS Plugin snapshot for run configuration

Similarly a ME_LP_PCy (y=0...7) register contains a bit for each low-power mode (HALT and STOP) which determines whether or not a peripheral has to be frozen with clock gated during the low-power mode.

The user settings in the McuLowPower form of the TRESOS plug-in provides the configuration information to fill the ME_LP_PC0...7 registers.

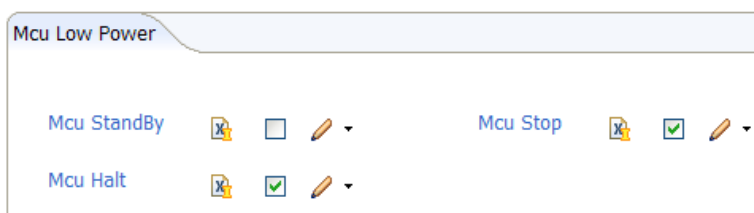


Figure 3-12. TRESOS Plugin snapshot for low-power configuration

A control register ME_PCTLz (z=0...143) associated to each peripheral selects the run and non-run configurations e.g the ME_RUN_PCx and ME_LP_PCy registers that determine the behaviour of the peripheral.

The user settings in the McuPeripheral form of the TRESOS plug-in provides the configuration information to fill the ME_PCTL0...143 registers.

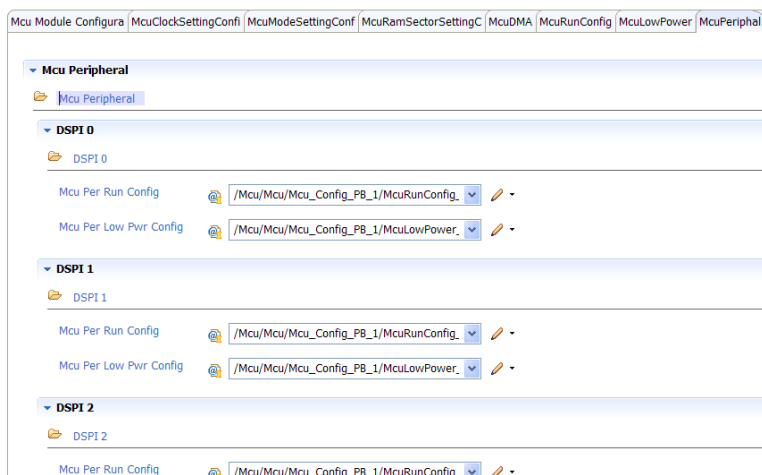


Figure 3-13. TRESOS Plugin snapshot for peripheral configuration

The `Mcu_Init` applies the user setting for determining the behaviour of the microcontroller when a reset request is generated by a reset source. In general when a reset is triggered the device proceeds through different steps hereinafter called phases, each one associated with a particular reset of some modules. In details depending on the reset source, one or two kinds of reset sequence can be applied to the microcontroller. The full sequence starting from PHASE0, ensures a safe-start up state for both digital and analog modules. Differently during a short sequence starting from PHASE1 or PHASE3, most digital modules are reset normally while analog modules state is preserved. Alternatively a reset source can be disabled to allow the microcontroller to generate a safe mode or an interrupt request instead of starting a full or partial sequence when the event occurs associated to the source. The `McuResetSource` form of the TRESOS plug-in contains the settings for enabling /disabling each configurable reset source and for specifying the type of reset sequence to be executed or alternatively the interrupt or safe request to be generated when the a reset request from the source is triggered.

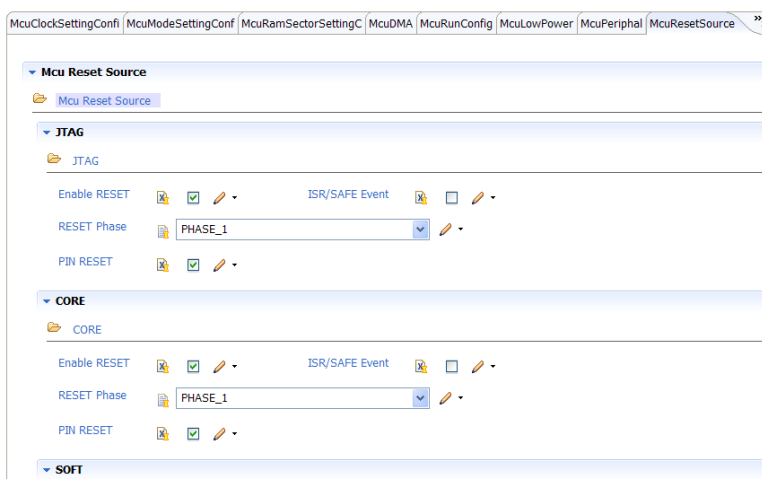


Figure 3-14. TRESOS Plugin snapshot for reset configuration

The checkbox "enable reset" enables or disables the reset source. It affects the FERD register. When it is checked the microcontroller performs a full or short reset sequence on a reset event depending on the value of the "reset phase" checkbox that affects the FESS register. Moreover the PIN reset checkbox enables the generation of an external reset when it is checked and the source generates a reset request. It affects the FBRE register. When the "enable reset" is disabled, the microcontroller generates a SAFE mode or an interrupt request depending on the value of the ISR/SAFE event checkbox that affects the FEAR register.

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the MCU Driver. The most of the parameters are described below.

4.1 Configuration elements of Mcu

Included forms :

- IMPLEMENTATION_CONFIG_VARIANT
- NonAutosar
- McuGeneralConfiguration
- McuPublishedInformation
- CommonPublishedInformation
- McuModuleConfiguration

Table 4-1. Revision table

Revision	Date
4.1.0	2010-12-03

4.2 Form IMPLEMENTATION_CONFIG_VARIANT

Table 4-2. Detailed description.

VariantPreCompile	Only precompile time configuration parameters. If Config Variant is equal to VariantPreCompile, the files Mcu_Cfg.h and Mcu_Cfg.c should be used.
VariantPostBuild	Mix of precompile and postbuild time configuration parameters. If Config Variant is equal to VariantPostBuild, the files Mcu_Cfg.h and Mcu_PBcfg.c should be used.

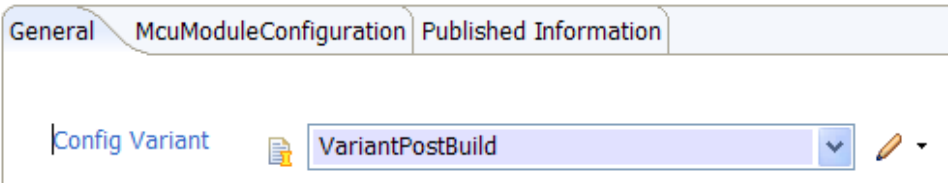


Figure 4-1. Tresos Plugin snapshot for IMPLEMENTATION_CONFIG_VARIANT form.

Table 4-3. Attribute IMPLEMENTATION_CONFIG_VARIANT detailed description

Property	Value
Label	Config Variant
Type	ENUMERATION
Default	VariantPostBuild
Range	VariantPostBuild VariantPreCompile

4.3 Form NonAutosar

NonAutosar

Autosar Requirements:

This container contains the global configuration parameters of the Non-Autosar Mcu driver. This container is a MultipleConfigurationContainer, i.e. this container and its sub-containers exist once per configuration set.

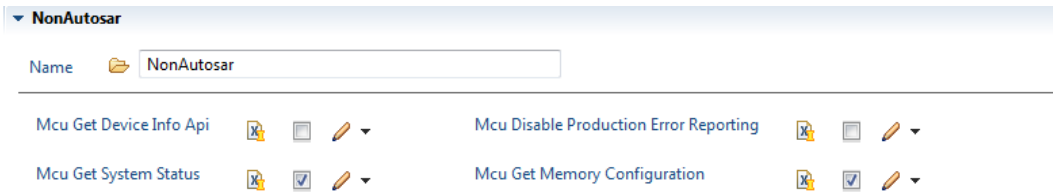


Figure 4-2. Tresos Plugin snapshot for NonAutosar form.

4.3.1 McuGetDeviceInfoApi (NonAutosar)

Table 4-4. Attribute McuGetDeviceInfoApi (NonAutosar) detailed description

Property	Value
Label	Mcu Get Device Info Api
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.3.2 McuDisableDemReportErrorStatus (NonAutosar)

McuDisableDemReportErrorStatus

Switches the Diagnostic Error Reporting and Notification OFF.

Table 4-5. Attribute McuDisableDemReportErrorStatus (NonAutosar) detailed description

Property	Value
Label	Mcu Disable Production Error Reporting
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.3.3 McuSscmGetSystemStatus (NonAutosar)

Table 4-6. Attribute McuSscmGetSystemStatus (NonAutosar) detailed description

Property	Value
Label	Mcu Get System Status
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.3.4 McuSscmGetMemConfig (NonAutosar)

Table 4-7. Attribute McuSscmGetMemConfig (NonAutosar) detailed description

Property	Value
Label	Mcu Get Memory Configuration
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.4 Form McuGeneralConfiguration

This container contains the configuration (parameters) of the MCU driver.

The screenshot shows the 'Mcu General Configuration' form in the Tresos Plugin. The form has a header bar with the title 'Mcu General Configuration' and a search bar. Below the header, there is a list of configuration parameters, each with a checkbox and a dropdown menu. The parameters are arranged in two columns. The first column includes: Mcu Dev Error Detect, Mcu Init Clock, Perform Reset API, Mcu Perform Reset Callout (set to NULL_PTR), Version Info API, Mcu DMA Init, Mcu Enter Low Power Mode, McueMiosConfigureGPRENapi, Mcu Clock Notification Callout (set to McuNotificationCallout), Mcu Configure Clock Output, and McuSscmRAE. The second column includes: Mcu Get Ram State Api, Mcu No PII, Mcu Callout Before PerformReset, Mcu Configure Flash, Mcu LP Transition Check, McuSkipFunction, Enable Notification Api, McuSscmPAE, and Mcu Disable all peripherals during init clock. Each parameter has a checkbox and a dropdown menu with a pencil icon.

Figure 4-3. Tresos Plugin snapshot for McuGeneralConfiguration form.

4.4.1 McuDevErrorDetect (McuGeneralConfiguration)

Pre-processor switch for enabling the development error detection and reporting to the DEM. The switch McuDevErrorDetect shall activate or deactivate the detection of all development errors. The detection of development errors is configurable (ON/OFF) at precompile time. The detection of production code errors cannot be switched off.

Table 4-8. Attribute McuDevErrorDetect (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu Dev Error Detect
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	false

4.4.2 McuGetRamStateApi (McuGeneralConfiguration)

Table 4-9. Attribute McuGetRamStateApi (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu Get Ram State Api
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	false

4.4.3 McuInitClock (McuGeneralConfiguration)

If this parameter is set to FALSE, the clock initialization has to be disabled from the MCU driver. This concept applies when there are some write once clock registers and a bootloader is present. If this parameter is set to TRUE, the MCU driver is responsible of the clock initialization

Table 4-10. Attribute McuInitClock (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu Init Clock
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.4.4 McuNoPll (McuGeneralConfiguration)

This parameter shall be set True, if the H/W does not have a PLL or the PLL circuitry is enabled after the power on without S/W intervention. In this case MCU_DistributePllClock has to be disabled and MCU_GetPllStatus has to return MCU_PLL_STATUS_UNDEFINED. Otherwise this parameters has to be set False

Table 4-11. Attribute McuNoPll (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu No Pll
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.4.5 McuPerformResetApi (McuGeneralConfiguration)

Pre-processor switch to enable/disable the use of the function Mcu_PerformReset().

Table 4-12. Detailed description.

0	Mcu_PerformReset() API is not used
1	Mcu_PerformReset() API is used

Table 4-13. Attribute McuPerformResetApi (McuGeneralConfiguration) detailed description

Property	Value
Label	Perform Reset API
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	false

4.4.6 McuCalloutBeforePerformReset (McuGeneralConfiguration)

Check this if you want a callout function, called by MCU right before Mcu_PerformReset().

Note:Implementation Specific Parameter

Table 4-14. Attribute McuCalloutBeforePerformReset (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu Callout Before PerformReset
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.4.7 McuPerformResetCallout (McuGeneralConfiguration)

Function name of callout. The field is editable only if McuCalloutBeforePerformReset is true.

Note:Implementation Specific Parameter

Table 4-15. Attribute McuPerformResetCallout (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu Perform Reset Callout
Type	FUNCTION-NAME
Origin	Custom
Symbolic Name	false
Default	NULL_PTR

4.4.8 McuVersionInfoApi (McuGeneralConfiguration)

Pre-processor switch to enable / disable the API to read out the modules version information.

Table 4-16. Attribute McuVersionInfoApi (McuGeneralConfiguration) detailed description

Property	Value
Label	Version Info API
Type	BOOLEAN
Origin	AUTOSAR_ECUC

Table continues on the next page...

Table 4-16. Attribute McuVersionInfoApi (McuGeneralConfiguration) detailed description (continued)

Property	Value
Symbolic Name	false
Default	false

4.4.9 McuConfigureFlash (McuGeneralConfiguration)

Table 4-17. Detailed description.

0	Global Flash Settings including latency timings are not configured by the driver (reset values will not be updated).
1	Global Flash Settings including latency timings are configured by the driver

Note:Implementation Specific Parameter

Table 4-18. Attribute McuConfigureFlash (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu Configure Flash
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.4.10 McuDmalnit (McuGeneralConfiguration)

If this parameter is set to FALSE, DMA initialization is not performed by MCU driver. This concept is to give possibility to application to initialize DMA by its own way.

If this parameter is set to TRUE, the MCU driver is responsible of the DMA initialization via container McuDMAMUXandEDMAConfig and McuDMA.

Table 4-19. Attribute McuDmalnit (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu DMA Init
Type	BOOLEAN

Table continues on the next page...

Table 4-19. Attribute McuDmalnit (McuGeneralConfiguration) detailed description (continued)

Property	Value
Origin	Custom
Symbolic Name	false
Default	true

4.4.11 McuLPTransitionCheck (McuGeneralConfiguration)

Table 4-20. Detailed description.

0	No check is performed for low power transition complete.
1	The transition to low power will be checked by driver. If DET is enabled, a DET error will be reported after the timeout loop waiting for the transition to complete will expire.

Note:Implementation Specific Parameter

Table 4-21. Attribute McuLPTransitionCheck (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu LP Transition Check
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.4.12 McuEnterLowPowerMode (McuGeneralConfiguration)

Table 4-22. Detailed description.

0	Transition to Mcu in LOW POWER mode is performed in customer function.
1	Transition to Mcu in LOW POWER mode is performed by Mcu_SetMode API.

Note:Implementation Specific Parameter

Table 4-23. Attribute McuEnterLowPowerMode (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu Enter Low Power Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.4.13 McuSkipFunction (McuGeneralConfiguration)

TRUE: Mcu_SkipFunction() is enabled FALSE: Mcu_SkipFunction() is disabled

Table 4-24. Attribute McuSkipFunction (McuGeneralConfiguration) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.4.14 McueMiosConfigureGPRENapi (McuGeneralConfiguration)

Pre-processor switch to enable/disable the use of the function
Mcu_eMios_Configure_GPREN().

Table 4-25. Detailed description.

0	Mcu_eMios_Configure_GPREN() API is not used
1	Mcu_eMios_Configure_GPREN() API is used

Table 4-26. Attribute McueMiosConfigureGPRENapi (McuGeneralConfiguration) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom

Table continues on the next page...

Table 4-26. Attribute McueMiosConfigureGPRENApi (McuGeneralConfiguration) detailed description (continued)

Property	Value
Symbolic Name	false
Default	false

4.4.15 McuEnableDisableNotificationApi (McuGeneralConfiguration)

TRUE: Notification Api is enabled FALSE: Notification Api is disabled

Table 4-27. Attribute McuEnableDisableNotificationApi (McuGeneralConfiguration) detailed description

Property	Value
Label	Enable Notification Api
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.4.16 McuNotification (McuGeneralConfiguration)

Function name of callout.

Prototype: void FunctionName(uint8 cmuIndex).

param[in] cmuIndex.

0 --> Select CMU.

The field is editable only if McuEnableDisableNotificationApi is true.

Note:Implementation Specific Parameter

Table 4-28. Attribute McuNotification (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu Clock Notification Callout
Type	FUNCTION-NAME
Origin	Custom

Table continues on the next page...

Table 4-28. Attribute McuNotification (McuGeneralConfiguration) detailed description (continued)

Property	Value
Symbolic Name	false
Default	NULL_PTR

4.4.17 McuConfClkOutput (McuGeneralConfiguration)

Configuration Flag to allow Enable/Disable Clock Out. Note: Clock output is configured or not in Mcu_InitClock/Mcu_SetMode API. Note: Implementation specific Parameter.

Table 4-29. Attribute McuConfClkOutput (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu Configure Clock Output
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.4.18 McuSscmPAE (McuGeneralConfiguration)

TRUE: SSCM ERROR Peripheral Bus Abort Enable(PAE) is enabled FALSE: SSCM ERROR Peripheral Bus Abort Enable(PAE) is disabled

Table 4-30. Attribute McuSscmPAE (McuGeneralConfiguration) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.4.19 McuSscmRAE (McuGeneralConfiguration)

TRUE: SSCM ERROR Register Bus Abort Enable(RAE) is enabled FALSE: SSCM ERROR Register Bus Abort Enable(RAE) is disabled

Table 4-31. Attribute McuSscmRAE (McuGeneralConfiguration) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.4.20 McuDisablePeripheralDuringInitclock (McuGeneralConfiguration)

This configuration is applicable also in case of fast transition mode.

Table 4-32. Detailed description.

0	Not recommended mode. In this case, there is no gate of peripherals (Mcu_Disable_Peripheral is not called) during Mcu_InitClock. This mode implies that it is up to application to ensure that new clock tree is applicable (see note bellowed)
1	It is default value: during execution of Mcu_InitClock, all peripherals will be gated via internal functions Mcu_Disable_Peripheral in order to apply new clock tree.

Warning: in the case that you want to remove the gate of peripheral during init clock procedure, be aware that it is up to application to check that such update of clock tree will not lead to unexpected behaviour. Indeed, hardware (and then MCU plugin) is not able to guarantee following use case due to new clock tree/configuration provided:

- Change of system clock in new clock tree.
- Need to change any clock dividers compared to previous clock configuration.
- Disable CMU/eMIOS clock is impossible during init clock. Ie user must ensure that this peripheral is well configured in new clock configuration provided.

Note:Implementation Specific Parameter

Table 4-33. Attribute McuDisablePeripheralDuringInitclock (McuGeneralConfiguration) detailed description

Property	Value
Label	Mcu Disable all peripherals during init clock
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.5 Form McuPublishedInformation

Container holding all MCU specific published information parameters.

Included forms :

- [Form McuResetReasonConf](#)

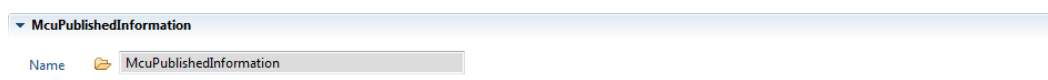


Figure 4-4. Tressos Plugin snapshot for McuPublishedInformation form.

4.5.1 Form McuResetReasonConf

This container contains the configuration for the different type of reset reason that can be retrieved from Mcu_GetResetReason Api.

Is included by form : [Form McuPublishedInformation](#)



Figure 4-5. Tressos Plugin snapshot for McuResetReasonConf form.

4.5.1.1 McuResetReason (McuResetReasonConf)

The parameter represents the different type of reset that a Micro supports. This parameter is referenced by the parameter EcuMResetReason in the ECU State manager module.

Table 4-34. Attribute McuResetReason (McuResetReasonConf) detailed description

Property	Value
Type	INTEGER_LABEL
Origin	AUTOSAR_ECUC
Symbolic Name	true
Default	0
Invalid	Range <=255 >=0

4.6 Form CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

The screenshot shows the 'Common Published Information' configuration form. At the top, there is a dropdown menu labeled 'Name' with 'CommonPublishedInformation' selected. Below this, there are several input fields, each preceded by a lightbulb icon, indicating they are required or have default values. The fields and their values are:

- Ar Release Major Version: 4
- Ar Release Minor Version: 0
- Ar Release Revision Version: 3
- Module Id: 101
- Sw Major Version: 1
- Sw Minor Version: 0
- Sw Patch Version: 1
- vendor Api Infix: (empty)
- Vendor Id: 43

Figure 4-6. Tresos Plugin snapshot for CommonPublishedInformation form.

4.6.1 ArReleaseMajorVersion (CommonPublishedInformation)

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Table 4-35. Attribute ArReleaseMajorVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Ar Release Major Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	4
Invalid	Range >=4 <=4

4.6.2 ArReleaseMinorVersion (CommonPublishedInformation)

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Table 4-36. Attribute ArReleaseMinorVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Ar Release Minor Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range >=0 <=0

4.6.3 ArReleaseRevisionVersion (CommonPublishedInformation)

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Table 4-37. Attribute ArReleaseRevisionVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Ar Release Revision Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	3
Invalid	Range >=3 <=3

4.6.4 ModuleId (CommonPublishedInformation)

Module ID of this module from Module List.

Table 4-38. Attribute ModuleId (CommonPublishedInformation) detailed description

Property	Value
Label	Module Id
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	101
Invalid	Range >=101 <=101

4.6.5 SwMajorVersion (CommonPublishedInformation)

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Table 4-39. Attribute SwMajorVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Sw Major Version
Type	INTEGER_LABEL
Origin	Custom

Table continues on the next page...

Table 4-39. Attribute SwMajorVersion (CommonPublishedInformation) detailed description (continued)

Property	Value
Symbolic Name	false
Default	1
Invalid	Range >=1 <=1

4.6.6 SwMinorVersion (CommonPublishedInformation)

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Table 4-40. Attribute SwMinorVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Sw Minor Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range >=0 <=0

4.6.7 SwPatchVersion (CommonPublishedInformation)

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Table 4-41. Attribute SwPatchVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Sw Patch Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	1

Table continues on the next page...

Table 4-41. Attribute SwPatchVersion (CommonPublishedInformation) detailed description (continued)

Property	Value
Invalid	Range >=1 <=1

4.6.8 VendorApiInfix (CommonPublishedInformation)

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name. This parameter is used to specify the vendor specific name. In total, the implementation specific name is generated as follows:

<ModuleName>_>VendorId>_<VendorApiInfix><Api name from SWS>. E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write. This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

Table 4-42. Attribute VendorApiInfix (CommonPublishedInformation) detailed description

Property	Value
Label	vendor Api Infix
Type	STRING_LABEL
Origin	Custom
Symbolic Name	false
Default	
Enable	false

4.6.9 VendorId (CommonPublishedInformation)

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Table 4-43. Attribute VendorId (CommonPublishedInformation) detailed description

Property	Value
Label	Vendor Id

Table continues on the next page...

Table 4-43. Attribute VendorId (CommonPublishedInformation) detailed description (continued)

Property	Value
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	43
Invalid	Range >=43 <=43

4.7 Form McuModuleConfiguration

This container contains the configuration (parameters) of the MCU driver.

Included forms :

- [Form McuEnableMode](#)
- [Form McuInterruptTransition](#)
- [Form McuPowerDomain](#)
- [Form McuDMAMUXandEDMAConfig](#)
- [Form McuFlashPFCR0](#)
- [Form McuFlashPFCR1](#)
- [Form McuFlashPFAPR](#)
- [Form McuDemEventParameterRefs](#)
- [Form McuDMA](#)
- [Form McuPeripheral](#)
- [Form McuResetSource](#)
- [Form McuClockSettingConfig](#)
- [Form McuModeSettingConf](#)
- [Form McuRamSectorSettingConf](#)
- [Form McuRunConfig](#)
- [Form McuLowPower](#)

The screenshot displays the 'Mcu Module Configur...' tab in the Tresos configuration tool. The settings are as follows:

- Mcu Clock Src Failure Notification:** A dropdown menu set to 'DISABLED'.
- Mcu Number Of Mcu Modes (1 -> 255):** A dropdown menu set to '3'.
- Mcu Ram Sectors (1 -> 4294967295):** A dropdown menu set to '1'.
- Reset Setting (1 -> 255):** A text input field set to '0'.
- External Crystal Frequency [Hz] (4000000 -> 16000000):** A text input field set to '8000000.0'.
- Loops TimeOut (0 -> 4294967295):** A text input field set to '100000000'.

Below these settings are two expandable sections:

- Mcu Enable Mode:** Contains checkboxes for various modes: STOP Mode, HALT Mode, RUN2 Mode, STANDBY Mode, RUN3 Mode, and RUN1 Mode. All checkboxes are currently checked.
- Mcu Interrupt Transition:** A section header for configuring interrupt transitions.

Figure 4-7. Tresos Plugin snapshot for McuModuleConfiguration form.

4.7.1 McuClockSrcFailureNotification (McuModuleConfiguration)

Enables/Disables clock failure notification. In case this feature is not supported by HW the setting should be disabled.

Table 4-44. Attribute McuClockSrcFailureNotification (McuModuleConfiguration) detailed description

Property	Value
Label	Mcu Clock Src Failure Notification
Type	ENUMERATION
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	ENABLED
Range	DISABLED ENABLED

4.7.2 McuNumberOfMcuModes (McuModuleConfiguration)

This parameter shall represent the number of Modes available for the MCU.
CalculationFormula = Number of configured McuModeSettingConf.

Table 4-45. Attribute McuNumberOfMcuModes (McuModuleConfiguration) detailed description

Property	Value
Label	Mcu Number Of Mcu Modes
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	false
Invalid	Range <=255 >=1

4.7.3 McuRamSectors (McuModuleConfiguration)

This parameter shall represent the number of RAM sectors available for the MCU.
ClculationFormula = Number of configured McuRamSectorSettingConf.

Table 4-46. Attribute McuRamSectors (McuModuleConfiguration) detailed description

Property	Value
Label	Mcu Ram Sectors
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	false
Invalid	Range <=4294967295 >=1

4.7.4 McuResetSetting (McuModuleConfiguration)

This parameters applies to the function Mcu_PerformReset, which performs a microcontroller reset using the hardware feature of the microcontroller.

Note: This parameter is not used by the current Implementation. Software Reset occurs when Mcu_PerformReset fuction is called. The Reset is not user configurable parameter.

Table 4-47. Attribute McuResetSetting (McuModuleConfiguration) detailed description

Property	Value
Label	Reset Setting
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	0
Enable	false
Invalid	Range <=255 >=1

4.7.5 FrequencyOfFXOSC (McuModuleConfiguration)

Frequency Of FXOSC [Hz]. FrequencyOfFXOSC has to be in range 4 MHz up to 16 MHz.

Note: Implementation specific Parameter.

Table 4-48. Attribute FrequencyOfFXOSC (McuModuleConfiguration) detailed description

Property	Value
Label	Frequency Of FXOSC (Fast external crystal oscillator) [Hz]
Type	FLOAT
Origin	Custom
Symbolic Name	false
Default	8000000
Invalid	Range <=16000000 >=4000000

4.7.6 McuTimeout (McuModuleConfiguration)

This parameter is used for internal loop, oscillator stability. It represents the maximum number of loops for blocking functionality. Choose a value of this parameter which is large enough to avoid early exits from loops.

Note: Implementation specific Parameter.

Table 4-49. Attribute McuTimeout (McuModuleConfiguration) detailed description

Property	Value
Label	Loops TimeOut
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	4294967294
Invalid	Range >=0 <=4294967294

4.7.7 Form McuEnableMode

This container allows a way to disable the device modes which are not required for a given device. RESET, SAFE, DRUN and RUN0 modes are always enabled.

Is included by form : [Form McuModuleConfiguration](#)

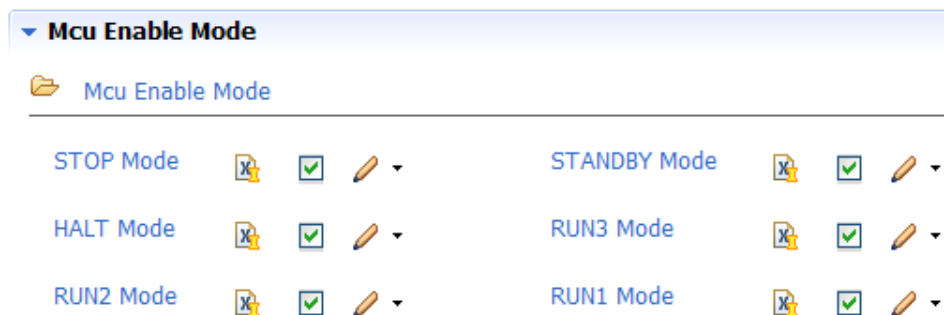


Figure 4-8. Tressos Plugin snapshot for McuEnableMode form.

4.7.7.1 McuModeStop (McuEnableMode)

Check if you want enable the STOP mode.

Note: Implementation specific Parameter .

Table 4-50. Attribute McuModeStop (McuEnableMode) detailed description

Property	Value
Label	STOP Mode
Type	BOOLEAN

Table continues on the next page...

Table 4-50. Attribute McuModeStop (McuEnableMode) detailed description (continued)

Property	Value
Origin	Custom
Symbolic Name	false
Default	true

4.7.7.2 McuModeSTBY (McuEnableMode)

Check if you want enable the StandBy mode.

Note: Implementation specific Parameter .

Table 4-51. Attribute McuModeSTBY (McuEnableMode) detailed description

Property	Value
Label	STANDBY Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.7.3 McuModeHalt (McuEnableMode)

Check if you want enable the HALT mode.

Note: Implementation specific Parameter .

Table 4-52. Attribute McuModeHalt (McuEnableMode) detailed description

Property	Value
Label	HALT Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.7.4 McuModeRun3 (McuEnableMode)

Check if you want enable the RUN3 mode.

Note: Implementation specific Parameter .

Table 4-53. Attribute McuModeRun3 (McuEnableMode) detailed description

Property	Value
Label	RUN3 Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.7.5 McuModeRun2 (McuEnableMode)

Check if you want enable the RUN2 mode

Note: Implementation specific Parameter .

Table 4-54. Attribute McuModeRun2 (McuEnableMode) detailed description

Property	Value
Label	RUN2 Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.7.6 McuModeRun1 (McuEnableMode)

Check if you want enable the RUN1 mode

Note: Implementation specific Parameter .

Table 4-55. Attribute McuModeRun1 (McuEnableMode) detailed description

Property	Value
Label	RUN1 Mode
Type	BOOLEAN
Origin	Custom

Table continues on the next page...

Table 4-55. Attribute McuModeRun1 (McuEnableMode) detailed description (continued)

Property	Value
Symbolic Name	false
Default	true

4.7.8 Form McuInterruptTransition

Note: Implementation specific Container.

Is included by form : [Form McuModuleConfiguration](#)

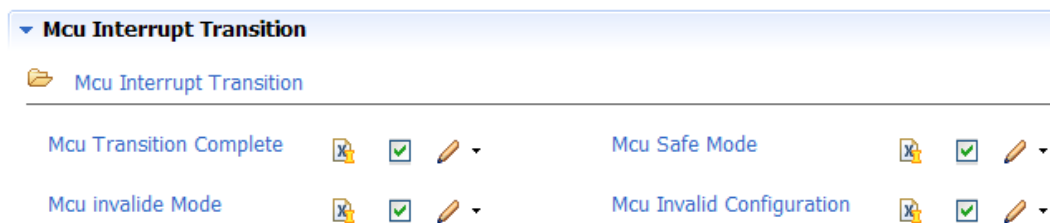


Figure 4-9. Tresos Plugin snapshot for McuInterruptTransition form.

4.7.8.1 McuTransitionComplete (McuInterruptTransition)

Check if you want enable the ISR on Transition Complete

Note: Implementation specific Parameter .

Table 4-56. Attribute McuTransitionComplete (McuInterruptTransition) detailed description

Property	Value
Label	Mcu Transition Complete
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.8.2 McuSafeMode (McuInterruptTransition)

Check if you want enable Safe Mode Interrupt.

Note: Implementation specific Parameter.

Table 4-57. Attribute McuSafeMode (McuInterruptTransition) detailed description

Property	Value
Label	Mcu Safe Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.8.3 McuInvalidMode (McuInterruptTransition)

Check if you want enable the Invalid Mode Interrupt.

Note: Implementation specific Parameter.

Table 4-58. Attribute McuInvalidMode (McuInterruptTransition) detailed description

Property	Value
Label	Mcu invalide Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.8.4 McuInvalidConfiguration (McuInterruptTransition)

Check if you want enable the Invalid Configuration Interrupt.

Note: Implementation specific Parameter.

Table 4-59. Attribute McuInvalidConfiguration (McuInterruptTransition) detailed description

Property	Value
Label	Mcu Invalid Configuration
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.9 Form McuPowerDomain

Note: Implementation specific Container.

Is included by form : [Form McuModuleConfiguration](#)

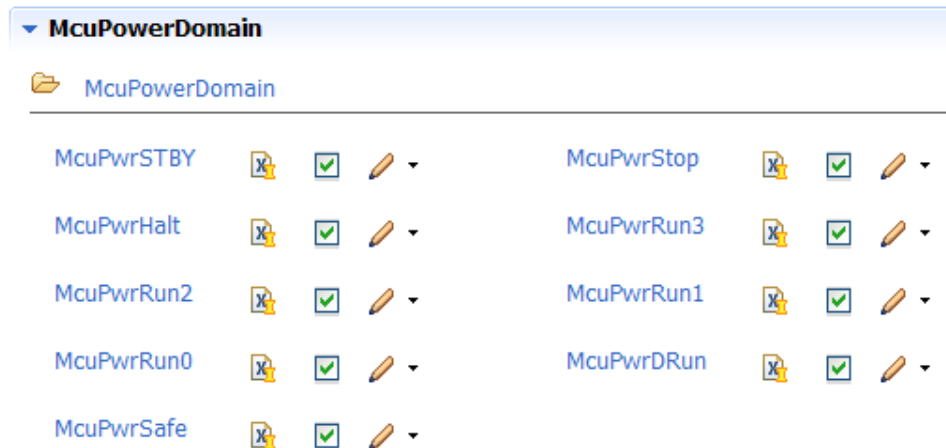


Figure 4-10. Tressos Plugin snapshot for McuPowerDomain form.

4.7.9.1 McuPwrSTBY (McuPowerDomain)

Check if you want to keep in Power On the peripherals located in PD2 when STANDBY Mode occurs. Note: Implementation specific Parameter.

Table 4-60. Attribute McuPwrSTBY (McuPowerDomain) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.9.2 McuPwrStop (McuPowerDomain)

Check if you want to keep in Power On the peripherals located in PD2 when STOP Mode occurs. Note: Implementation specific Parameter.

Table 4-61. Attribute McuPwrStop (McuPowerDomain) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.9.3 McuPwrHalt (McuPowerDomain)

Check if you want to keep in Power On the peripherals located in PD2 when HALT Mode occurs. Note: Implementation specific Parameter.

Table 4-62. Attribute McuPwrHalt (McuPowerDomain) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.9.4 McuPwrRun3 (McuPowerDomain)

Check if you want to keep in Power On the peripherals located in PD2 when RUN3 Mode occurs. Note: Implementation specific Parameter.

Table 4-63. Attribute McuPwrRun3 (McuPowerDomain) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.9.5 McuPwrRun2 (McuPowerDomain)

Check if you want to keep in Power On the peripherals located in PD2 when RUN2 Mode occurs. Note: Implementation specific Parameter.

Table 4-64. Attribute McuPwrRun2 (McuPowerDomain) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.9.6 McuPwrRun1 (McuPowerDomain)

Check if you want to keep in Power On the peripherals located in PD2 when RUN1 Mode occurs, Note: Implementation specific Parameter.

Table 4-65. Attribute McuPwrRun1 (McuPowerDomain) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.9.7 McuPwrRun0 (McuPowerDomain)

Check if you want to keep in Power On the peripherals located in PD2 when RUN0 Mode occurs. Note: Implementation specific Parameter.

Table 4-66. Attribute McuPwrRun0 (McuPowerDomain) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.9.8 McuPwrDRun (McuPowerDomain)

Check if you want to keep in Power On the peripherals located in PD2 when DRUN Mode occurs. Note: Implementation specific Parameter.

Table 4-67. Attribute McuPwrDRun (McuPowerDomain) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.9.9 McuPwrSafe (McuPowerDomain)

Check if you want to keep in Power On the peripherals located in PD2 when SAFE Mode occurs. Note: Implementation specific Parameter.

Table 4-68. Attribute McuPwrSafe (McuPowerDomain) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.10 Form McuDMAMUXandEDMAConfig

This container contains the SPECIFIC configuration (parameters) of the DMAMUXandEDMA Configuration. Click on RED button,to enable the container.

Note: implementation specific Container.

Is included by form : [Form McuModuleConfiguration](#)

Mcu DMAMUX and EDMA Config

Mcu DMAMUX and EDMA Config

eDMA Channel Priority 0 (0 -> 4294967295)

eDMA Channel Priority 4 (0 -> 4294967295)

eDMA Channel Priority 8 (0 -> 4294967295)

eDMA Channel Priority 12 (0 -> 4294967295)

EMLM Enable minor loop mapping CLM Continuous link mode

HALT DMA operations HOE Halt on error

ERGA Enable Round-Robin Group Arbitration ERCA

EDBG EBW

GRPOPRI (0 -> 1)

Figure 4-11. TresoS Plugin snapshot for McuDAMUXandEDMAConfig form.

4.7.10.1 eDMAChannelPriority0 (McuDMAMUXandEDMAConfig)

eDMA Channel Priority 0,1,2 and 3 register value (DCHPRI0 up to DCHPRI3). First byte corresponds to DCHPRI0 register. Example: 0x80010203UL corresponds to

Table 4-69. Detailed description.

DCHPRI0	0x80
DCHPRI1	0x01
DCHPRI2	0x02
DCHPRI3	0x03

Note: each eDMA channel must have different priority.

DCHPRIIn[0] represents ECP bit of (DCHPRIIn) register corresponds to Enable Channel Preemption.

Table 4-70. Detailed description.

0	Channel n cannot be suspended by a higher priority channels service request.
1	Channel n can be temporarily suspended by the service request of a higher priority channel.

DCHPRIIn[1] represents DPA bit of (DCHPRIIn) register corresponds to Disable Preempt Ability.

Table 4-71. Detailed description.

0	Channel n can suspend a lower priority channel.
1	Channel n cannot suspend any channel, regardless of channel priority.

DCHPRIIn[4_7] represents CHPRI bit of (DCHPRIIn) register corresponds to channel priority. Value ranges from 0 to 15.

Note: Implementation specific Parameter.

Table 4-72. Attribute eDMAChannelPriority0 (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	eDMA Channel Priority 0
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	66051
Invalid	Range <=4294967295 >=0

4.7.10.2 eDMAChannelPriority4 (McuDMAMUXandEDMAConfig)

eDMA Channel Priority 4,5,6 and 7 register value (DCHPRI4 up to DCHPRI7). First byte corresponds to DCHPRI4 register. Example: 0x80010203UL corresponds to

Table 4-73. Detailed description.

DCHPRI4	0x80
DCHPRI5	0x01
DCHPRI6	0x02
DCHPRI7	0x03

Note: each eDMA channel must have different priority.

DCHPRIn[0] represents ECP bit of (DCHPRIn) register corresponds to Enable Channel Preemption.

Table 4-74. Detailed description.

0	Channel n cannot be suspended by a higher priority channels service request.
1	Channel n can be temporarily suspended by the service request of a higher priority channel.

DCHPRIn[1] represents DPA bit of (DCHPRIn) register corresponds to Disable Preempt Ability.

Table 4-75. Detailed description.

0	Channel n can suspend a lower priority channel.
1	Channel n cannot suspend any channel, regardless of channel priority.

DCHPRIn[4_7] represents CHPRI bit of (DCHPRIn) register corresponds to channel priority. Value ranges from 0 to 15.

Note: Implementation specific Parameter.

Table 4-76. Attribute eDMAChannelPriority4 (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	eDMA Channel Priority 4
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	67438087
Invalid	Range <=4294967295 >=0

4.7.10.3 eDMAChannelPriority8 (McuDMAMUXandEDMAConfig)

eDMA Channel Priority 8,9,10 and 11 register value (DCHPRI8 up to DCHPRI11). First byte corresponds to DCHPRI8 register. Example: 0x80010203UL corresponds to

Table 4-77. Detailed description.

DCHPRI8	0x80
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Table continues on the next page...

Table 4-77. Detailed description. (continued)

DCHPRI9	0x01
DCHPRI10	0x02
DCHPRI11	0x03

Note: each eDMA channel must have different priority.

DCHPRIIn[0] represents ECP bit of (DCHPRIIn) register corresponds to Enable Channel Preemption.

Table 4-78. Detailed description.

0	Channel n cannot be suspended by a higher priority channels service request.
1	Channel n can be temporarily suspended by the service request of a higher priority channel.

DCHPRIIn[1] represents DPA bit of (DCHPRIIn) register corresponds to Disable Preempt Ability.

Table 4-79. Detailed description.

0	Channel n can suspend a lower priority channel.
1	Channel n cannot suspend any channel, regardless of channel priority.

DCHPRIIn[4_7] represents CHPRI bit of (DCHPRIIn) register corresponds to channel priority. Value ranges from 0 to 15.

Note: Implementation specific Parameter.

**Table 4-80. Attribute eDMAChannelPriority8
(McuDMAMUXandEDMAConfig) detailed description**

Property	Value
Label	eDMA Channel Priority 8
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	134810123
Invalid	Range <=4294967295 >=0

4.7.10.4 eDMAChannelPriority12 (McuDMAMUXandEDMAConfig)

eDMA Channel Priority 12,13,14 and 15 register value (DCHPRI12 up to DCHPRI15). First byte corresponds to DCHPRI12 register. Example: 0x00010203UL corresponds to

Table 4-81. Detailed description.

DCHPRI12	0x80
DCHPRI13	0x01
DCHPRI14	0x02
DCHPRI15	0x03

Note: each eDMA channel must have different priority.

DCHPRIIn[0] represents ECP bit of (DCHPRIIn) register corresponds to Enable Channel Preemption.

Table 4-82. Detailed description.

0	Channel n cannot be suspended by a higher priority channels service request.
1	Channel n can be temporarily suspended by the service request of a higher priority channel.

DCHPRIIn[1] represents DPA bit of (DCHPRIIn) register corresponds to Disable Preempt Ability.

Table 4-83. Detailed description.

0	Channel n can suspend a lower priority channel.
1	Channel n cannot suspend any channel, regardless of channel priority.

DCHPRIIn[4_7] represents CHPRI bit of (DCHPRIIn) register corresponds to channel priority. Value ranges from 0 to 15.

Note: Implementation specific Parameter.

Table 4-84. Attribute eDMAChannelPriority12 (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	eDMA Channel Priority 12
Type	INTEGER
Origin	Custom

Table continues on the next page...

Table 4-84. Attribute eDMAChannelPriority12 (McuDMAMUXandEDMAConfig) detailed description (continued)

Property	Value
Symbolic Name	false
Default	202182159
Invalid	Range <=4294967295 >=0

4.7.10.5 EMLM (McuDMAMUXandEDMAConfig)

Enable minor loop mapping

Table 4-85. Detailed description.

0	Minor loop mapping disabled. TCD Word 2 is defined as a 32-bit nbytes field.
1	Minor loop mapping enabled. When set, TCDn Word 2 is redefined to include individual enable fields, an offset field and the NBYTES field. The individual enable fields allow the minor loop offset to be applied to the source address, the destination address, or both. The NBYTES field is reduced when either offset is enabled

Note: Implementation specific Parameter.

Table 4-86. Attribute EMLM (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	EMLM Enable minor loop mapping
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.10.6 CLM (McuDMAMUXandEDMAConfig)

Continuous link mode

Table 4-87. Detailed description.

0	A minor loop channel link made to itself goes through channel arbitration before being activated again.
1	A minor loop channel link made to itself does not go through channel arbitration before being activated again. Upon minor loop completion, the channel is active again if that channel has a minor loop channel link enabled and the link channel is itself. This effectively applies the minor loop offsets and restarts the next minor loop.

Note: Implementation specific Parameter.

Table 4-88. Attribute CLM (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	CLM Continuous link mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.10.7 HALT (McuDMAMUXandEDMAConfig)

HALT DMA operations

Table 4-89. Detailed description.

0	Normal operation.
1	Stall the start of any new channels. Executing channels are allowed to complete. Channel execution resumes when the HALT bit is cleared.

Note: Implementation specific Parameter.

Table 4-90. Attribute HALT (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	HALT DMA operations

Table continues on the next page...

Table 4-90. Attribute HALT (McuDMAMUXandEDMAConfig) detailed description (continued)

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.10.8 HOE (McuDMAMUXandEDMAConfig)

Halt on error

Table 4-91. Detailed description.

0	Normal operation.
1	Any error causes the HALT bit to be set. Subsequently, all service requests are ignored until the HALT bit is cleared

Note: Implementation specific Parameter.

Table 4-92. Attribute HOE (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	HOE Halt on error
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.10.9 ERGA (McuDMAMUXandEDMAConfig)

Enable Round-Robin Group Arbitration.

Table 4-93. Detailed description.

0	Fixed-priority arbitration is used for selection among the groups.
1	Round-robin arbitration is used for selection among the groups.

Note: Implementation specific Parameter.

Table 4-94. Attribute ERGA (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	ERGA Enable Round-Robin Group Arbitration
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.10.10 ERCA (McuDMAMUXandEDMAConfig)

Enable Round-Robin Channel Arbitration.

Table 4-95. Detailed description.

0	Fixed-priority arbitration is used for channel selection.
1	Round-robin arbitration is used for channel selection.

Note: Implementation specific Parameter.

Table 4-96. Attribute ERCA (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	ERCA
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.10.11 EDBG (McuDMAMUXandEDMAConfig)

Enable Debug.

Table 4-97. Detailed description.

0	The assertion of the system debug control input is ignored.
1	The assertion of the system debug control input causes the eDMA to stall the start of a new channel. Executing channels are allowed to complete. Channel execution will resume when either the system debug control input is negated or the EDBG bit is cleared.

Note: Implementation specific Parameter.

Table 4-98. Attribute EDBG (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	EDBG
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.10.12 EBW (McuDMAMUXandEDMAConfig)

Enable Buffered Writes.

Table 4-99. Detailed description.

0	The bufferable write signal (hprot[2]) is not asserted during AMBA AHB writes.
1	The bufferable write signal (hprot[2]) is asserted on all AMBA AHB writes except for the last write sequence.

Note: Implementation specific Parameter.

Table 4-100. Attribute EBW (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	EBW
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.10.13 GRP0PRI (McuDMAMUXandEDMAConfig)

Note

Group 0 priority level when fixed priority group arbitration is enabled.

Implementation specific Parameter.

Table 4-101. Attribute GRP0PRI (McuDMAMUXandEDMAConfig) detailed description

Property	Value
Label	GRP0PRI
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range <=1 >=0

4.7.11 Form McuFlashPFCR0

This container contains the SPECIFIC configuration (parameters) of the flash memory array.

Note: Implementation specific Container.

Is included by form : [Form McuModuleConfiguration](#)

The screenshot shows the 'Mcu Flash PFCR0' configuration form. It includes a folder icon and the text 'Mcu Flash PFCR0'. Below this, there are several configuration items, each with a document icon, a text input field, and an edit icon (pencil):

- Read While Write Control (0 -> 7): Input field contains '7'.
- Port 0 Page Buffer Configuration (0 -> 3): Input field contains '3'.
- Port 0 Data Prefetch Enable: Input field contains a checked checkbox.
- Port 0 Instruction Prefetch Enable: Input field contains a checked checkbox.
- Port 0 Prefetch Limit (0 -> 3): Input field contains '2'.
- Port 0 Buffer Enable: Input field contains a checked checkbox.

Figure 4-12. Tresos Plugin snapshot for McuFlashPFCR0 form.

4.7.11.1 ReadWhileWriteControl (McuFlashPFCR0)

Bank0 Read-While-Write Control. This 3-bit field defines the controller response to flash reads while the array is busy with a program (write) or erase operation. This field is set to 7 by hardware reset enabling the stall-while-write/erase and disabling the abort and notification interrupts.

Table 4-102. Detailed description.

0..3	Terminate any attempted read while write/erase with an error response
7	Generate a bus stall for a read while write/erase, disable the stall notification interrupt, disable the abort + abort notification interrupt.
6	Generate a bus stall for a read while write/erase, enable the stall notification interrupt, disable the abort + abort notification interrupt
5	Generate a bus stall for a read while write/erase, enable the operation abort, disable the abort notification interrupt.
4	Generate a bus stall for a read while write/erase, enable the operation abort and the abort notification interrupt

This field is set to 7 by hardware reset enabling the stall-while-write/erase and disabling the abort and notification interrupts.

Note:: implementation specific parameter.

Table 4-103. Attribute ReadWhileWriteControl (McuFlashPFCR0) detailed description

Property	Value
Label	Read While Write Control
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	7

4.7.11.2 Port0PageBufferConfiguration (McuFlashPFCR0)

Bank0, Port 0 Page Buffer Configuration. This field controls the configuration of the four page buffers in the PFLASH controller. The buffers can be organized as a "pool" of available resources, or with a fixed partition between instruction and data buffers. If enabled, when a buffer miss occurs, it is allocated to the least-recently-used buffer within

the group and the just-fetched entry then marked as most-recently-used. If the flash access is for the next-sequential line, the buffer is not marked as most-recently-used until the given address produces a buffer hit.

Table 4-104. Detailed description.

0	All four buffers are available for any flash access, that is, there is no partitioning of the buffers based on the access type.
1	Reserved.
2	The buffers are partitioned into two groups with buffers 0 and 1 allocated for instruction fetches and buffers 2 and 3 for data accesses.
3	The buffers are partitioned into two groups with buffers 0,1,2 allocated for instruction fetches and buffer 3 for data accesses.

This field is set to 3 by hardware reset.

Note:: implementation specific parameter.

Table 4-105. Attribute Port0PageBufferConfiguration (McuFlashPFCR0) detailed description

Property	Value
Label	Port 0 Page Buffer Configuration
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	3

4.7.11.3 Port0DataPrefetchEnable (McuFlashPFCR0)

Bank0, Port 0 Data Prefetch Enable. This field enables or disables prefetching initiated by a data read access. This field is set by hardware reset.

Table 4-106. Detailed description.

0	No prefetching is triggered by a data read access.
1	If page buffers are enabled (B0_P0_BFE = 1), prefetching is triggered by any data read access.

Note:: implementation specific parameter.

Table 4-107. Attribute Port0DataPrefetchEnable (McuFlashPFCR0) detailed description

Property	Value
Label	Port 0 Data Prefetch Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.11.4 Port0InstructionPrefetchEnable (McuFlashPFCR0)

Bank0, Port 0 Instruction Prefetch Enable. This field enables or disables prefetching initiated by an instruction fetch read access. This field is cleared by hardware reset.

Table 4-108. Detailed description.

0	No prefetching is triggered by an instruction fetch read access
1	If page buffers are enabled (B0_P0_BFE = 1), prefetching is triggered by any instruction fetch read access.

Note:: implementation specific parameter.

Table 4-109. Attribute Port0InstructionPrefetchEnable (McuFlashPFCR0) detailed description

Property	Value
Label	Port 0 Instruction Prefetch Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.11.5 Port0PrefetchLimit (McuFlashPFCR0)

Bank0, Port 0 Prefetch Limit. This field controls the prefetch algorithm used by the PFLASH controller. This field defines the prefetch behavior. In all situations when enabled, only a single prefetch is initiated on each buffer miss or hit. This field is set to 2 by hardware reset.

Table 4-110. Detailed description.

0	No prefetching is performed.
1	The referenced line is prefetched on a buffer miss, that is, prefetch on miss.
2 and 3	The referenced line is prefetched on a buffer miss, or the next sequential page is prefetched on a buffer hit (if not already present), that is, prefetch on miss or hit.

Note:: implementation specific parameter.

Table 4-111. Attribute Port0PrefetchLimit (McuFlashPFCR0) detailed description

Property	Value
Label	Port 0 Prefetch Limit
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	2

4.7.11.6 Port0BufferEnable (McuFlashPFCR0)

Bank0, Port 0 Buffer Enable. This bit enables or disables page buffer read hits. It is also used to invalidate the buffers. This bit is set by hardware reset, enabling the page buffers.

Table 4-112. Detailed description.

0	The page buffers are disabled from satisfying read requests, and all buffer valid bits are cleared.
1	The page buffers are enabled to satisfy read requests on hits. Buffer valid bits may be set when the buffers are successfully filled.

Note:: implementation specific parameter.

Table 4-113. Attribute Port0BufferEnable (McuFlashPFCR0) detailed description

Property	Value
Label	Port 0 Buffer Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.12 Form McuFlashPFCR1

This container contains the SPECIFIC configuration (parameters) of the flash memory array.

Note: Implementation specific Container.

Is included by form : [Form McuModuleConfiguration](#)

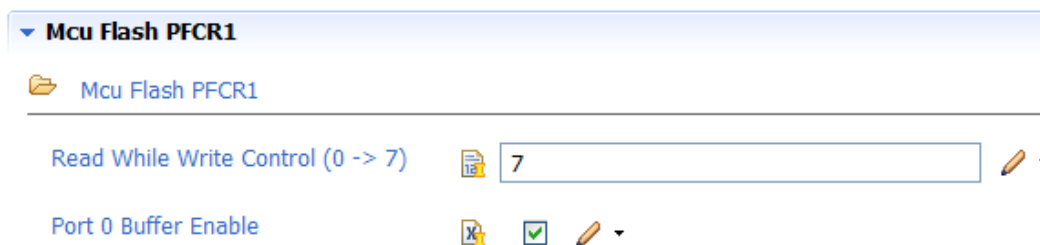


Figure 4-13. Tresos Plugin snapshot for McuFlashPFCR1 form.

4.7.12.1 ReadWhileWriteControl (McuFlashPFCR1)

Bank1 Read-While-Write Control. This 3-bit field defines the controller response to flash reads while the array is busy with a program (write) or erase operation. This field is set to 7 by hardware reset enabling the stall-while-write/erase and disabling the abort and notification interrupts.

Table 4-114. Detailed description.

0..3	Terminate any attempted read while write/erase with an error response
7	Generate a bus stall for a read while write/erase, disable the stall notification interrupt, disable the abort + abort notification interrupt.

Table continues on the next page...

Table 4-114. Detailed description. (continued)

6	Generate a bus stall for a read while write/erase, enable the stall notification interrupt, disable the abort + abort notification interrupt
5	Generate a bus stall for a read while write/erase, enable the operation abort, disable the abort notification interrupt.
4	Generate a bus stall for a read while write/erase, enable the operation abort and the abort notification interrupt

This field is set to 7 by hardware reset enabling the stall-while-write/erase and disabling the abort and notification interrupts.

Note:: implementation specific parameter.

Table 4-115. Attribute ReadWhileWriteControl (McuFlashPFCR1) detailed description

Property	Value
Label	Read While Write Control
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	7

4.7.12.2 Port0BufferEnable (McuFlashPFCR1)

Bank1, Port 0 Buffer Enable. This bit enables or disables page buffer read hits. It is also used to invalidate the buffers. This bit is set by hardware reset, enabling the page buffers.

Table 4-116. Detailed description.

0	The page buffers are disabled from satisfying read requests, and all buffer valid bits are cleared.
1	The page buffers are enabled to satisfy read requests on hits. Buffer valid bits may be set when the buffers are successfully filled.

Note:: implementation specific parameter.

Table 4-117. Attribute Port0BufferEnable (McuFlashPFCR1) detailed description

Property	Value
Label	Port 0 Buffer Enable

Table continues on the next page...

Table 4-117. Attribute Port0BufferEnable (McuFlashPFCR1) detailed description (continued)

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.13 Form McuFlashPFAPR

This container contains the SPECIFIC configuration (parameters) of the read and write accesses to the flash based on the system master number.

Note:: Implementation specific Container.

Is included by form : [Form McuModuleConfiguration](#)

Figure 4-14. Tresos Plugin snapshot for McuFlashPFAPR form.

4.7.13.1 Master0PrefetchDisable (McuFlashPFAPR)

Master x Prefetch Disable (x = 0,2). This bit controls whether prefetching may be triggered based on the master number of the requesting AHB master. This field is further qualified by the PFCRn[B02_Px_DPFE, B02_Px_IPFE, Bx_Py_BFE] bits.

Table 4-118. Detailed description.

0	Prefetching may be triggered by this master
1	No prefetching may be triggered by this master.

Note:: implementation specific parameter.

Table 4-119. Attribute Master0PrefetchDisable (McuFlashPFAPR) detailed description

Property	Value
Label	Master 0 Prefetch Disable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.13.2 Master2PrefetchDisable (McuFlashPFAPR)

Master x Prefetch Disable (x = 0,2). This bit controls whether prefetching may be triggered based on the master number of the requesting AHB master. This field is further qualified by the PFCRn[B02_Px_DPFE, B02_Px_IPFE, Bx_Py_BFE] bits.

Table 4-120. Detailed description.

0	Prefetching may be triggered by this master
1	No prefetching may be triggered by this master.

Note:: implementation specific parameter.

Table 4-121. Attribute Master2PrefetchDisable (McuFlashPFAPR) detailed description

Property	Value
Label	Master 2 Prefetch Disable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.13.3 Master0AccessProtection (McuFlashPFAPR)

Master x Access Protection (x = 0,2). This field controls whether read and write accesses to the flash are allowed based on the master number of the initiating module.

Table 4-122. Detailed description.

0	No accesses may be performed by this master
1	Only read accesses may be performed by this master.

Table continues on the next page...

Table 4-122. Detailed description. (continued)

2	Only write accesses may be performed by this master.
3	Both read and write accesses may be performed by this master.

Note:: implementation specific parameter.

Table 4-123. Attribute Master0AccessProtection (McuFlashPFAPR) detailed description

Property	Value
Label	Master 0 Access Protection
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	3

4.7.13.4 Master2AccessProtection (McuFlashPFAPR)

Master x Access Protection (x = 0,2). This field controls whether read and write accesses to the flash are allowed based on the master number of the initiating module.

Table 4-124. Detailed description.

0	No accesses may be performed by this master
1	Only read accesses may be performed by this master.
2	Only write accesses may be performed by this master.
3	Both read and write accesses may be performed by this master.

Note:: implementation specific parameter.

Table 4-125. Attribute Master2AccessProtection (McuFlashPFAPR) detailed description

Property	Value
Label	Master 2 Access Protection
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	3

4.7.14 Form McuDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus API in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Is included by form : [Form McuModuleConfiguration](#)

The screenshot shows a configuration window titled "Mcu Dem Event Parameter Refs". Below the title bar, there is a folder icon and the text "Mcu Dem Event Parameter Refs". The main area contains a list of four error events, each with a label, an icon, a dropdown menu, and an edit/delete icon.

Error Event	Path
MCU_E_CLOCK_FAILURE	/Dem/Dem/DemConfigSet_0/MCU_E_CLOCK
MCU_E_TIMEOUT_TRANSITION	/Dem/Dem/DemConfigSet_0/MCU_E_TIMEO
MCU_E_QUARTZ_FAILURE	/Dem/Dem/DemConfigSet_0/MCU_E_QUART
MCU_E_LOCK_FAILURE	/Dem/Dem/DemConfigSet_0/MCU_E_LOCK_

Figure 4-15. Tressos Plugin snapshot for McuDemEventParameterRefs form.

4.7.14.1 MCU_E_CLOCK_FAILURE (McuDemEventParameterRefs)

Reference to configured DEM event to report "Clock source failure".

Table 4-126. Attribute MCU_E_CLOCK_FAILURE (McuDemEventParameterRefs) detailed description

Property	Value
Label	MCU_E_CLOCK_FAILURE
Type	SYMBOLIC-NAME-REFERENCE
Origin	AUTOSAR_ECUC
Enable	true

4.7.14.2 MCU_E_TIMEOUT_TRANSITION (McuDemEventParameterRefs)

Reference to configured DEM event to report "Timeout expired".

**Table 4-127. Attribute MCU_E_TIMEOUT_TRANSITION
(McuDemEventParameterRefs) detailed description**

Property	Value
Label	MCU E_TIMEOUT_TRANSITION
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom
Enable	true

4.7.14.3 MCU_E_QUARTZ_FAILURE (McuDemEventParameterRefs)

Reference to configured DEM event to report "Oscillator frequency lower than RC frequency event."

**Table 4-128. Attribute MCU_E_QUARTZ_FAILURE
(McuDemEventParameterRefs) detailed description**

Property	Value
Label	MCU E_QUARTZ_FAILURE
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom
Enable	true

4.7.14.4 MCU_E_LOCK_FAILURE (McuDemEventParameterRefs)

Reference to configured DEM event to report "PLL loss the lock."

**Table 4-129. Attribute MCU_E_LOCK_FAILURE
(McuDemEventParameterRefs) detailed description**

Property	Value
Label	MCU E_LOCK_FAILURE
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom
Enable	true

4.7.15 Form McuDMA

This container contains the configuration parameters of eDMA channel multiplexer (DMA_MUX) that allows to route DMA sources (slots) to eDMA channels.

Note: Implementation specific Parameter.

Is included by form : [Form McuModuleConfiguration](#)

Figure 4-16. Tresos Plugin snapshot for McuDMA form.

4.7.15.1 Form DSPIO_TX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

DSPI0 TX

DSPI0 TX

DMA Channel Id (0 -> 35) 0

Mcu DMA Source DSPI0_TX

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-17. Tresos Plugin snapshot for DSPI0_TX form.

4.7.15.1.1 DMAChannelId (DSPI0_TX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-130. Attribute DMAChannelId (DSPI0_TX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range <=35 >=0

4.7.15.1.2 McuDMASource (DSPI0_TX)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-131. Attribute McuDMASource (DSPI0_TX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI0_TX
Range	DSPI0_TX

4.7.15.1.3 DMAChannelEnable (DSPIO_TX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-132. Attribute DMAChannelEnable (DSPIO_TX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.1.4 DMAChannelTriggerEnable (DSPIO_TX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-133. Attribute DMAChannelTriggerEnable (DSPIO_TX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.2 Form DSPIO_RX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration interface for the DSPI0_RX module. It includes a folder icon and the label 'DSPI0_RX'. Below this, there are three configuration items: 'DMA Channel Id (0 -> 35)' with a value of '1', 'Mcu DMA Source' with a value of 'DSPI0_RX', and 'DMA Channel Enable' and 'DMA Channel Trigger Enable' both set to 'false'.

Figure 4-18. TRESOS Plugin snapshot for DSPI0_RX form.

4.7.15.2.1 DMAChannelId (DSPI0_RX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-134. Attribute DMAChannelId (DSPI0_RX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	1
Invalid	Range <=35 >=0

4.7.15.2.2 McuDMASource (DSPI0_RX)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-135. Attribute McuDMASource (DSPI0_RX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI0_RX
Range	DSPI0_RX

4.7.15.2.3 DMAChannelEnable (DSPI0_RX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-136. Attribute DMAChannelEnable (DSPI0_RX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.2.4 DMAChannelTriggerEnable (DSPI0_RX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-137. Attribute DMAChannelTriggerEnable (DSPI0_RX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.3 Form DSPI1_TX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

DSPI1 TX

DSPI1 TX

DMA Channel Id (0 -> 35) 2

Mcu DMA Source DSPI1_TX

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-19. Tresos Plugin snapshot for DSPI1_TX form.

4.7.15.3.1 DMACHannelId (DSPI1_TX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-138. Attribute DMACHannelId (DSPI1_TX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	2
Invalid	Range <=35 >=0

4.7.15.3.2 McuDMASource (DSPI1_TX)

This parameter specifies the DMA Source for the DMA channel(bit SRC in DMAMUX_CHCONFIGx register)

Table 4-139. Attribute McuDMASource (DSPI1_TX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI1_TX
Range	DSPI1_TX

4.7.15.3.3 DMAChannelEnable (DSPI1_TX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-140. Attribute DMAChannelEnable (DSPI1_TX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.3.4 DMAChannelTriggerEnable (DSPI1_TX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-141. Attribute DMAChannelTriggerEnable (DSPI1_TX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.4 Form DSPI1_RX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnableA; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration interface for the DSPI1 RX module. It includes a title bar 'DSPI1 RX', a folder icon, and three main configuration rows. The first row is 'DMA Channel Id (0 -> 35)' with a dropdown menu showing '3'. The second row is 'Mcu DMA Source' with a dropdown menu showing 'DSPI1_RX'. The third row contains two checkboxes: 'DMA Channel Enable' and 'DMA Channel Trigger Enable', both of which are currently unchecked.

Figure 4-20. Tresos Plugin snapshot for DSPI1_RX form.

4.7.15.4.1 DMACHannelId (DSPI1_RX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-142. Attribute DMACHannelId (DSPI1_RX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	3
Invalid	Range <=35 >=0

4.7.15.4.2 McuDMASource (DSPI1_RX)

This parameter specifies the DMA Source for the DMA channel(bit SRC in DMAMUX_CHCONFIGx register)

Table 4-143. Attribute McuDMASource (DSPI1_RX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI1_RX
Range	DSPI1_RX

4.7.15.4.3 DMAChannelEnable (DSPI1_RX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-144. Attribute DMAChannelEnable (DSPI1_RX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.4.4 DMAChannelTriggerEnable (DSPI1_RX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-145. Attribute DMAChannelTriggerEnable (DSPI1_RX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.5 Form DSPI2_TX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnableA; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

DSPI2 TX

DSPI2 TX

DMA Channel Id (0 -> 35)

Mcu DMA Source

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-21. Tresos Plugin snapshot for DSPI2_TX form.

4.7.15.5.1 DMACHannelId (DSPI2_TX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-146. Attribute DMACHannelId (DSPI2_TX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	4
Invalid	Range <=35 >=0

4.7.15.5.2 McuDMASource (DSPI2_TX)

This parameter specifies the DMA Source for the DMA channel(bit SRC in DMAMUX_CHCONFIGx register)

Table 4-147. Attribute McuDMASource (DSPI2_TX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI2_TX
Range	DSPI2_TX

4.7.15.5.3 DMAChannelEnable (DSPI2_TX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-148. Attribute DMAChannelEnable (DSPI2_TX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.5.4 DMAChannelTriggerEnable (DSPI2_TX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-149. Attribute DMAChannelTriggerEnable (DSPI2_TX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.6 Form DSPI2_RX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnableA; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the 'DSPI2 RX' configuration form. It includes a folder icon and the text 'DSPI2 RX'. Below this, there are three configuration items: 'DMA Channel Id (0 -> 35)' with a dropdown menu showing '5', 'Mcu DMA Source' with a dropdown menu showing 'DSPI2_RX', and 'DMA Channel Enable' with an unchecked checkbox. To the right of these, there is a label 'DMA Channel Trigger Enable' followed by another unchecked checkbox. Each dropdown and checkbox has a small edit icon (pencil) next to it.

Figure 4-22. Tresos Plugin snapshot for DSPI2_RX form.

4.7.15.6.1 DMAChannelId (DSPI2_RX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-150. Attribute DMAChannelId (DSPI2_RX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	5
Invalid	Range <=35 >=0

4.7.15.6.2 McuDMASource (DSPI2_RX)

This parameter specifies the DMA Source for the DMA channel(bit SRC in DMAMUX_CHCONFIGx register)

Table 4-151. Attribute McuDMASource (DSPI2_RX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI2_RX
Range	DSPI2_RX

4.7.15.6.3 DMAChannelEnable (DSPI2_RX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-152. Attribute DMAChannelEnable (DSPI2_RX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.6.4 DMAChannelTriggerEnable (DSPI2_RX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-153. Attribute DMAChannelTriggerEnable (DSPI2_RX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.7 Form DSPI3_TX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnableA; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

▼ DSPI3 TX

DSPI3 TX

DMA Channel Id (0 -> 35) 6

Mcu DMA Source DSPI3_TX

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-23. Tressos Plugin snapshot for DSPI3_TX form.

4.7.15.7.1 DMACHannelId (DSPI3_TX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-154. Attribute DMACHannelId (DSPI3_TX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	6
Invalid	Range <=35 >=0

4.7.15.7.2 McuDMASource (DSPI3_TX)

This parameter specifies the DMA Source for the DMA channel(bit SRC in DMAMUX_CHCONFIGx register)

Table 4-155. Attribute McuDMASource (DSPI3_TX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI3_TX
Range	DSPI3_TX

4.7.15.7.3 DMAChannelEnable (DSPI3_TX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-156. Attribute DMAChannelEnable (DSPI3_TX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.7.4 DMAChannelTriggerEnable (DSPI3_TX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-157. Attribute DMAChannelTriggerEnable (DSPI3_TX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.8 Form DSPI3_RX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnableA; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

DSPI3 RX

DMA Channel Id (0 -> 35) 7

Mcu DMA Source DSPI3_RX

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-24. Tresos Plugin snapshot for DSPI3_RX form.

4.7.15.8.1 DMAChannelId (DSPI3_RX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-158. Attribute DMAChannelId (DSPI3_RX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	7
Invalid	Range <=35 >=0

4.7.15.8.2 McuDMASource (DSPI3_RX)

This parameter specifies the DMA Source for the DMA channel(bit SRC in DMAMUX_CHCONFIGx register)

Table 4-159. Attribute McuDMASource (DSPI3_RX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI3_RX
Range	DSPI3_RX

4.7.15.8.3 DMAChannelEnable (DSPI3_RX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-160. Attribute DMAChannelEnable (DSPI3_RX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.8.4 DMAChannelTriggerEnable (DSPI3_RX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-161. Attribute DMAChannelTriggerEnable (DSPI3_RX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.9 Form DSPI4_TX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnableA; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

DSPI4 TX

DMA Channel Id (0 -> 35) 8

Mcu DMA Source DSPI4_TX

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-25. Tressos Plugin snapshot for DSPI4_TX form.

4.7.15.9.1 DMACHannelId (DSPI4_TX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-162. Attribute DMACHannelId (DSPI4_TX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	8
Invalid	Range <=35 >=0

4.7.15.9.2 McuDMASource (DSPI4_TX)

This parameter specifies the DMA Source for the DMA channel(bit SRC in DMAMUX_CHCONFIGx register)

Table 4-163. Attribute McuDMASource (DSPI4_TX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI4_TX
Range	DSPI4_TX

4.7.15.9.3 DMAChannelEnable (DSPI4_TX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-164. Attribute DMAChannelEnable (DSPI4_TX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.9.4 DMAChannelTriggerEnable (DSPI4_TX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-165. Attribute DMAChannelTriggerEnable (DSPI4_TX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.10 Form DSPI4_RX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnableA; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration interface for the DSPI4 RX module. It includes a header 'DSPI4 RX' with a folder icon. Below it, there are three main configuration rows: 'DMA Channel Id (0 -> 35)' with a dropdown menu showing '9', 'Mcu DMA Source' with a dropdown menu showing 'DSPI4_RX', and 'DMA Channel Enable' with an unchecked checkbox. To the right of the 'DMA Channel Enable' row is the 'DMA Channel Trigger Enable' row, also with an unchecked checkbox. Each row has a small icon (a document with a blue 'P' or a pencil) to its right, indicating edit or save functionality.

Figure 4-26. Tresos Plugin snapshot for DSPI4_RX form.

4.7.15.10.1 DMACHannelId (DSPI4_RX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-166. Attribute DMACHannelId (DSPI4_RX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	9
Invalid	Range <=35 >=0

4.7.15.10.2 McuDMASource (DSPI4_RX)

This parameter specifies the DMA Source for the DMA channel(bit SRC in DMAMUX_CHCONFIGx register)

Table 4-167. Attribute McuDMASource (DSPI4_RX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI4_RX
Range	DSPI4_RX

4.7.15.10.3 DMAChannelEnable (DSPI4_RX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-168. Attribute DMAChannelEnable (DSPI4_RX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.10.4 DMAChannelTriggerEnable (DSPI4_RX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-169. Attribute DMAChannelTriggerEnable (DSPI4_RX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.11 Form DSPI5_TX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnableA; Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

Figure 4-27. Tresos Plugin snapshot for DSPI5_TX form.

4.7.15.11.1 DMACHannelId (DSPI5_TX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-170. Attribute DMACHannelId (DSPI5_TX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	10
Invalid	Range <=35 >=0

4.7.15.11.2 McuDMASource (DSPI5_TX)

This parameter specifies the DMA Source for the DMA channel(bit SRC in DMAMUX_CHCONFIGx register)

Table 4-171. Attribute McuDMASource (DSPI5_TX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI5_TX
Range	DSPI5_TX

4.7.15.11.3 DMAChannelEnable (DSPI5_TX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-172. Attribute DMAChannelEnable (DSPI5_TX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.11.4 DMAChannelTriggerEnable (DSPI5_TX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-173. Attribute DMAChannelTriggerEnable (DSPI5_TX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.12 Form DSPI5_RX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration interface for the DSPI5_RX module. It includes a header 'DSPI5 RX' and a sub-header 'DSPI5 RX'. Below these, there are four configuration fields: 'DMA Channel Id (0 -> 35)' with a value of 11, 'Mcu DMA Source' with a value of DSPI5_RX, 'DMA Channel Enable' with a checkbox that is unchecked, and 'DMA Channel Trigger Enable' with a checkbox that is unchecked. Each field has a small icon to its left and a pencil icon to its right for editing.

Figure 4-28. TRESOS Plugin snapshot for DSPI5_RX form.

4.7.15.12.1 DMACHannelId (DSPI5_RX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-174. Attribute DMACHannelId (DSPI5_RX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	11
Invalid	Range <=35 >=0

4.7.15.12.2 McuDMASource (DSPI5_RX)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-175. Attribute McuDMASource (DSPI5_RX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	DSPI5_RX
Range	DSPI5_RX

4.7.15.12.3 DMAChannelEnable (DSPI5_RX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-176. Attribute DMAChannelEnable (DSPI5_RX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.12.4 DMAChannelTriggerEnable (DSPI5_RX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-177. Attribute DMAChannelTriggerEnable (DSPI5_RX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.13 Form EMIOS0_CH0

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

Figure 4-29. TRESOS Plugin snapshot for EMIOS0_CH0 form.

4.7.15.13.1 DMACHannelId (EMIOS0_CH0)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-178. Attribute DMACHannelId (EMIOS0_CH0) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	12
Invalid	Range <=35 >=0

4.7.15.13.2 McuDMASource (EMIOS0_CH0)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-179. Attribute McuDMASource (EMIOS0_CH0) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS0_CH0
Range	EMIOS0_CH0

4.7.15.13.3 DMAChannelEnable (EMIOS0_CH0)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-180. Attribute DMAChannelEnable (EMIOS0_CH0) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.13.4 DMAChannelTriggerEnable (EMIOS0_CH0)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-181. Attribute DMAChannelTriggerEnable (EMIOS0_CH0) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.14 Form EMIOS0_CH1

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

Figure 4-30. Tresos Plugin snapshot for EMIOS0_CH1 form.

4.7.15.14.1 DMACHannelId (EMIOS0_CH1)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-182. Attribute DMACHannelId (EMIOS0_CH1) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	13
Invalid	Range <=35 >=0

4.7.15.14.2 McuDMASource (EMIOS0_CH1)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-183. Attribute McuDMASource (EMIOS0_CH1) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS0_CH1
Range	EMIOS0_CH1

4.7.15.14.3 DMAChannelEnable (EMIOS0_CH1)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-184. Attribute DMAChannelEnable (EMIOS0_CH1) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.14.4 DMAChannelTriggerEnable (EMIOS0_CH1)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-185. Attribute DMAChannelTriggerEnable (EMIOS0_CH1) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.15 Form EMIOS0_CH9

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the 'EMIOS0 CH9' configuration form. It includes a folder icon and the label 'EMIOS0 CH9'. Below this, there are three main configuration rows:

- DMA Channel Id (0 -> 35)**: A dropdown menu showing the value '14'.
- Mcu DMA Source**: A dropdown menu showing the value 'EMIOS0_CH9'.
- DMA Channel Enable**: A checkbox that is currently unchecked.
- DMA Channel Trigger Enable**: A checkbox that is currently unchecked.

 Each row has an edit icon (pencil) to its right.

Figure 4-31. Tresos Plugin snapshot for EMIOS0_CH9 form.

4.7.15.15.1 DMAChannelId (EMIOS0_CH9)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-186. Attribute DMAChannelId (EMIOS0_CH9) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	14
Invalid	Range <=35 >=0

4.7.15.15.2 McuDMASource (EMIOS0_CH9)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-187. Attribute McuDMASource (EMIOS0_CH9) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS0_CH9
Range	EMIOS0_CH9

4.7.15.15.3 DMAChannelEnable (EMIOS0_CH9)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-188. Attribute DMAChannelEnable (EMIOS0_CH9) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.15.4 DMAChannelTriggerEnable (EMIOS0_CH9)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-189. Attribute DMAChannelTriggerEnable (EMIOS0_CH9) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.16 Form EMIOS0_CH18

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration interface for the EMIOS0_CH18 module. At the top, there's a header 'EMIOS0 CH18'. Below it, a folder icon is followed by 'EMIOS0 CH18'. The main configuration area includes three rows: 'DMA Channel Id (0 -> 35)' with a dropdown menu showing '15', 'Mcu DMA Source' with a dropdown menu showing 'EMIOS0_CH18', and 'DMA Channel Enable' with an unchecked checkbox. To the right, 'DMA Channel Trigger Enable' also has an unchecked checkbox. Each input field has a small icon for saving or editing.

Figure 4-32. Tresos Plugin snapshot for EMIOS0_CH18 form.

4.7.15.16.1 DMAChannelId (EMIOS0_CH18)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-190. Attribute DMAChannelId (EMIOS0_CH18) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	15
Invalid	Range <=35 >=0

4.7.15.16.2 McuDMASource (EMIOS0_CH18)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-191. Attribute McuDMASource (EMIOS0_CH18) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS0_CH18
Range	EMIOS0_CH18

4.7.15.16.3 DMAChannelEnable (EMIOS0_CH18)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-192. Attribute DMAChannelEnable (EMIOS0_CH18) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.16.4 DMAChannelTriggerEnable (EMIOS0_CH18)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-193. Attribute DMAChannelTriggerEnable (EMIOS0_CH18) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.17 Form EMIOS0_CH25

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

Figure 4-33. TRESOS Plugin snapshot for EMIOS0_CH25 form.

4.7.15.17.1 DMACHannelId (EMIOS0_CH25)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-194. Attribute DMACHannelId (EMIOS0_CH25) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	16
Invalid	Range <=35 >=0

4.7.15.17.2 McuDMASource (EMIOS0_CH25)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-195. Attribute McuDMASource (EMIOS0_CH25) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS0_CH25
Range	EMIOS0_CH25

4.7.15.17.3 DMAChannelEnable (EMIOS0_CH25)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-196. Attribute DMAChannelEnable (EMIOS0_CH25) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.17.4 DMAChannelTriggerEnable (EMIOS0_CH25)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-197. Attribute DMAChannelTriggerEnable (EMIOS0_CH25) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.18 Form EMIOS0_CH26

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

EMIOS0 CH26

EMIOS0 CH26

DMA Channel Id (0 -> 35)

Mcu DMA Source

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-34. Tresos Plugin snapshot for EMIOS0_CH26 form.

4.7.15.18.1 DMACHannelId (EMIOS0_CH26)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-198. Attribute DMACHannelId (EMIOS0_CH26) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	17
Invalid	Range <=35 >=0

4.7.15.18.2 McuDMASource (EMIOS0_CH26)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-199. Attribute McuDMASource (EMIOS0_CH26) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS0_CH26
Range	EMIOS0_CH26

4.7.15.18.3 DMAChannelEnable (EMIOS0_CH26)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-200. Attribute DMAChannelEnable (EMIOS0_CH26) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.18.4 DMAChannelTriggerEnable (EMIOS0_CH26)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-201. Attribute DMAChannelTriggerEnable (EMIOS0_CH26) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.19 Form EMIOS1_CH0

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration interface for the EMIOS1_CH0 module. It includes a dropdown for DMA Channel Id (0 to 35) with the value 18 selected, a dropdown for Mcu DMA Source with EMIOS1_CH0 selected, and two checkboxes for DMA Channel Enable and DMA Channel Trigger Enable, both of which are currently disabled.

Figure 4-35. TRESOS Plugin snapshot for EMIOS1_CH0 form.

4.7.15.19.1 DMAChannelId (EMIOS1_CH0)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-202. Attribute DMAChannelId (EMIOS1_CH0) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	18
Invalid	Range <=35 >=0

4.7.15.19.2 McuDMASource (EMIOS1_CH0)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-203. Attribute McuDMASource (EMIOS1_CH0) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS1_CH0
Range	EMIOS1_CH0

4.7.15.19.3 DMAChannelEnable (EMIOS1_CH0)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-204. Attribute DMAChannelEnable (EMIOS1_CH0) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.19.4 DMAChannelTriggerEnable (EMIOS1_CH0)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-205. Attribute DMAChannelTriggerEnable (EMIOS1_CH0) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.20 Form EMIOS1_CH9

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the 'EMIOS1 CH9' configuration form. It includes a folder icon and the text 'EMIOS1 CH9'. Below this, there are four configuration items: 'DMA Channel Id (0 -> 35)' with a dropdown menu showing '19', 'Mcu DMA Source' with a dropdown menu showing 'EMIOS1_CH9', 'DMA Channel Enable' with an unchecked checkbox, and 'DMA Channel Trigger Enable' with an unchecked checkbox. Each item has a document icon and an edit icon.

Figure 4-36. TRESOS Plugin snapshot for EMIOS1_CH9 form.

4.7.15.20.1 DMACHannelId (EMIOS1_CH9)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-206. Attribute DMACHannelId (EMIOS1_CH9) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	19
Invalid	Range <=35 >=0

4.7.15.20.2 McuDMASource (EMIOS1_CH9)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-207. Attribute McuDMASource (EMIOS1_CH9) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS1_CH9
Range	EMIOS1_CH9

4.7.15.20.3 DMAChannelEnable (EMIOS1_CH9)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-208. Attribute DMAChannelEnable (EMIOS1_CH9) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.20.4 DMAChannelTriggerEnable (EMIOS1_CH9)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-209. Attribute DMAChannelTriggerEnable (EMIOS1_CH9) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.21 Form EMIOS1_CH17

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the 'EMIOS1 CH17' configuration form. It includes a folder icon and the text 'EMIOS1 CH17'. Below this, there are three main configuration rows: 'DMA Channel Id (0 -> 35)' with a dropdown menu showing '20', 'Mcu DMA Source' with a dropdown menu showing 'EMIOS1_CH17', and 'DMA Channel Enable' with an unchecked checkbox. To the right of the 'DMA Channel Enable' row is another row for 'DMA Channel Trigger Enable' with an unchecked checkbox. Each row has a small icon (a document with a pencil) to its right, indicating it is a configurable field.

Figure 4-37. Tresos Plugin snapshot for EMIOS1_CH17 form.

4.7.15.21.1 DMACHannelId (EMIOS1_CH17)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-210. Attribute DMACHannelId (EMIOS1_CH17) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	20
Invalid	Range <=35 >=0

4.7.15.21.2 McuDMASource (EMIOS1_CH17)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-211. Attribute McuDMASource (EMIOS1_CH17) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS1_CH17
Range	EMIOS1_CH17

4.7.15.21.3 DMAChannelEnable (EMIOS1_CH17)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-212. Attribute DMAChannelEnable (EMIOS1_CH17) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.21.4 DMAChannelTriggerEnable (EMIOS1_CH17)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-213. Attribute DMAChannelTriggerEnable (EMIOS1_CH17) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.22 Form EMIOS1_CH18

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration interface for the EMIOS1_CH18 module. It includes a header bar with the module name, a folder icon, and a list of configuration parameters. The parameters are: DMA Channel Id (0 -> 35) with a value of 21, Mcu DMA Source with a value of EMIOS1_CH18, DMA Channel Enable with a checkbox that is unchecked, and DMA Channel Trigger Enable with a checkbox that is unchecked. Each parameter has a small icon to its left and a pencil icon to its right for editing.

Figure 4-38. TRESOS Plugin snapshot for EMIOS1_CH18 form.

4.7.15.22.1 DMACHannelId (EMIOS1_CH18)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-214. Attribute DMACHannelId (EMIOS1_CH18) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	21
Invalid	Range <=35 >=0

4.7.15.22.2 McuDMASource (EMIOS1_CH18)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-215. Attribute McuDMASource (EMIOS1_CH18) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS1_CH18
Range	EMIOS1_CH18

4.7.15.22.3 DMAChannelEnable (EMIOS1_CH18)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-216. Attribute DMAChannelEnable (EMIOS1_CH18) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.22.4 DMAChannelTriggerEnable (EMIOS1_CH18)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-217. Attribute DMAChannelTriggerEnable (EMIOS1_CH18) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.23 Form EMIOS1_CH25

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration interface for the EMIOS1_CH25 module. It includes a title bar 'EMIOS1 CH25', a folder icon, and several configuration fields: 'DMA Channel Id (0 -> 35)' with a value of 22, 'Mcu DMA Source' with a dropdown set to 'EMIOS1_CH25', and two checkboxes, 'DMA Channel Enable' and 'DMA Channel Trigger Enable', both of which are currently unchecked.

Figure 4-39. Tresos Plugin snapshot for EMIOS1_CH25 form.

4.7.15.23.1 DMACHannelId (EMIOS1_CH25)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-218. Attribute DMACHannelId (EMIOS1_CH25) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	22
Invalid	Range <=35 >=0

4.7.15.23.2 McuDMASource (EMIOS1_CH25)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-219. Attribute McuDMASource (EMIOS1_CH25) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS1_CH25
Range	EMIOS1_CH25

4.7.15.23.3 DMAChannelEnable (EMIOS1_CH25)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-220. Attribute DMAChannelEnable (EMIOS1_CH25) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.23.4 DMAChannelTriggerEnable (EMIOS1_CH25)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-221. Attribute DMAChannelTriggerEnable (EMIOS1_CH25) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.24 Form EMIOS1_CH26

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration interface for EMIOS1_CH26. It includes a header 'EMIOS1 CH26' with a folder icon. Below are three configuration rows: 'DMA Channel Id (0 -> 35)' with a dropdown menu showing '23', 'Mcu DMA Source' with a dropdown menu showing 'EMIOS1_CH26', and 'DMA Channel Enable' with an unchecked checkbox. To the right, 'DMA Channel Trigger Enable' also has an unchecked checkbox. Each input field has a small icon (document with a lock) and a pencil icon for editing.

Figure 4-40. Tresos Plugin snapshot for EMIOS1_CH26 form.

4.7.15.24.1 DMAChannelId (EMIOS1_CH26)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-222. Attribute DMAChannelId (EMIOS1_CH26) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	23
Invalid	Range <=35 >=0

4.7.15.24.2 McuDMASource (EMIOS1_CH26)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-223. Attribute McuDMASource (EMIOS1_CH26) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	EMIOS1_CH26
Range	EMIOS1_CH26

4.7.15.24.3 DMAChannelEnable (EMIOS1_CH26)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-224. Attribute DMAChannelEnable (EMIOS1_CH26) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.24.4 DMAChannelTriggerEnable (EMIOS1_CH26)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-225. Attribute DMAChannelTriggerEnable (EMIOS1_CH26) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.25 Form ADC0_EOC

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

ADC0_EOC

DMA Channel Id (0 -> 35) 24

Mcu DMA Source ADC0_EOC

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-41. Tresos Plugin snapshot for ADC0_EOC form.

4.7.15.25.1 DMAChannelId (ADC0_EOC)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-226. Attribute DMAChannelId (ADC0_EOC) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	24
Invalid	Range <=35 >=0

4.7.15.25.2 McuDMASource (ADC0_EOC)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-227. Attribute McuDMASource (ADC0_EOC) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	ADC0_EOC
Range	ADC0_EOC

4.7.15.25.3 DMAChannelEnable (ADC0_EOC)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-228. Attribute DMAChannelEnable (ADC0_EOC) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.25.4 DMAChannelTriggerEnable (ADC0_EOC)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-229. Attribute DMAChannelTriggerEnable (ADC0_EOC) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.26 Form ADC1_EOC

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

Figure 4-42. Tresos Plugin snapshot for ADC1_EOC form.

4.7.15.26.1 DMAChannelId (ADC1_EOC)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-230. Attribute DMAChannelId (ADC1_EOC) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	25
Invalid	Range <=35 >=0

4.7.15.26.2 McuDMASource (ADC1_EOC)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-231. Attribute McuDMASource (ADC1_EOC) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	ADC1_EOC
Range	ADC1_EOC

4.7.15.26.3 DMAChannelEnable (ADC1_EOC)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-232. Attribute DMAChannelEnable (ADC1_EOC) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.26.4 DMAChannelTriggerEnable (ADC1_EOC)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-233. Attribute DMAChannelTriggerEnable (ADC1_EOC) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.27 Form IIC_RX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the 'IIC RX' configuration form. It includes a folder icon and the label 'IIC RX'. Below this, there are three main configuration rows:

- DMA Channel Id (0 -> 35)**: A text input field containing the value '26'.
- Mcu DMA Source**: A dropdown menu with 'IIC_RX' selected.
- DMA Channel Enable**: A checkbox that is currently unchecked.
- DMA Channel Trigger Enable**: A checkbox that is currently unchecked.

 Each input field has a small icon (a document with a blue 'i') and a pencil icon for editing.

Figure 4-43. Tresos Plugin snapshot for IIC_RX form.

4.7.15.27.1 DMAChannelId (IIC_RX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-234. Attribute DMAChannelId (IIC_RX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	26
Invalid	Range <=35 >=0

4.7.15.27.2 McuDMASource (IIC_RX)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-235. Attribute McuDMASource (IIC_RX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	IIC_RX
Range	IIC_RX

4.7.15.27.3 DMAChannelEnable (IIC_RX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-236. Attribute DMAChannelEnable (IIC_RX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.27.4 DMAChannelTriggerEnable (IIC_RX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-237. Attribute DMAChannelTriggerEnable (IIC_RX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.28 Form IIC_TX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the 'IIC TX' configuration form. At the top, there's a folder icon and the text 'IIC TX'. Below this, there are four configuration items:

- DMA Channel Id (0 -> 35)**: A text input field containing the value '27'.
- Mcu DMA Source**: A dropdown menu with 'IIC_TX' selected.
- DMA Channel Enable**: A checkbox that is currently unchecked.
- DMA Channel Trigger Enable**: A checkbox that is currently unchecked.

Figure 4-44. Tresos Plugin snapshot for IIC_TX form.

4.7.15.28.1 DMAChannelId (IIC_TX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-238. Attribute DMAChannelId (IIC_TX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	27
Invalid	Range <=35 >=0

4.7.15.28.2 McuDMASource (IIC_TX)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-239. Attribute McuDMASource (IIC_TX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	IIC_TX
Range	IIC_TX

4.7.15.28.3 DMAChannelEnable (IIC_TX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-240. Attribute DMAChannelEnable (IIC_TX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.28.4 DMAChannelTriggerEnable (IIC_TX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-241. Attribute DMAChannelTriggerEnable (IIC_TX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.29 Form LINFLEXD0_RX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

LINFLEXD0 RX

LINFLEXD0 RX

DMA Channel Id (0 -> 35) 28

Mcu DMA Source LINFLEXD0_RX

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-45. Tresos Plugin snapshot for LINFLEXD0_RX form.

4.7.15.29.1 DMACHannelId (LINFLEXD0_RX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-242. Attribute DMACHannelId (LINFLEXD0_RX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	28
Invalid	Range <=35 >=0

4.7.15.29.2 McuDMASource (LINFLEXD0_RX)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-243. Attribute McuDMASource (LINFLEXD0_RX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	LINFLEXD0_RX
Range	LINFLEXD0_RX

4.7.15.29.3 DMAChannelEnable (LINFLEXD0_RX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-244. Attribute DMAChannelEnable (LINFLEXD0_RX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.29.4 DMAChannelTriggerEnable (LINFLEXD0_RX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-245. Attribute DMAChannelTriggerEnable (LINFLEXD0_RX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.30 Form LINFLEXD0_TX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration interface for the LINFLEXD0_TX module. It includes a title bar 'LINFLEXD0 TX' and a folder icon. Below are three configuration rows: 'DMA Channel Id (0 -> 35)' with a dropdown menu showing '29', 'Mcu DMA Source' with a dropdown menu showing 'LINFLEXD0_TX', and 'DMA Channel Enable' with an unchecked checkbox. To the right, 'DMA Channel Trigger Enable' also has an unchecked checkbox. Each input field has a small icon for saving or editing.

Figure 4-46. TRESOS Plugin snapshot for LINFLEXD0_TX form.

4.7.15.30.1 DMACHannelId (LINFLEXD0_TX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-246. Attribute DMACHannelId (LINFLEXD0_TX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	29
Invalid	Range <=35 >=0

4.7.15.30.2 McuDMASource (LINFLEXD0_TX)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-247. Attribute McuDMASource (LINFLEXD0_TX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	LINFLEXD0_TX
Range	LINFLEXD0_TX

4.7.15.30.3 DMAChannelEnable (LINFLEXD0_TX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-248. Attribute DMAChannelEnable (LINFLEXD0_TX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.30.4 DMAChannelTriggerEnable (LINFLEXD0_TX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-249. Attribute DMAChannelTriggerEnable (LINFLEXD0_TX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.31 Form LINFLEXD1_RX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

LINFLEXD1 RX

LINFLEXD1 RX

DMA Channel Id (0 -> 35)

30

Mcu DMA Source

LINFLEXD1_RX

DMA Channel Enable

☐

DMA Channel Trigger Enable

☐

Figure 4-47. Tresos Plugin snapshot for LINFLEXD1_RX form.

4.7.15.31.1 DMACHannelId (LINFLEXD1_RX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-250. Attribute DMACHannelId (LINFLEXD1_RX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	30
Invalid	Range <=35 >=0

4.7.15.31.2 McuDMASource (LINFLEXD1_RX)

This parameter specifies the DMA Source for the DMA channel
bit SRC in DMAMUX_CHCONFIGx register

Table 4-251. Attribute McuDMASource (LINFLEXD1_RX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	LINFLEXD1_RX
Range	LINFLEXD1_RX

4.7.15.31.3 DMAChannelEnable (LINFLEXD1_RX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-252. Attribute DMAChannelEnable (LINFLEXD1_RX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.31.4 DMAChannelTriggerEnable (LINFLEXD1_RX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-253. Attribute DMAChannelTriggerEnable (LINFLEXD1_RX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.32 Form LINFLEXD1_TX

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

The screenshot shows the configuration for LINFLEXD1_TX. It includes a dropdown for DMA Channel Id (0 -> 35) with the value 31, a dropdown for Mcu DMA Source with the value LINFLEXD1_TX, and two checkboxes for DMA Channel Enable and DMA Channel Trigger Enable, both of which are currently unchecked.

Figure 4-48. Tresos Plugin snapshot for LINFLEXD1_TX form.

4.7.15.32.1 DMAChannelId (LINFLEXD1_TX)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-254. Attribute DMAChannelId (LINFLEXD1_TX) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	31
Invalid	Range <=35 >=0

4.7.15.32.2 McuDMASource (LINFLEXD1_TX)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-255. Attribute McuDMASource (LINFLEXD1_TX) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	LINFLEXD1_TX
Range	LINFLEXD1_TX

4.7.15.32.3 DMAChannelEnable (LINFLEXD1_TX)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-256. Attribute DMAChannelEnable (LINFLEXD1_TX) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.32.4 DMAChannelTriggerEnable (LINFLEXD1_TX)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-257. Attribute DMAChannelTriggerEnable (LINFLEXD1_TX) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.33 Form ALWAYS_ENABLED_0

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

▼ ALWAYS_ENABLED_0

ALWAYS_ENABLED_0

DMA Channel Id (0 -> 35) 32

Mcu DMA Source ALWAYS_ENABLED_0

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-49. Tresos Plugin snapshot for ALWAYS_ENABLED_0 form.

4.7.15.33.1 DMACHannelId (ALWAYS_ENABLED_0)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-258. Attribute DMACHannelId (ALWAYS_ENABLED_0) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	32
Invalid	Range <=35 >=0

4.7.15.33.2 McuDMASource (ALWAYS_ENABLED_0)

This parameter specifies the DMA Source for the DMA channel bit SRC in DMAMUX_CHCONFIGx register

Table 4-259. Attribute McuDMASource (ALWAYS_ENABLED_0) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	ALWAYS_ENABLED_0
Range	ALWAYS_ENABLED_0

4.7.15.33.3 DMAChannelEnable (ALWAYS_ENABLED_0)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-260. Attribute DMAChannelEnable (ALWAYS_ENABLED_0) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.33.4 DMAChannelTriggerEnable (ALWAYS_ENABLED_0)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-261. Attribute DMAChannelTriggerEnable (ALWAYS_ENABLED_0) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.34 Form ALWAYS_ENABLED_1

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

Figure 4-50. TRESOS Plugin snapshot for ALWAYS_ENABLED_1 form.

4.7.15.34.1 DMACHannelId (ALWAYS_ENABLED_1)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-262. Attribute DMACHannelId (ALWAYS_ENABLED_1) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	33
Invalid	Range <=35 >=0

4.7.15.34.2 McuDMASource (ALWAYS_ENABLED_1)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-263. Attribute McuDMASource (ALWAYS_ENABLED_1) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	ALWAYS_ENABLED_1
Range	ALWAYS_ENABLED_1

4.7.15.34.3 DMAChannelEnable (ALWAYS_ENABLED_1)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-264. Attribute DMAChannelEnable (ALWAYS_ENABLED_1) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.34.4 DMAChannelTriggerEnable (ALWAYS_ENABLED_1)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-265. Attribute DMAChannelTriggerEnable (ALWAYS_ENABLED_1) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.35 Form ALWAYS_ENABLED_2

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

▼ ALWAYS_ENABLED_2

ALWAYS_ENABLED_2

DMA Channel Id (0 -> 35) 34

Mcu DMA Source ALWAYS_ENABLED_2

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-51. Tresos Plugin snapshot for ALWAYS_ENABLED_2 form.

4.7.15.35.1 DMACHannelId (ALWAYS_ENABLED_2)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-266. Attribute DMACHannelId (ALWAYS_ENABLED_2) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	34
Invalid	Range ≤35 ≥0

4.7.15.35.2 McuDMASource (ALWAYS_ENABLED_2)

This parameter specifies the DMA Source for the DMA channel bit SRC in DMAMUX_CHCONFIGx register

Table 4-267. Attribute McuDMASource (ALWAYS_ENABLED_2) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	ALWAYS_ENABLED_2
Range	ALWAYS_ENABLED_2

4.7.15.35.3 DMAChannelEnable (ALWAYS_ENABLED_2)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-268. Attribute DMAChannelEnable (ALWAYS_ENABLED_2) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.35.4 DMAChannelTriggerEnable (ALWAYS_ENABLED_2)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-269. Attribute DMAChannelTriggerEnable (ALWAYS_ENABLED_2) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.36 Form ALWAYS_ENABLED_3

This register contains the combination for DMAChannelId, McuDMASource, DMAChannelEnable, and DMAChannelTriggerEnable Note that the column 'Name' is automatically generated, so the user must not edit this.

Is included by form : [Form McuDMA](#)

▼ ALWAYS_ENABLED_3

ALWAYS_ENABLED_3

DMA Channel Id (0 -> 35) 35

Mcu DMA Source ALWAYS_ENABLED_3

DMA Channel Enable ☐ DMA Channel Trigger Enable ☐

Figure 4-52. TRESOS Plugin snapshot for ALWAYS_ENABLED_3 form.

4.7.15.36.1 DMACHannelId (ALWAYS_ENABLED_3)

Channel Id of the DMAchannel.

Note: Implementation specific Parameter.

Table 4-270. Attribute DMACHannelId (ALWAYS_ENABLED_3) detailed description

Property	Value
Label	DMA Channel Id
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	35
Invalid	Range ≤35 ≥0

4.7.15.36.2 McuDMASource (ALWAYS_ENABLED_3)

This parameter specifies the DMA Source for the DMA channel

bit SRC in DMAMUX_CHCONFIGx register

Table 4-271. Attribute McuDMASource (ALWAYS_ENABLED_3) detailed description

Property	Value
Label	Mcu DMA Source
Type	STRING
Origin	Custom
Symbolic Name	false
Default	ALWAYS_ENABLED_3
Range	ALWAYS_ENABLED_3

4.7.15.36.3 DMAChannelEnable (ALWAYS_ENABLED_3)

DMA channel enable (bit ENBL in DMAMUX_CHCONFIGx register).

Note: Implementation specific Parameter.

Table 4-272. Attribute DMAChannelEnable (ALWAYS_ENABLED_3) detailed description

Property	Value
Label	DMA Channel Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.15.36.4 DMAChannelTriggerEnable (ALWAYS_ENABLED_3)

DMA channel trigger enable (bit TRIG in DMAMUX_CHCONFIGx register)

Note: Implementation specific Parameter.

Table 4-273. Attribute DMAChannelTriggerEnable (ALWAYS_ENABLED_3) detailed description

Property	Value
Label	DMA Channel Trigger Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.16 Form McuPeriphal

This contains the combination for current peripheral in Run and LowPower Mode.

Note:: Implementation specific Parameter.

Is included by form : [Form McuModuleConfiguration](#)

Included forms :

- [Form DSPIO](#)

- [Form DSPI1](#)
- [Form DSPI2](#)
- [Form DSPI3](#)
- [Form DSPI4](#)
- [Form DSPI5](#)
- [Form LIN_FLEX8](#)
- [Form LIN_FLEX9](#)
- [Form FlexCAN0](#)
- [Form FlexCAN1](#)
- [Form FlexCAN2](#)
- [Form FlexCAN3](#)
- [Form FlexCAN4](#)
- [Form FlexCAN5](#)
- [Form DMA_CH_MUX](#)
- [Form ADC0](#)
- [Form ADC1](#)
- [Form I2C0](#)
- [Form LIN_FLEX0](#)
- [Form LIN_FLEX1](#)
- [Form LIN_FLEX2](#)
- [Form LIN_FLEX3](#)
- [Form LIN_FLEX4](#)
- [Form LIN_FLEX5](#)
- [Form LIN_FLEX6](#)
- [Form LIN_FLEX7](#)
- [Form CTU](#)
- [Form CANSampler](#)
- [Form SIUL](#)
- [Form WKPU](#)
- [Form eMIOS0](#)
- [Form eMIOS1](#)
- [Form RTC_API](#)
- [Form PIT_RTI](#)
- [Form CMU0](#)

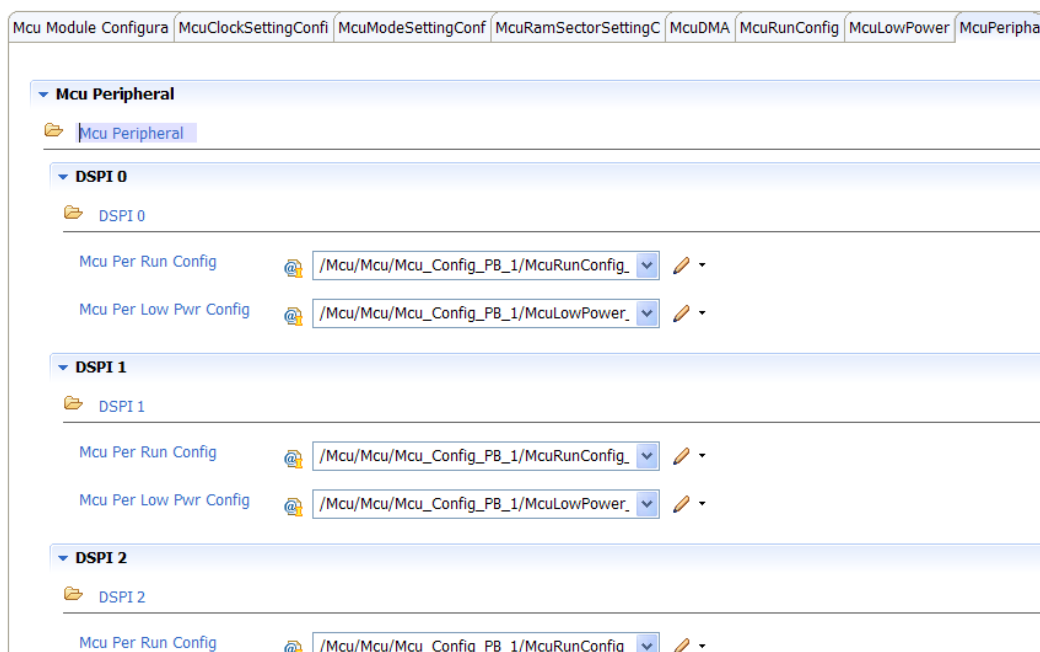


Figure 4-53. Tressos Plugin snapshot for McuPeripheral form.

4.7.16.1 Form DSPI0

This contains the combination for current peripheral in Run and LowPower Mode. Note: Implementation specific Parameter.

Is included by form : [Form McuPeripheral](#)

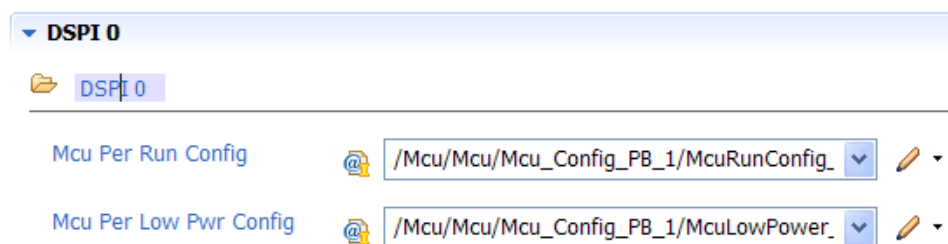


Figure 4-54. Tressos Plugin snapshot for DSPI0 form.

4.7.16.1.1 McuPerRunConfig (DSPI0)

Table 4-274. Attribute McuPerRunConfig (DSPI0) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.1.2 McuPerLowPwrConfig (DSPI0)

Table 4-275. Attribute McuPerLowPwrConfig (DSPI0) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.2 Form DSPI1

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

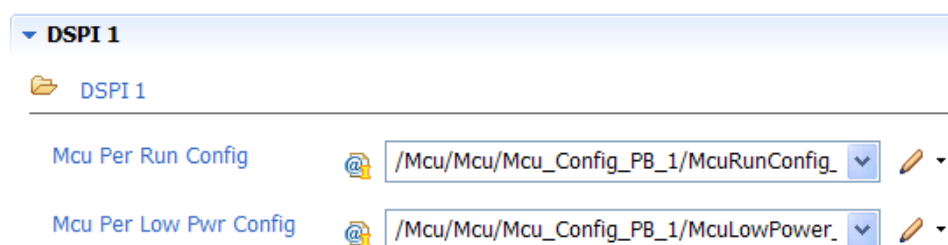


Figure 4-55. Tresos Plugin snapshot for DSPI1 form.

4.7.16.2.1 McuPerRunConfig (DSPI1)

Table 4-276. Attribute McuPerRunConfig (DSPI1) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.2.2 McuPerLowPwrConfig (DSPI1)

Table 4-277. Attribute McuPerLowPwrConfig (DSPI1) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.3 Form DSPI2

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeripheral](#)

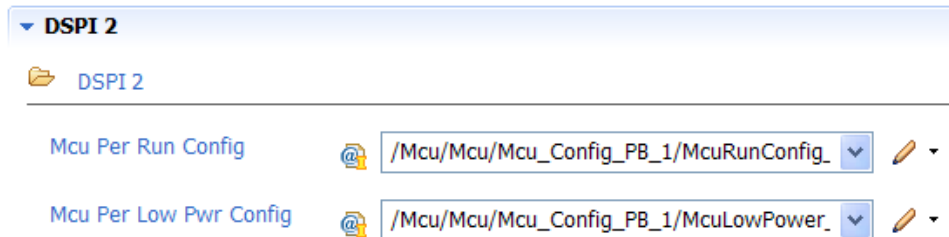


Figure 4-56. Tressos Plugin snapshot for DSPI2 form.

4.7.16.3.1 McuPerRunConfig (DSPI2)

Table 4-278. Attribute McuPerRunConfig (DSPI2) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.3.2 McuPerLowPwrConfig (DSPI2)

Table 4-279. Attribute McuPerLowPwrConfig (DSPI2) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.4 Form DSPI3

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeripheral](#)

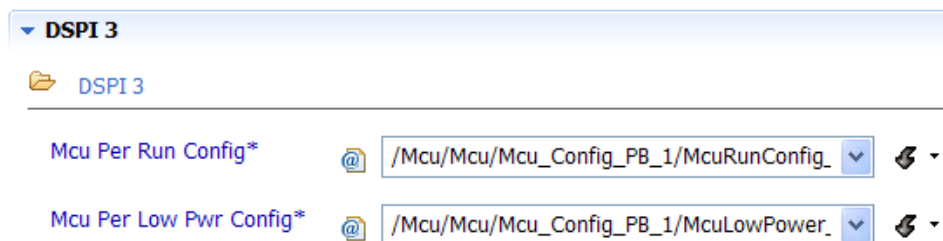


Figure 4-57. Tresos Plugin snapshot for DSPI3 form.

4.7.16.4.1 McuPerRunConfig (DSPI3)

Table 4-280. Attribute McuPerRunConfig (DSPI3) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.4.2 McuPerLowPwrConfig (DSPI3)

Table 4-281. Attribute McuPerLowPwrConfig (DSPI3) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.5 Form DSPI4

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

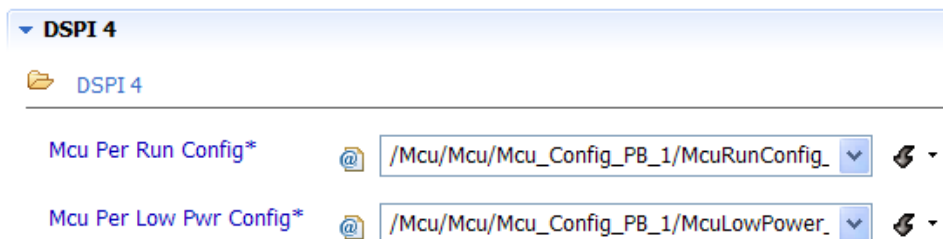


Figure 4-58. Tresos Plugin snapshot for DSPI4 form.

4.7.16.5.1 McuPerRunConfig (DSPI4)

Table 4-282. Attribute McuPerRunConfig (DSPI4) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.5.2 McuPerLowPwrConfig (DSPI4)

Table 4-283. Attribute McuPerLowPwrConfig (DSPI4) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.6 Form DSPI5

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeripheral](#)

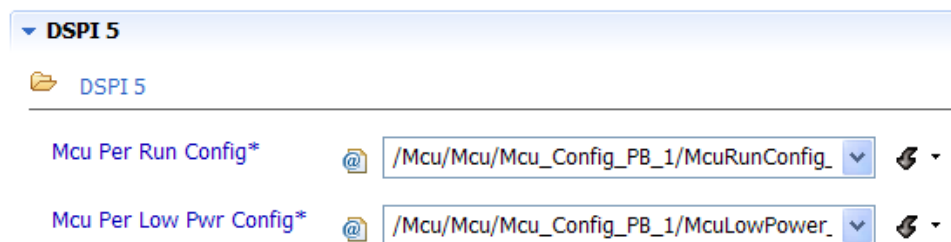


Figure 4-59. Tresos Plugin snapshot for DSPI5 form.

4.7.16.6.1 McuPerRunConfig (DSPI5)

Table 4-284. Attribute McuPerRunConfig (DSPI5) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.6.2 McuPerLowPwrConfig (DSPI5)

Table 4-285. Attribute McuPerLowPwrConfig (DSPI5) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.7 Form LIN_FLEX8

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

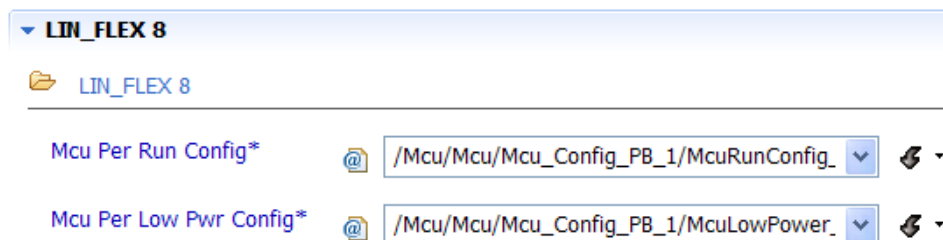


Figure 4-60. TRESOS Plugin snapshot for LIN_FLEX8 form.

4.7.16.7.1 McuPerRunConfig (LIN_FLEX8)

Table 4-286. Attribute McuPerRunConfig (LIN_FLEX8) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.7.2 McuPerLowPwrConfig (LIN_FLEX8)

Table 4-287. Attribute McuPerLowPwrConfig (LIN_FLEX8) detailed description

Property	Value
Label	Mcu Per Low Pwr Config

Table continues on the next page...

Table 4-287. Attribute McuPerLowPwrConfig (LIN_FLEX8) detailed description (continued)

Property	Value
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.8 Form LIN_FLEX9

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

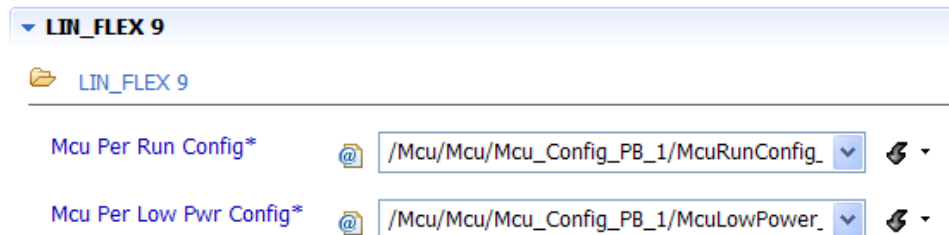


Figure 4-61. Tresos Plugin snapshot for LIN_FLEX9 form.

4.7.16.8.1 McuPerRunConfig (LIN_FLEX9)

Table 4-288. Attribute McuPerRunConfig (LIN_FLEX9) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.8.2 McuPerLowPwrConfig (LIN_FLEX9)

Table 4-289. Attribute McuPerLowPwrConfig (LIN_FLEX9) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.9 Form FlexCAN0

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

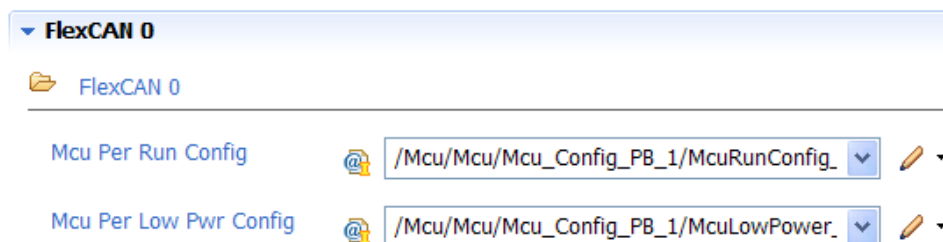


Figure 4-62. Tresos Plugin snapshot for FlexCAN0 form.

4.7.16.9.1 McuPerRunConfig (FlexCAN0)

Table 4-290. Attribute McuPerRunConfig (FlexCAN0) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.9.2 McuPerLowPwrConfig (FlexCAN0)

Table 4-291. Attribute McuPerLowPwrConfig (FlexCAN0) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.10 Form FlexCAN1

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeripheral](#)

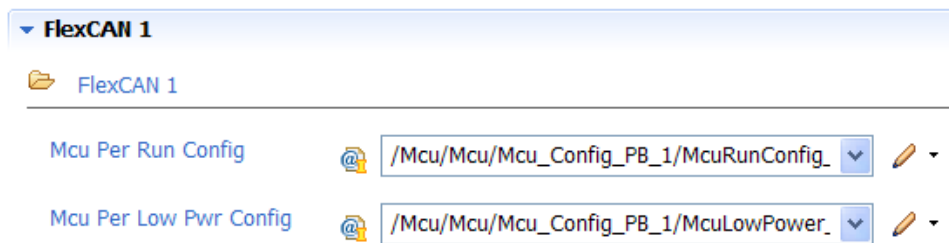


Figure 4-63. Tressos Plugin snapshot for FlexCAN1 form.

4.7.16.10.1 McuPerRunConfig (FlexCAN1)

Table 4-292. Attribute McuPerRunConfig (FlexCAN1) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.10.2 McuPerLowPwrConfig (FlexCAN1)

Table 4-293. Attribute McuPerLowPwrConfig (FlexCAN1) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.11 Form FlexCAN2

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeripheral](#)

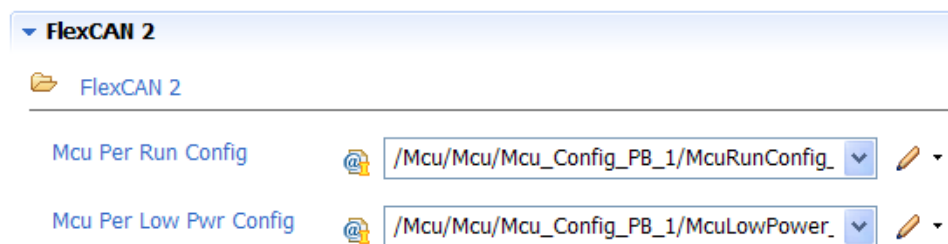


Figure 4-64. Tressos Plugin snapshot for FlexCAN2 form.

4.7.16.11.1 McuPerRunConfig (FlexCAN2)

Table 4-294. Attribute McuPerRunConfig (FlexCAN2) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.11.2 McuPerLowPwrConfig (FlexCAN2)

Table 4-295. Attribute McuPerLowPwrConfig (FlexCAN2) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.12 Form FlexCAN3

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

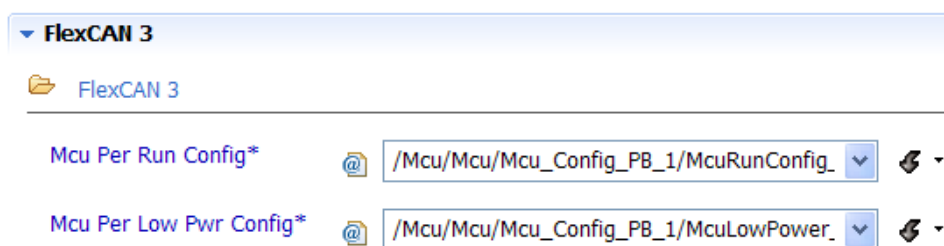


Figure 4-65. Tresos Plugin snapshot for FlexCAN3 form.

4.7.16.12.1 McuPerRunConfig (FlexCAN3)

Table 4-296. Attribute McuPerRunConfig (FlexCAN3) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.12.2 McuPerLowPwrConfig (FlexCAN3)

Table 4-297. Attribute McuPerLowPwrConfig (FlexCAN3) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.13 Form FlexCAN4

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

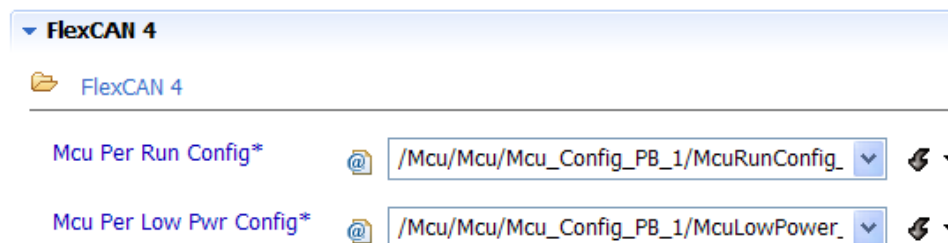


Figure 4-66. Tresos Plugin snapshot for FlexCAN4 form.

4.7.16.13.1 McuPerRunConfig (FlexCAN4)

Table 4-298. Attribute McuPerRunConfig (FlexCAN4) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.13.2 McuPerLowPwrConfig (FlexCAN4)

Table 4-299. Attribute McuPerLowPwrConfig (FlexCAN4) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.14 Form FlexCAN5

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

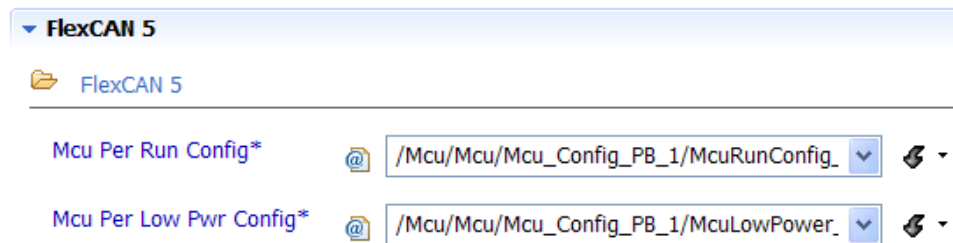


Figure 4-67. Tressos Plugin snapshot for FlexCAN5 form.

4.7.16.14.1 McuPerRunConfig (FlexCAN5)

Table 4-300. Attribute McuPerRunConfig (FlexCAN5) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.14.2 McuPerLowPwrConfig (FlexCAN5)

Table 4-301. Attribute McuPerLowPwrConfig (FlexCAN5) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.15 Form DMA_CH_MUX

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

DMA CH MUX

DMA CH MUX

Mcu Per Run Config* @ /Mcu/Mcu/Mcu_Config_PB_1/McuRunConfig_ ↕

Mcu Per Low Pwr Config* @ /Mcu/Mcu/Mcu_Config_PB_1/McuLowPower_ ↕

Figure 4-68. Tresos Plugin snapshot for DMA_CH_MUX form.

4.7.16.15.1 McuPerRunConfig (DMA_CH_MUX)

Table 4-302. Attribute McuPerRunConfig (DMA_CH_MUX) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.15.2 McuPerLowPwrConfig (DMA_CH_MUX)

Table 4-303. Attribute McuPerLowPwrConfig (DMA_CH_MUX) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.16 Form ADC0

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

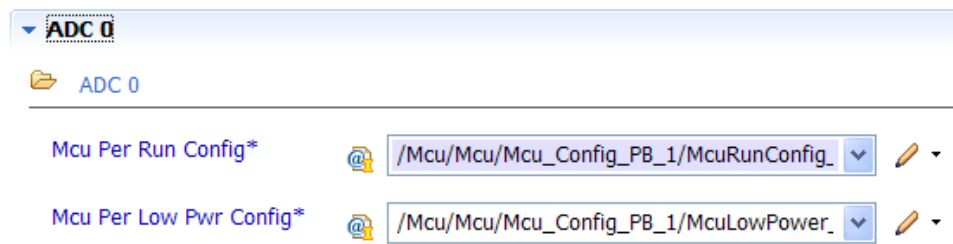


Figure 4-69. Tressos Plugin snapshot for ADC0 form.

4.7.16.16.1 McuPerRunConfig (ADC0)

Table 4-304. Attribute McuPerRunConfig (ADC0) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.16.2 McuPerLowPwrConfig (ADC0)

Table 4-305. Attribute McuPerLowPwrConfig (ADC0) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.17 Form ADC1

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeripheral](#)

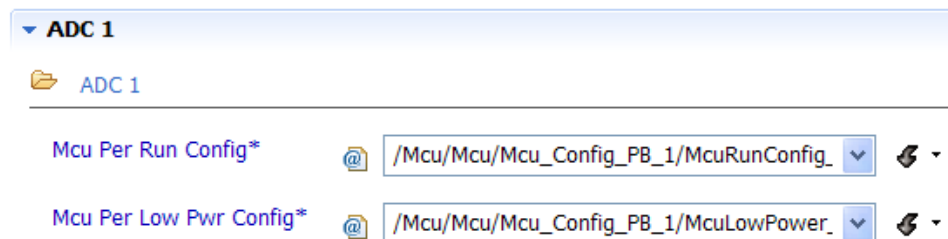


Figure 4-70. Tressos Plugin snapshot for ADC1 form.

4.7.16.17.1 McuPerRunConfig (ADC1)

Table 4-306. Attribute McuPerRunConfig (ADC1) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.17.2 McuPerLowPwrConfig (ADC1)

Table 4-307. Attribute McuPerLowPwrConfig (ADC1) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.18 Form I2C0

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeripheral](#)

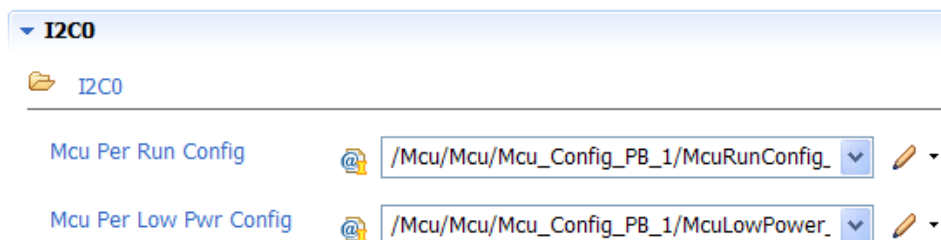


Figure 4-71. Tresos Plugin snapshot for I2C0 form.

4.7.16.18.1 McuPerRunConfig (I2C0)

Table 4-308. Attribute McuPerRunConfig (I2C0) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.18.2 McuPerLowPwrConfig (I2C0)

Table 4-309. Attribute McuPerLowPwrConfig (I2C0) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.19 Form LIN_FLEX0

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

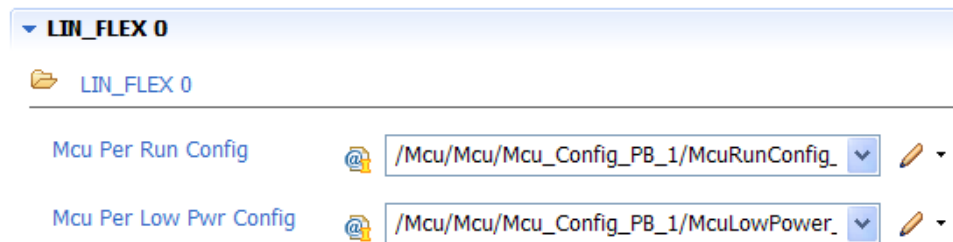


Figure 4-72. Tresos Plugin snapshot for LIN_FLEX0 form.

4.7.16.19.1 McuPerRunConfig (LIN_FLEX0)

Table 4-310. Attribute McuPerRunConfig (LIN_FLEX0) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.19.2 McuPerLowPwrConfig (LIN_FLEX0)

Table 4-311. Attribute McuPerLowPwrConfig (LIN_FLEX0) detailed description

Property	Value
Label	Mcu Per Low Pwr Config

Table continues on the next page...

Table 4-311. Attribute McuPerLowPwrConfig (LIN_FLEX0) detailed description (continued)

Property	Value
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.20 Form LIN_FLEX1

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

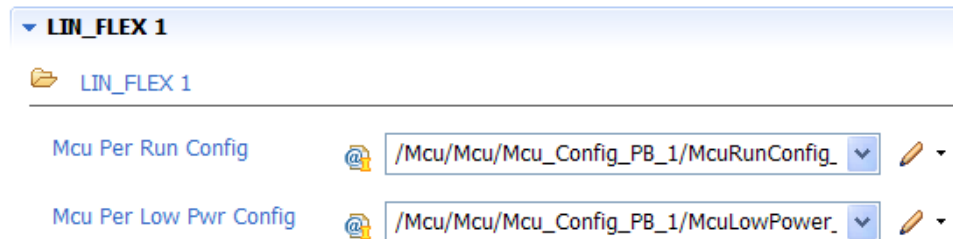


Figure 4-73. TRESOS Plugin snapshot for LIN_FLEX1 form.

4.7.16.20.1 McuPerRunConfig (LIN_FLEX1)

Table 4-312. Attribute McuPerRunConfig (LIN_FLEX1) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.20.2 McuPerLowPwrConfig (LIN_FLEX1)

Table 4-313. Attribute McuPerLowPwrConfig (LIN_FLEX1) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.21 Form LIN_FLEX2

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

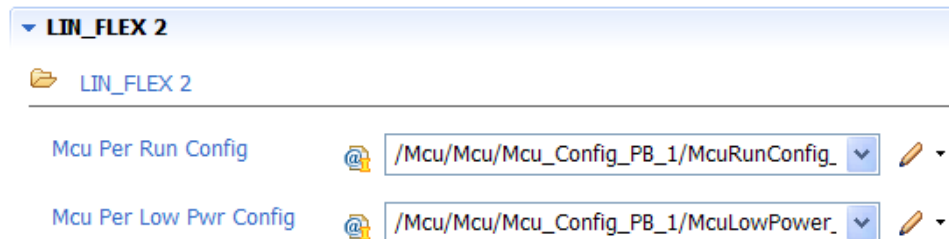


Figure 4-74. Tressos Plugin snapshot for LIN_FLEX2 form.

4.7.16.21.1 McuPerRunConfig (LIN_FLEX2)

Table 4-314. Attribute McuPerRunConfig (LIN_FLEX2) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.21.2 McuPerLowPwrConfig (LIN_FLEX2)

Table 4-315. Attribute McuPerLowPwrConfig (LIN_FLEX2) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.22 Form LIN_FLEX3

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

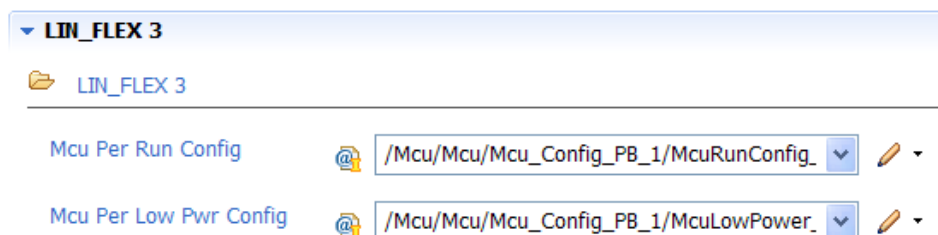


Figure 4-75. TRESOS Plugin snapshot for LIN_FLEX3 form.

4.7.16.22.1 McuPerRunConfig (LIN_FLEX3)

Table 4-316. Attribute McuPerRunConfig (LIN_FLEX3) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.22.2 McuPerLowPwrConfig (LIN_FLEX3)

Table 4-317. Attribute McuPerLowPwrConfig (LIN_FLEX3) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.23 Form LIN_FLEX4

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

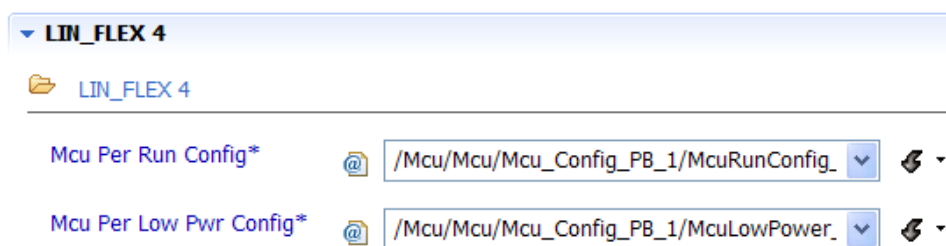


Figure 4-76. Tressos Plugin snapshot for LIN_FLEX4 form.

4.7.16.23.1 McuPerRunConfig (LIN_FLEX4)

Table 4-318. Attribute McuPerRunConfig (LIN_FLEX4) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.23.2 McuPerLowPwrConfig (LIN_FLEX4)

Table 4-319. Attribute McuPerLowPwrConfig (LIN_FLEX4) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.24 Form LIN_FLEX5

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

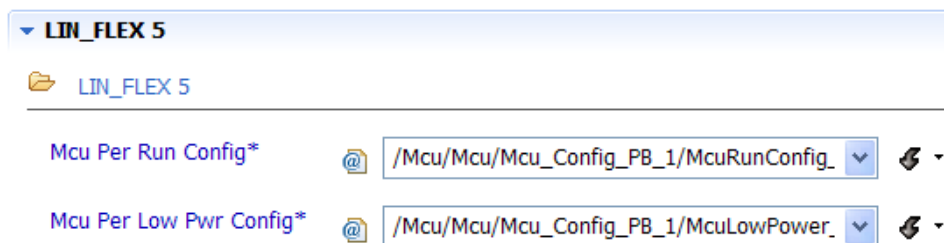


Figure 4-77. TRESOS Plugin snapshot for LIN_FLEX5 form.

4.7.16.24.1 McuPerRunConfig (LIN_FLEX5)

Table 4-320. Attribute McuPerRunConfig (LIN_FLEX5) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.24.2 McuPerLowPwrConfig (LIN_FLEX5)

Table 4-321. Attribute McuPerLowPwrConfig (LIN_FLEX5) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.25 Form LIN_FLEX6

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

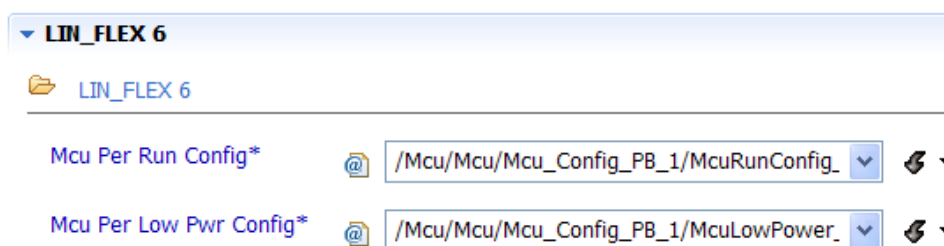


Figure 4-78. Tressos Plugin snapshot for LIN_FLEX6 form.

4.7.16.25.1 McuPerRunConfig (LIN_FLEX6)

Table 4-322. Attribute McuPerRunConfig (LIN_FLEX6) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.25.2 McuPerLowPwrConfig (LIN_FLEX6)

Table 4-323. Attribute McuPerLowPwrConfig (LIN_FLEX6) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.26 Form LIN_FLEX7

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

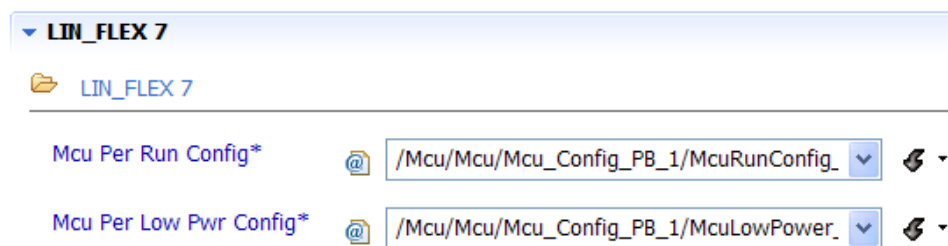


Figure 4-79. TRESOS Plugin snapshot for LIN_FLEX7 form.

4.7.16.26.1 McuPerRunConfig (LIN_FLEX7)

Table 4-324. Attribute McuPerRunConfig (LIN_FLEX7) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.26.2 McuPerLowPwrConfig (LIN_FLEX7)

Table 4-325. Attribute McuPerLowPwrConfig (LIN_FLEX7) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.27 Form CTU

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

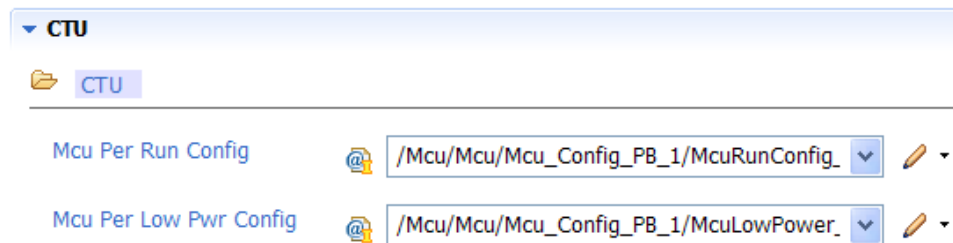


Figure 4-80. Tressos Plugin snapshot for CTU form.

4.7.16.27.1 McuPerRunConfig (CTU)

Table 4-326. Attribute McuPerRunConfig (CTU) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.27.2 McuPerLowPwrConfig (CTU)

Table 4-327. Attribute McuPerLowPwrConfig (CTU) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.28 Form CANSampler

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

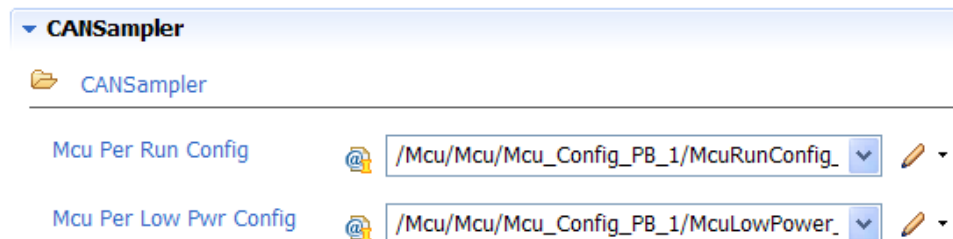


Figure 4-81. Tressos Plugin snapshot for CANSampler form.

4.7.16.28.1 McuPerRunConfig (CANSampler)

Table 4-328. Attribute McuPerRunConfig (CANSampler) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.28.2 McuPerLowPwrConfig (CANSampler)

Table 4-329. Attribute McuPerLowPwrConfig (CANSampler) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.29 Form SIUL

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

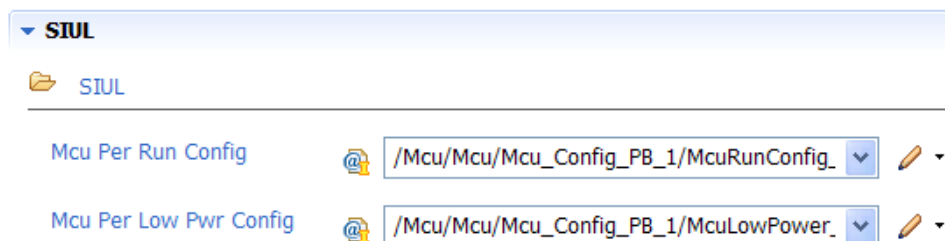


Figure 4-82. Tresos Plugin snapshot for SIUL form.

4.7.16.29.1 McuPerRunConfig (SIUL)

Table 4-330. Attribute McuPerRunConfig (SIUL) detailed description

Property	Value
Label	Mcu Per Run Config

Table continues on the next page...

Table 4-330. Attribute McuPerRunConfig (SIUL) detailed description (continued)

Property	Value
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.29.2 McuPerLowPwrConfig (SIUL)

Table 4-331. Attribute McuPerLowPwrConfig (SIUL) detailed description

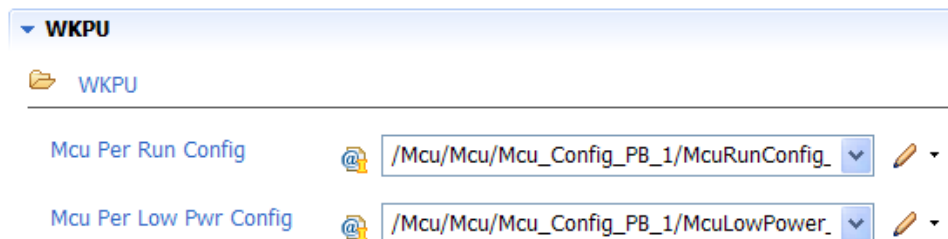
Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.30 Form WKPU

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

**Figure 4-83. Tresos Plugin snapshot for WKPU form.**

4.7.16.30.1 McuPerRunConfig (WKPU)

Table 4-332. Attribute McuPerRunConfig (WKPU) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.30.2 McuPerLowPwrConfig (WKPU)

Table 4-333. Attribute McuPerLowPwrConfig (WKPU) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.31 Form eMIOS0

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

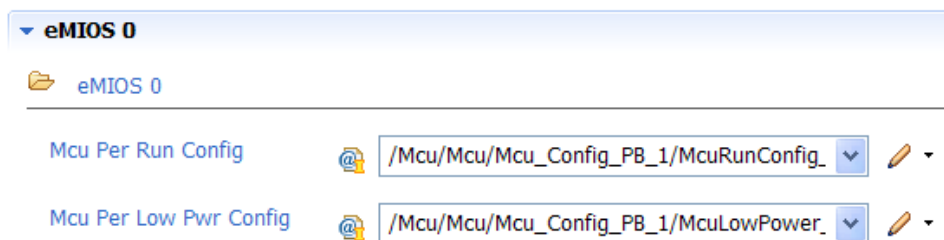


Figure 4-84. Tresos Plugin snapshot for eMIOS0 form.

4.7.16.31.1 McuPerRunConfig (eMIOS0)

Table 4-334. Attribute McuPerRunConfig (eMIOS0) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.31.2 McuPerLowPwrConfig (eMIOS0)

Table 4-335. Attribute McuPerLowPwrConfig (eMIOS0) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.32 Form eMIOS1

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

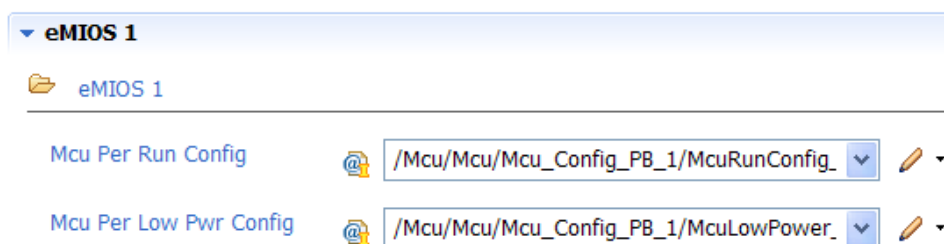


Figure 4-85. Tressos Plugin snapshot for eMIOS1 form.

4.7.16.32.1 McuPerRunConfig (eMIOS1)

Table 4-336. Attribute McuPerRunConfig (eMIOS1) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.32.2 McuPerLowPwrConfig (eMIOS1)

Table 4-337. Attribute McuPerLowPwrConfig (eMIOS1) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.33 Form RTC_API

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

▼ **RTC API**

RTC API

Mcu Per Run Config /Mcu/Mcu/Mcu_Config_PB_1/McuRunConfig_@

Mcu Per Low Pwr Config /Mcu/Mcu/Mcu_Config_PB_1/McuLowPower_@

Figure 4-86. TRESOS Plugin snapshot for RTC_API form.

4.7.16.33.1 McuPerRunConfig (RTC_API)

Table 4-338. Attribute McuPerRunConfig (RTC_API) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.33.2 McuPerLowPwrConfig (RTC_API)

Table 4-339. Attribute McuPerLowPwrConfig (RTC_API) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.34 Form PIT_RTI

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriphal](#)

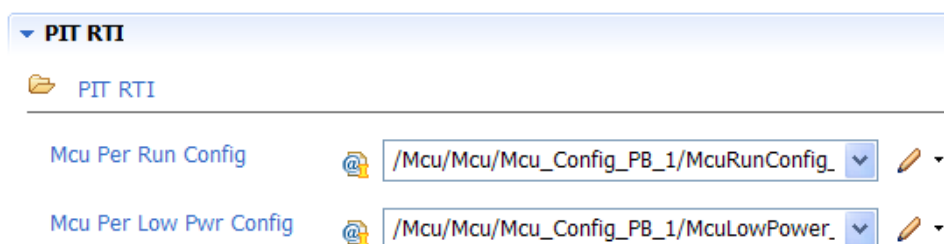


Figure 4-87. Tressos Plugin snapshot for PIT_RTI form.

4.7.16.34.1 McuPerRunConfig (PIT_RTI)

Table 4-340. Attribute McuPerRunConfig (PIT_RTI) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.34.2 McuPerLowPwrConfig (PIT_RTI)

Table 4-341. Attribute McuPerLowPwrConfig (PIT_RTI) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.35 Form CMU0

This contains the combination for current peripheral in Run and LowPower Mode.

Note: Implementation specific Parameter.

Is included by form : [Form McuPeriph](#)

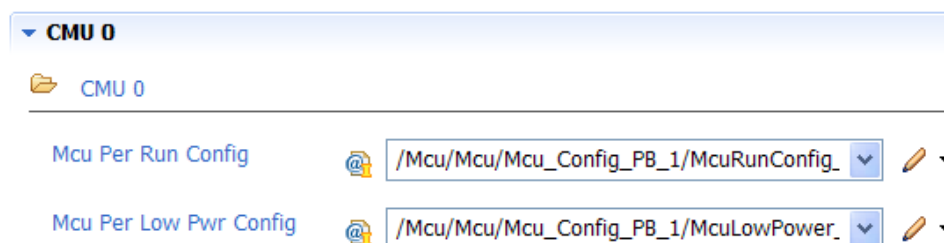


Figure 4-88. Tressos Plugin snapshot for CMU0 form.

4.7.16.35.1 McuPerRunConfig (CMU0)

Table 4-342. Attribute McuPerRunConfig (CMU0) detailed description

Property	Value
Label	Mcu Per Run Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.16.35.2 McuPerLowPwrConfig (CMU0)

Table 4-343. Attribute McuPerLowPwrConfig (CMU0) detailed description

Property	Value
Label	Mcu Per Low Pwr Config
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom

4.7.17 Form McuResetSource

This name is automatically generated.

Is included by form : [Form McuModuleConfiguration](#)

Included forms :

- [Form JTAG](#)
- [Form CORE](#)
- [Form SOFT](#)
- [Form CHKSTOP](#)
- [Form PLL](#)
- [Form CMU0_OLR](#)
- [Form CMU0_FHL](#)
- [Form LVD45](#)
- [Form FLASH](#)
- [Form EXR](#)
- [Form LVD12PD0](#)
- [Form LVD12PD1](#)
- [Form SWT](#)
- [Form LVD27](#)
- [Form LVD27_VREG](#)

The screenshot shows the Tresos Plugin configuration window for the **McuResetSource** form. The window has a tabbed interface at the top with the following tabs: **McuClockSettingConf**, **McuModeSettingConf**, **McuRamSectorSettingC**, **McuDMA**, **McuRunConfig**, **McuLowPower**, **McuPeripheral**, and **McuResetSource** (selected). The main content area is divided into sections: **Mcu Reset Source**, **JTAG**, **CORE**, and **SOFT**. The **JTAG** section is expanded, showing a folder icon and the following settings: **Enable RESET** (checked), **ISR/SAFE Event** (unchecked), **RESET Phase** (set to **PHASE_1**), and **PIN RESET** (checked). The **CORE** section is also expanded, showing similar settings. The **SOFT** section is collapsed.

Figure 4-89. Tresos Plugin snapshot for McuResetSource form.

4.7.17.1 Form JTAG

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

The screenshot shows the Tresos Plugin configuration window for the **JTAG** form. The **JTAG** section is expanded, showing a folder icon and the following settings: **Enable RESET** (checked), **ISR/SAFE Event** (unchecked), **RESET Phase** (set to **PHASE_1**), and **PIN RESET** (checked).

Figure 4-90. Tresos Plugin snapshot for JTAG form.

4.7.17.1.1 McuEnableReset (JTAG)

Enabling Triggers a reset sequence

Table 4-344. Attribute McuEnableReset (JTAG) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.1.2 McuEnableInterrupt (JTAG)

If enabled the **ISR** event occurs else the **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-345. Attribute McuEnableInterrupt (JTAG) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.1.3 McuResetPhase (JTAG)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-346. Attribute McuResetPhase (JTAG) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.1.4 McuResetPin (JTAG)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-347. Attribute McuResetPin (JTAG) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.2 Form CORE

Reset/ISR/Safe source..

Is included by form : [Form McuResetSource](#)

Figure 4-91. Tresos Plugin snapshot for CORE form.

4.7.17.2.1 McuEnableReset (CORE)

Enabling Triggers a reset sequence

Table 4-348. Attribute McuEnableReset (CORE) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.2.2 McuEnableInterrupt (CORE)

If enabled the **ISR** event occurs else the **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-349. Attribute McuEnableInterrupt (CORE) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false
Enable	true

4.7.17.2.3 McuResetPhase (CORE)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-350. Attribute McuResetPhase (CORE) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.2.4 McuResetPin (CORE)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-351. Attribute McuResetPin (CORE) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.3 Form SOFT

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

Figure 4-92. Tresos Plugin snapshot for SOFT form.

4.7.17.3.1 McuEnableReset (SOFT)

Enabling Triggers a reset sequence

Table 4-352. Attribute McuEnableReset (SOFT) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.3.2 McuEnableInterrupt (SOFT)

A **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-353. Attribute McuEnableInterrupt (SOFT) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.3.3 McuResetPhase (SOFT)

The reset sequence will start from **PHASE1**.

Table 4-354. Attribute McuResetPhase (SOFT) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.3.4 McuResetPin (SOFT)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-355. Attribute McuResetPin (SOFT) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.4 Form CHKSTOP

Reset/ISR/Safe source..

Is included by form : [Form McuResetSource](#)

▼ **CHKSTOP**

CHKSTOP

Enable RESET ☒ ISR/SAFE Event ☐

RESET Phase PHASE_1

PIN RESET ☒

Figure 4-93. TRESOS Plugin snapshot for CHKSTOP form.

4.7.17.4.1 McuEnableReset (CHKSTOP)

If flagged the **RESET** occurs, else no reset occurs and **ISR** or **SAFE** (it depends from McuEnableInterrupt) should be activated.

Table 4-356. Attribute McuEnableReset (CHKSTOP) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.4.2 McuEnableInterrupt (CHKSTOP)

A **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-357. Attribute McuEnableInterrupt (CHKSTOP) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.4.3 McuResetPhase (CHKSTOP)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-358. Attribute McuResetPhase (CHKSTOP) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.4.4 McuResetPin (CHKSTOP)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-359. Attribute McuResetPin (CHKSTOP) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.5 Form PLL

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

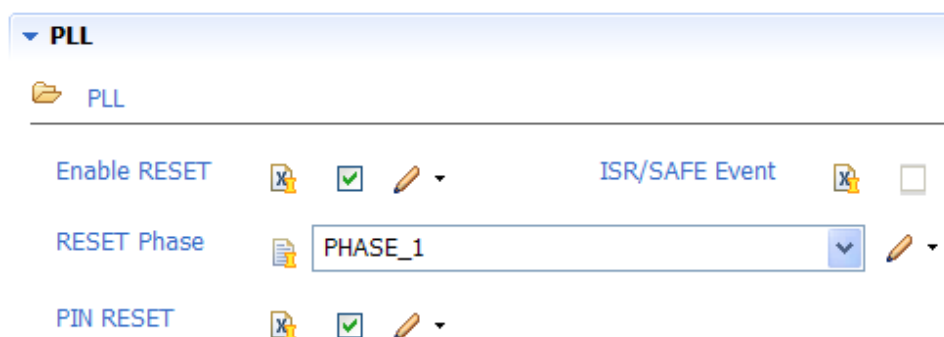


Figure 4-94. Tresos Plugin snapshot for PLL form.

4.7.17.5.1 McuEnableReset (PLL)

If flagged the **RESET** occurs, else no reset occurs and **ISR** or **SAFE** (it depends from McuEnableInterrupt) should be activated.

Table 4-360. Attribute McuEnableReset (PLL) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.5.2 McuEnableInterrupt (PLL)

If enabled the **ISR** event occurs else the **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-361. Attribute McuEnableInterrupt (PLL) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.5.3 McuResetPhase (PLL)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-362. Attribute McuResetPhase (PLL) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.5.4 McuResetPin (PLL)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-363. Attribute McuResetPin (PLL) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.6 Form CMU0_OLR

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

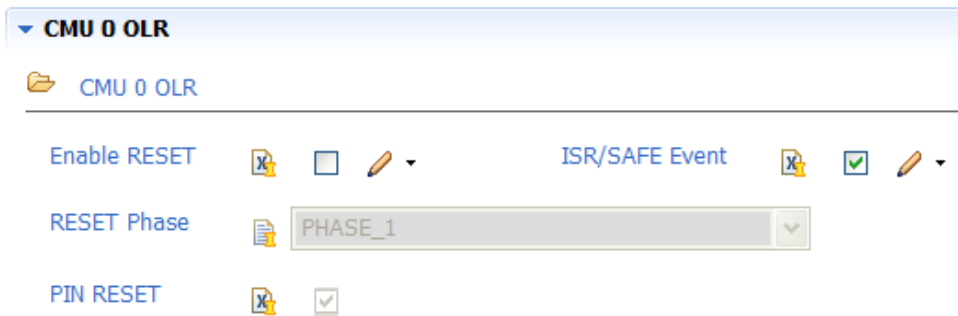


Figure 4-95. Tresos Plugin snapshot for CMU0_OLR form.

4.7.17.6.1 McuEnableReset (CMU0_OLR)

If flagged the **RESET** occurs, else no reset occurs and **ISR** or **SAFE** (it depends from McuEnableInterrupt) should be activated.

Table 4-364. Attribute McuEnableReset (CMU0_OLR) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.6.2 McuEnableInterrupt (CMU0_OLR)

If enabled the **ISR** event occurs else the **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-365. Attribute McuEnableInterrupt (CMU0_OLR) detailed description

Property	Value
Label	ISR/SAFE Event

Table continues on the next page...

Table 4-365. Attribute McuEnableInterrupt (CMU0_OLR) detailed description (continued)

Property	Value
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.6.3 McuResetPhase (CMU0_OLR)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-366. Attribute McuResetPhase (CMU0_OLR) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.6.4 McuResetPin (CMU0_OLR)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-367. Attribute McuResetPin (CMU0_OLR) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.7 Form CMU0_FHL

Reset/ISR/Safe source..

Is included by form : [Form McuResetSource](#)

CMU 0 FHL

CMU 0 FHL

Enable RESET ☐

ISR/SAFE Event ☒

RESET Phase PHASE_1

PIN RESET ☒

Figure 4-96. TRESOS Plugin snapshot for CMU0_FHL form.

4.7.17.7.1 McuEnableReset (CMU0_FHL)

If flagged the **RESET** occurs, else no reset occurs and **ISR** or **SAFE** (it depends from McuEnableInterrupt) should be activated.

Table 4-368. Attribute McuEnableReset (CMU0_FHL) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.7.2 McuEnableInterrupt (CMU0_FHL)

If enabled the **ISR** event occurs else the **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-369. Attribute McuEnableInterrupt (CMU0_FHL) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.7.3 McuResetPhase (CMU0_FHL)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-370. Attribute McuResetPhase (CMU0_FHL) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.7.4 McuResetPin (CMU0_FHL)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-371. Attribute McuResetPin (CMU0_FHL) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.8 Form LVD45

Reset/ISR/Safe source..

Is included by form : [Form McuResetSource](#)

▼ LVD45

📁 LVD45

Enable RESET ☒

ISR/SAFE Event ☐

RESET Phase

PIN RESET ☒

Figure 4-97. Tressos Plugin snapshot for LVD45 form.

4.7.17.8.1 McuEnableReset (LVD45)

Enabling Triggers a reset sequence

Table 4-372. Attribute McuEnableReset (LVD45) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.8.2 McuEnableInterrupt (LVD45)

If enabled the **ISR** event occurs else the **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-373. Attribute McuEnableInterrupt (LVD45) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.8.3 McuResetPhase (LVD45)

The reset sequence will start from **PHASE1**.

Table 4-374. Attribute McuResetPhase (LVD45) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.8.4 McuResetPin (LVD45)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-375. Attribute McuResetPin (LVD45) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.9 Form FLASH

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

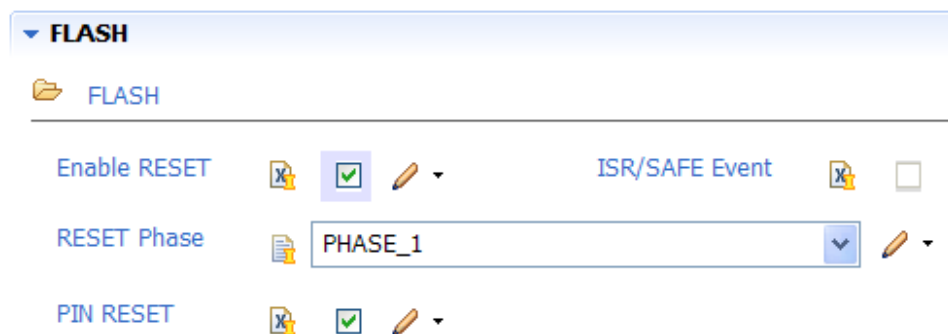


Figure 4-98. Tresos Plugin snapshot for FLASH form.

4.7.17.9.1 McuEnableReset (FLASH)

If flagged the **RESET** occurs, else no reset occurs and **ISR** or **SAFE** (it depends from McuEnableInterrupt) should be activated.

Table 4-376. Attribute McuEnableReset (FLASH) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.9.2 McuEnableInterrupt (FLASH)

If enabled the **ISR** event occurs else the **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-377. Attribute McuEnableInterrupt (FLASH) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.9.3 McuResetPhase (FLASH)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-378. Attribute McuResetPhase (FLASH) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.9.4 McuResetPin (FLASH)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-379. Attribute McuResetPin (FLASH) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.10 Form EXR

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

Figure 4-99. Tresos Plugin snapshot for EXR form.

4.7.17.10.1 McuEnableReset (EXR)

Enabling Triggers a reset sequence

Table 4-380. Attribute McuEnableReset (EXR) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.10.2 McuEnableInterrupt (EXR)

A **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-381. Attribute McuEnableInterrupt (EXR) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN

Table continues on the next page...

Table 4-381. Attribute McuEnableInterrupt (EXR) detailed description (continued)

Property	Value
Origin	Custom
Default	false

4.7.17.10.3 McuResetPhase (EXR)

The reset sequence will start from **PHASE1**.

Table 4-382. Attribute McuResetPhase (EXR) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.10.4 McuResetPin (EXR)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-383. Attribute McuResetPin (EXR) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.11 Form LVD12PD0

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

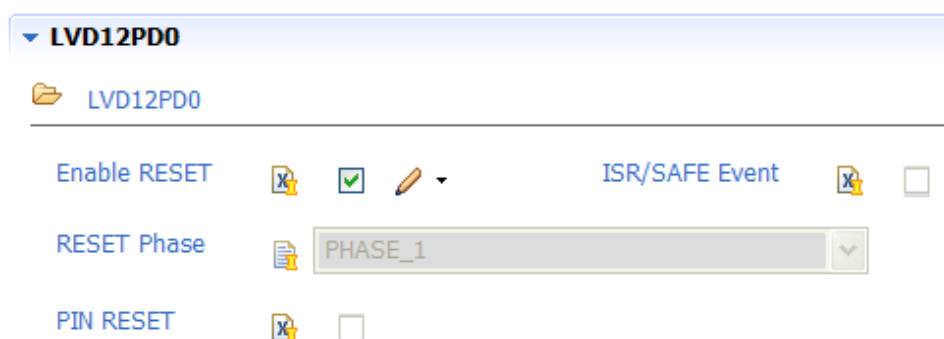


Figure 4-100. Tressos Plugin snapshot for LVD12PD0 form.

4.7.17.11.1 McuEnableReset (LVD12PD0)

Enabling Triggers a reset sequence

Table 4-384. Attribute McuEnableReset (LVD12PD0) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.11.2 McuEnableInterrupt (LVD12PD0)

A **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-385. Attribute McuEnableInterrupt (LVD12PD0) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.11.3 McuResetPhase (LVD12PD0)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-386. Attribute McuResetPhase (LVD12PD0) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.11.4 McuResetPin (LVD12PD0)

Enable the source, managing reset, to generate a reset on relative **pin out**.

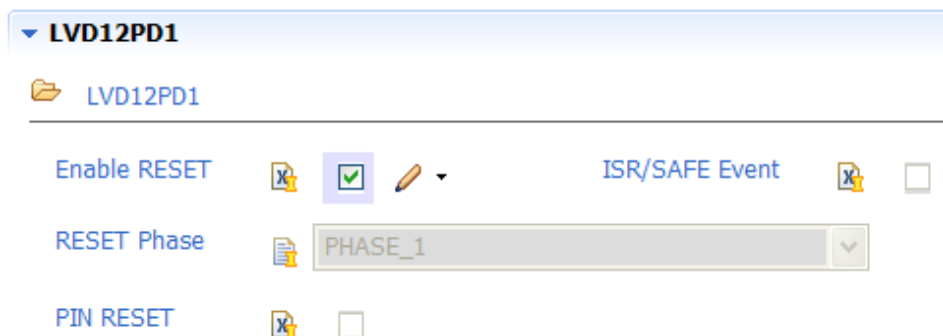
Table 4-387. Attribute McuResetPin (LVD12PD0) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.12 Form LVD12PD1

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

**Figure 4-101. Tresos Plugin snapshot for LVD12PD1 form.**

4.7.17.12.1 McuEnableReset (LVD12PD1)

Enabling Triggers a reset sequence

Table 4-388. Attribute McuEnableReset (LVD12PD1) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.12.2 McuEnableInterrupt (LVD12PD1)

A **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-389. Attribute McuEnableInterrupt (LVD12PD1) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.12.3 McuResetPhase (LVD12PD1)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-390. Attribute McuResetPhase (LVD12PD1) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.12.4 McuResetPin (LVD12PD1)

Enable the source, managing reset, to generate a reset on relative **pin out**.

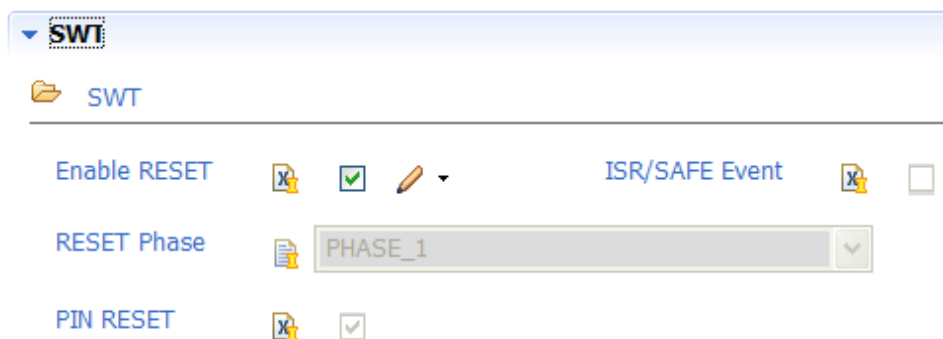
Table 4-391. Attribute McuResetPin (LVD12PD1) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.13 Form SWT

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

**Figure 4-102. Tresos Plugin snapshot for SWT form.**

4.7.17.13.1 McuEnableReset (SWT)

Enabling Triggers a reset sequence

Table 4-392. Attribute McuEnableReset (SWT) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.13.2 McuEnableInterrupt (SWT)

A **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-393. Attribute McuEnableInterrupt (SWT) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.13.3 McuResetPhase (SWT)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-394. Attribute McuResetPhase (SWT) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.13.4 McuResetPin (SWT)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-395. Attribute McuResetPin (SWT) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.14 Form LVD27

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

Figure 4-103. Tresos Plugin snapshot for LVD27 form.

4.7.17.14.1 McuEnableReset (LVD27)

Enabling Triggers a reset sequence

Table 4-396. Attribute McuEnableReset (LVD27) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.14.2 McuEnableInterrupt (LVD27)

A **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-397. Attribute McuEnableInterrupt (LVD27) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.14.3 McuResetPhase (LVD27)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-398. Attribute McuResetPhase (LVD27) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.14.4 McuResetPin (LVD27)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-399. Attribute McuResetPin (LVD27) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.15 Form LVD27_VREG

Reset/ISR/Safe source.

Is included by form : [Form McuResetSource](#)

Figure 4-104. Tressos Plugin snapshot for LVD27_VREG form.

4.7.17.15.1 McuEnableReset (LVD27_VREG)

Enabling Triggers a reset sequence

Table 4-400. Attribute McuEnableReset (LVD27_VREG) detailed description

Property	Value
Label	Enable RESET
Type	BOOLEAN
Origin	Custom
Default	true

4.7.17.15.2 McuEnableInterrupt (LVD27_VREG)

A **SAFE** event occurs.

Note: be aware that in the case of Bolero 256K, it is not recommended to select to enter in safe mode if you intend to select external oscillator as your system clock source. In this case, it is possible to have deadlock situation as detailed in errata e3220.

Table 4-401. Attribute McuEnableInterrupt (LVD27_VREG) detailed description

Property	Value
Label	ISR/SAFE Event
Type	BOOLEAN
Origin	Custom
Default	false

4.7.17.15.3 McuResetPhase (LVD27_VREG)

If McuEnableReset is flagged the selected phase of reset sequence is applied.

Table 4-402. Attribute McuResetPhase (LVD27_VREG) detailed description

Property	Value
Label	RESET Phase
Type	ENUMERATION
Origin	Custom
Default	PHASE_1
Range	PHASE_1 PHASE_3

4.7.17.15.4 McuResetPin (LVD27_VREG)

Enable the source, managing reset, to generate a reset on relative **pin out**.

Table 4-403. Attribute McuResetPin (LVD27_VREG) detailed description

Property	Value
Label	PIN RESET
Type	BOOLEAN
Origin	Custom
Default	false

4.7.18 Form McuClockSettingConfig

This container contains the configuration (parameters) for the Clock settings of the MCU.

Note: Please see MCU031 for more information on the MCU clock settings.

Is included by form : [Form McuModuleConfiguration](#)

Included forms :

- [Form GeneralClockSettings](#)
- [Form McuXtal_A](#)
- [Form McuXtal_B_32KHz](#)
- [Form McuRC_16MHz](#)
- [Form McuRC_128KHz](#)
- [Form McuClkOutput](#)
- [Form McuPll_0](#)
- [Form McuEMIOSSettings_A](#)
- [Form McuEMIOSSettings_B](#)
- [Form McuPeripheralClkSet_0](#)
- [Form McuPeripheralClkSet_1](#)
- [Form McuPeripheralClkSet_2](#)
- [Form McuClkMonitor](#)
- [Form McuFlashPFCR0](#)
- [Form McuFlashPFCR1](#)
- [Form McuClockReferencePoint](#)

Mcu Clock Setting Co

McuClockReferencePoi

McuXtal_A

McuXtal_B_32KHz

McuRC_16MHz

McuRC_128KHz

McuClkOutput

Mcu Clock Setting Id (1 -> 255)

1

Mcu System Clock Frequency (125000 -> 64000000)*

6.4E7

General Clock Settings

General Clock Settings

System Clock Select*

McuPLL

RUN Mode

RUN0

XOSC

☒

McuPLLStatus

☒

Voltage Regulator

☒

Code Flash Status

NormalMode

Data Flash Status

NormalMode

RC 16MHz

☒

Output Power Down Control

☐

Figure 4-105. Tresos Plugin snapshot for McuClockSettingConfig form.

4.7.18.1 McuClockSettingId (McuClockSettingConfig)

The Id of this McuClockSettingConfig to be used as argument for the API call "Mcu_InitClock".

Table 4-404. Attribute McuClockSettingId (McuClockSettingConfig) detailed description

Property	Value
Label	Mcu Clock Setting Id
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	true
Invalid	Range <=255 >=1

4.7.18.2 McuSystemClockFrequency (McuClockSettingConfig)

This is the system clock frequency. It shall be given in Hz.

Table 4-405. Attribute McuSystemClockFrequency (McuClockSettingConfig) detailed description

Property	Value
Label	Mcu System Clock Frequency
Type	FLOAT
Origin	Custom
Symbolic Name	false

4.7.18.3 Form GeneralClockSettings

This container contains the SPECIFIC configuration (parameters) of the GeneralClockSettings Configuration.

Note: Implementation specific Container.

Is included by form : [Form McuClockSettingConfig](#)

The screenshot shows the 'General Clock Settings' form. It has a title bar 'General Clock Settings' and a folder icon. Below are several configuration items, each with a document icon, a text field, and a pencil icon for editing:

- System Clock Select**: McuExtQuartzOsc
- RUN Mode**: RUN0
- XOSC**: [checked]
- Voltage Regulator**: [checked]
- Code Flash Status**: NormalMode
- Data Flash Status**: NormalMode
- RC 16MHz**: [checked]
- McuPLLStatus**: [checked]
- Output Power Down Control**: [unchecked]

Figure 4-106. Tresos Plugin snapshot for GeneralClockSettings form.

4.7.18.3.1 SystemClockSelect (GeneralClockSettings)

System Clock Select. The system clock is either:

Table 4-406. Detailed description.

Internal RC oscillator 16Mhz IRC
Internal RC oscillator 16 MHz divided IRC Div

Table continues on the next page...

Table 4-406. Detailed description. (continued)

Fast external crystal oscillator XOSC
Fast external crystal oscillator divided XOSC div
PLL
SYSCLOCK OFF : configurable only in STOP mode, mandatory in STANDBY mode

Note: Implementation specific Parameter.

Note: For Bolero 256K , in the case that you select Fast external crystal oscillator XOSC or Fast external crystal oscillator divided XOSC as your system clock, it will not be possible to enter in safe mode (see errata e3220 for details) .

Table 4-407. Attribute SystemClockSelect (GeneralClockSettings) detailed description

Property	Value
Label	System Clock Select
Type	ENUMERATION
Origin	Custom
Symbolic Name	false

4.7.18.3.2 McuStartRunMode (GeneralClockSettings)

This field contains the operative mode where apply the Clock selected.

This parameter will not be used when the McuOperatingMode is configured as HALT or STOP mode and the LowPowerClockSettingConfig uses the Clock configuration of the same container.

Note:Implementation specific Parameter .

Table 4-408. Attribute McuStartRunMode (GeneralClockSettings) detailed description

Property	Value
Label	RUN Mode
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	DRUN
Range	RUN3 RUN2 RUN1 RUN0

Table 4-408. Attribute McuStartRunMode (GeneralClockSettings) detailed description

Property	Value
	DRUN SAFE HALT STOP STANDBY

4.7.18.3.3 McuOSC (GeneralClockSettings)

External Oscillator in ON or OFF state.

Note: Implementation specific Parameter .

Table 4-409. Attribute McuOSC (GeneralClockSettings) detailed description

Property	Value
Label	XOSC
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.3.4 McuPLLStatus (GeneralClockSettings)

Put the PLL0 in ON or OFF state.

Note: Implementation specific Parameter.

Table 4-410. Attribute McuPLLStatus (GeneralClockSettings) detailed description

Property	Value
Label	McuPLLStatus
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.3.5 VoltageRegulator (GeneralClockSettings)

Put the **Main Voltage Regulator** in **ON** or **OFF** state.

Table 4-411. Attribute VoltageRegulator (GeneralClockSettings) detailed description

Property	Value
Label	Voltage Regulator
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.3.6 McuCodeFlashStatus (GeneralClockSettings)

Select the Code Flash Status in new operative Mode.

Note: Implementation specific Parameter .

Table 4-412. Attribute McuCodeFlashStatus (GeneralClockSettings) detailed description

Property	Value
Label	Code Flash Status
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	NormalMode
Range	PowerDown LowPower NormalMode

4.7.18.3.7 McuDataFlashStatus (GeneralClockSettings)

Select the Data Flash Status in new operative Mode. Note: Implementation specific Parameter.

Table 4-413. Attribute McuDataFlashStatus (GeneralClockSettings) detailed description

Property	Value
Label	Data Flash Status
Type	ENUMERATION

Table continues on the next page...

Table 4-413. Attribute McuDataFlashStatus (GeneralClockSettings) detailed description (continued)

Property	Value
Origin	Custom
Symbolic Name	false
Default	NormalMode
Range	PowerDown LowPower NormalMode

4.7.18.3.8 McuRC16MHzOscillator (GeneralClockSettings)

Put the RC Oscillator in ON or OFF state. This parameter is configurable when the McuStartRunModeMode is configured as 'HALT' or 'STANDBY' or 'STOP' modes.

Note: Implementation specific Parameter .

Table 4-414. Attribute McuRC16MHzOscillator (GeneralClockSettings) detailed description

Property	Value
Label	RC 16MHz
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.3.9 McuOutputPowerDownControl (GeneralClockSettings)

Output power-down status: This bit controls the output power-down of I/Os.

This parameter is configurable only when McuStartRunMode is configured as SAFE/TEST/ STOP0 mode.

Disabled - No automatic safe gating of I/Os used and pads power sequence driver is enabled.

Enabled - In SAFE/TEST modes, outputs of pads are forced to high impedance state and pads power sequence driver is disabled. The inputs are level unchanged. In STOP0 mode, only the pad power sequence driver is disabled, but the state of the output remains functional.

Note: Implementation specific Parameter .

Table 4-415. Attribute McuOutputPowerDownControl (GeneralClockSettings) detailed description

Property	Value
Label	Output Power Down Control
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.4 Form McuXtal_A

This container contains the SPECIFIC configuration (parameters) of the MCU XOSC (External Oscillator) Configuration.

Note: Implementation specific Container.

Is included by form : [Form McuClockSettingConfig](#)

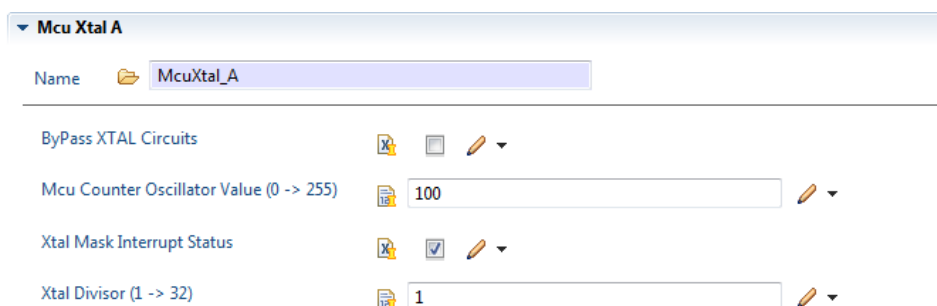


Figure 4-107. TRESOS Plugin snapshot for McuXtal_A form.

4.7.18.4.1 McuXtalByPass (McuXtal_A)

If flagged Bypass the XTAL internal circuit.

Note: Implementation specific Parameter .

Table 4-416. Attribute McuXtalByPass (McuXtal_A) detailed description

Property	Value
Label	ByPass XTAL Circuits
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.4.2 McuXtalCounterOscillator (McuXtal_A)

When the internal counter reaches the value the oscillator is stable.

Note: Implementation specific Parameter . This parameter value will not be considered when theMcuXtalByPass is true.

Table 4-417. Attribute McuXtalCounterOscillator (McuXtal_A) detailed description

Property	Value
Label	Mcu Counter Oscillator Value
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	100

4.7.18.4.3 McuXtalMaskInterrupt (McuXtal_A)

Check if want to enable Interrupts for XTAL_A. This parameter value (true/false) will not be considered when theMcuXtalByPass is true.

Note: Implementation specific Parameter .

Table 4-418. Attribute McuXtalMaskInterrupt (McuXtal_A) detailed description

Property	Value
Label	Xtal Mask Interrupt Status
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.4.4 McuXtalDivisor (McuXtal_A)

Set the Divisor factor for XTAL signal

Note: Implementation specific Parameter .

Table 4-419. Attribute McuXtalDivisor (McuXtal_A) detailed description

Property	Value
Label	Xtal Divisor
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	1

4.7.18.5 Form McuXtal_B_32KHz

This container contains the SPECIFIC configuration (parameters) of the MCU XOSC (External Oscillator) Configuration.

32KHz oscillator is not present for 256K derivatives. These configurations will not be effective for 256K derivatives.

Note: Implementation specific Container.

Is included by form : [Form McuClockSettingConfig](#)

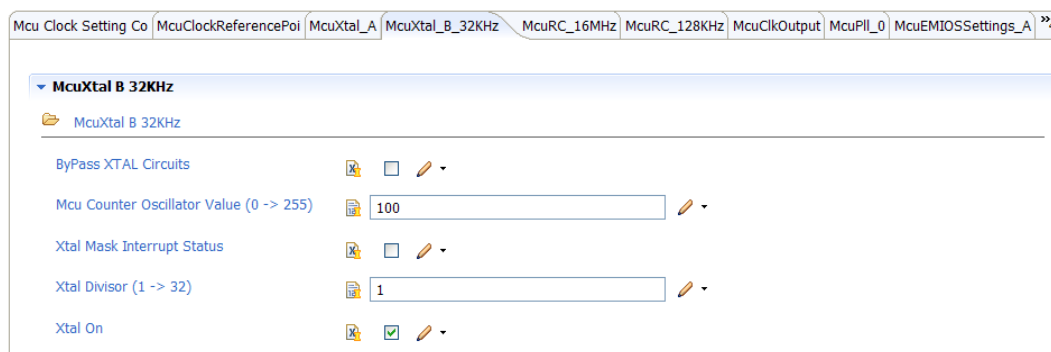


Figure 4-108. Tressos Plugin snapshot for McuXtal_B_32KHz form.

4.7.18.5.1 McuXtalByPass (McuXtal_B_32KHz)

If flagged Bypass the XTAL internal circuit.

Note: Implementation specific Parameter .

Table 4-420. Attribute McuXtalByPass (McuXtal_B_32KHz) detailed description

Property	Value
Label	ByPass XTAL Circuits

Table continues on the next page...

Table 4-420. Attribute McuXtalByPass (McuXtal_B_32KHz) detailed description (continued)

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.5.2 McuXtalCounterOscillator (McuXtal_B_32KHz)

When the internal counter reaches the value the oscillator is stable.

Note: Implementation specific Parameter . This parameter value will not be considered when theMcuXtalByPass is true.

Table 4-421. Attribute McuXtalCounterOscillator (McuXtal_B_32KHz) detailed description

Property	Value
Label	Mcu Counter Oscillator Value
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	100

4.7.18.5.3 McuXtalMaskInterrupt (McuXtal_B_32KHz)

Check if want to enable Interrupts for XTAL_A. This parameter value (true/false) will not be considered when theMcuXtalByPass is true.

Note: Implementation specific Parameter .

Table 4-422. Attribute McuXtalMaskInterrupt (McuXtal_B_32KHz) detailed description

Property	Value
Label	Xtal Mask Interrupt Status
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.5.4 McuXtalDivisor (McuXtal_B_32KHz)

Set the Divisor factor for the slow external XTAL clock source

Note: Implementation specific Parameter .

Table 4-423. Attribute McuXtalDivisor (McuXtal_B_32KHz) detailed description

Property	Value
Label	Xtal Divisor
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	1

4.7.18.5.5 McuXtalOn (McuXtal_B_32KHz)

The XTAL circuit configurable for ON or OFF

Note: Implementation specific Parameter .

Table 4-424. Attribute McuXtalOn (McuXtal_B_32KHz) detailed description

Property	Value
Label	Xtal On
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.6 Form McuRC_16MHz

This container contains the SPECIFIC configuration (parameters) of the Internal 16MHz Oscillator(MIRC16M) Configuration.

Note: Implementation specific Container.

Is included by form : [Form McuClockSettingConfig](#)

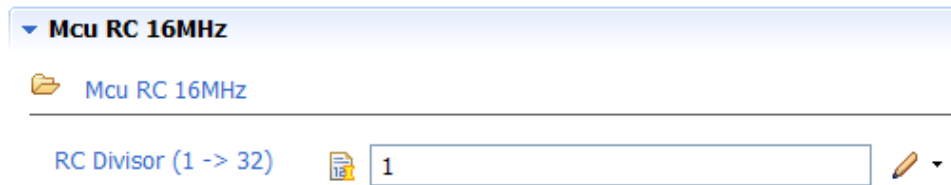


Figure 4-109. Tresos Plugin snapshot for McuRC_16MHz form.

4.7.18.6.1 McuRCDivisor (McuRC_16MHz)

Set the divisor factor for RC16MHz signal

Note: Implementation specific Parameter .

Table 4-425. Attribute McuRCDivisor (McuRC_16MHz) detailed description

Property	Value
Label	RC Divisor
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	1

4.7.18.7 Form McuRC_128KHz

This container contains the SPECIFIC configuration (parameters) of the Internal 16MHz Oscillator(MIRC16M) Configuration.

Note: Implementation specific Container.

Is included by form : [Form McuClockSettingConfig](#)

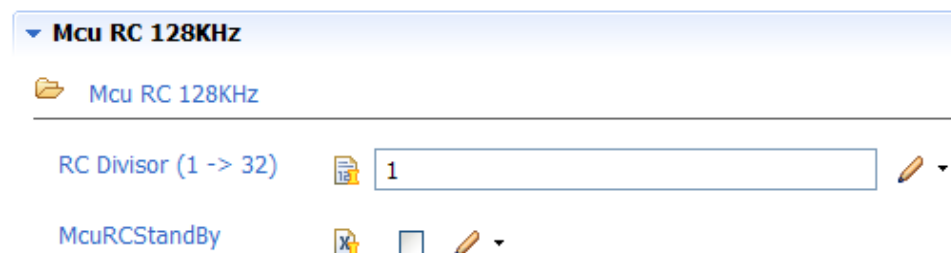


Figure 4-110. Tresos Plugin snapshot for McuRC_128KHz form.

4.7.18.7.1 McuRCDivisor (McuRC_128KHz)

Set the divisor factor for the slow internal RC oscillator 128KHz clock source

Note: Implementation specific Parameter .

Table 4-426. Attribute McuRCDivisor (McuRC_128KHz) detailed description

Property	Value
Label	RC Divisor
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	1

4.7.18.7.2 McuRCStandBy (McuRC_128KHz)

Check if you want enable th Oscillator in Standby mode.

Note: Implementation specific Parameter .

Table 4-427. Attribute McuRCStandBy (McuRC_128KHz) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.8 Form McuClkOutput

This container contains the SPECIFIC configuration (parameters) of the Clock Output Configuration.

Note: Implementation specific Container.

Is included by form : [Form McuClockSettingConfig](#)

Figure 4-111. Tresos Plugin snapshot for McuClkOutput form.

4.7.18.8.1 McuClkOutput (McuClkOutput)

Enable/Disable Clock Out.

The configured value of the McuClkDivisor and McuClkSource parameters will be written into the Output Clock Division Select Register only if McuClkOutput is enabled.

Note: Implementation specific Parameter.

Table 4-428. Attribute McuClkOutput (McuClkOutput) detailed description

Property	Value
Label	Clock Output
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.8.2 McuClkDivisor (McuClkOutput)

Put the divisor value for Clock Out.

Note: Implementation specific Parameter .

Table 4-429. Attribute McuClkDivisor (McuClkOutput) detailed description

Property	Value
Label	Clock Divisor
Type	ENUMERATION
Origin	Custom

Table continues on the next page...

Table 4-429. Attribute McuClkDivisor (McuClkOutput) detailed description (continued)

Property	Value
Symbolic Name	false
Default	DIV_1

4.7.18.8.3 McuClkSource (McuClkOutput)

Select the clock Out source.

Note: Implementation specific Parameter.

Table 4-430. Attribute McuClkSource (McuClkOutput) detailed description

Property	Value
Label	Clock Source
Type	ENUMERATION
Origin	Custom
Symbolic Name	false

4.7.18.9 Form McuPll_0

This container contains the SPECIFIC configuration (parameters) of the MCU PLL Configuration.

Note: Implementation specific Container.

Is included by form : [Form McuClockSettingConfig](#)

Included forms :

- [Form McuPll0_Parameter](#)
- [Form McuFmPll0_Parameter](#)

The screenshot shows the Tresos Configuration Plug-in interface for the McuPll_0 form. The top navigation bar includes tabs for McuXtal_A, McuXtal_B_32KHz, McuRC_16MHz, McuRC_128KHz, McuClkOutput, McuPll_0 (selected), McuEMIOSSettings_A, and McuEMIOSSettings_B. The main content area is divided into three sections:

- Mcu Pll 0**: Contains a folder icon and a dropdown menu for PLL0 Mode, currently set to FM.
- McuPll0_Parameter**: Contains a folder icon and several configuration fields:
 - PLL0 Frequency (16000000 -> 64000000): 64000000
 - Output Divisor Factor (ODF) (2 -> 16): 4
 - Input Divisor Factor (IDF) (1 -> 15): 3
 - Multiplier Factor (NDIV) (32 -> 96): 96
 - Progressive Clock Switching: ☐
 - Fail Interrupt: ☐
- Mcu Fm Pll 0 Parameter**: Contains a folder icon and a field for Modulation Frequency (0 -> 64000000) set to 10000.

Figure 4-112. Tresos Plugin snapshot for McuPll_0 form.

4.7.18.9.1 McuPllMode_0 (McuPll_0)

Select the PLL Mode.

Note: Implementation specific Parameter .

Table 4-431. Attribute McuPllMode_0 (McuPll_0) detailed description

Property	Value
Label	PLL0 Mode
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	Normal

4.7.18.9.2 Form McuPll0_Parameter

This container contains the configuration parameters of the MCU PLL Configuration to generate System clock both in Normal and FM mode. Note: Implementation specific Container.

Is included by form : [Form McuPll_0](#)

McuPll0_Parameter

McuPll0_Parameter

PLL0 Frequency (16000000 -> 64000000)

64000000

Output Divisor Factor (ODF) (2 -> 16)

4

Input Divisor Factor (IDF) (1 -> 15)

3

Multiplicator Factor (NDIV) (32 -> 96)

96

Progressive Clock Switching

Fail Interrupt

Figure 4-113. Tresos Plugin snapshot for McuPll0_Parameter form.

4.7.18.9.2.1 PLL_0_Frequency (McuPll0_Parameter)

The Clock for Pll_0 .The range value is [16..120]MHz . It should not be edited.

Note: implementation specific parameter.

Table 4-432. Attribute PLL_0_Frequency (McuPll0_Parameter) detailed description

Property	Value
Label	PLL0 Frequency
Type	INTEGER
Origin	Custom
Symbolic Name	false

4.7.18.9.2.2 McuOutputDivisionFactor (McuPll0_Parameter)

Put the ODF factor. It is the PLL output division factor.User can provied values only 2/4/8/16.

Note: Implementation specific Parameter.

Table 4-433. Attribute McuOutputDivisionFactor (McuPll0_Parameter) detailed description

Property	Value
Label	Output Divisor Factor (ODF)
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	4

Table continues on the next page...

Table 4-433. Attribute McuOutputDivisionFactor (McuPII0_Parameter) detailed description (continued)

Property	Value
Invalid	Range ≤ 16 ≥ 2

4.7.18.9.2.3 McuInputDivisionFactor (McuPII0_Parameter)

PLL Input Divisor factor(IDF): User has to provide value in the range of [1..15] that is (IDF+1). Internally this value will be written in PLL Control register in the range of [0..14].

Note: Implementation specific Parameter.

Table 4-434. Attribute McuInputDivisionFactor (McuPII0_Parameter) detailed description

Property	Value
Label	Input Divisor Factor (IDF)
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	3
Invalid	Range ≤ 15 ≥ 1

4.7.18.9.2.4 McuMultiplicationFactor (McuPII0_Parameter)

It sets the PLL loop division factor. Put the NDIV value in the range of [32..96]. Internally this value is written in the PLL control register.

Note: Implementation specific Parameter .

Table 4-435. Attribute McuMultiplicationFactor (McuPII0_Parameter) detailed description

Property	Value
Label	Multiplicator Factor (NDIV)
Type	INTEGER
Origin	Custom
Symbolic Name	false

Table continues on the next page...

Table 4-435. Attribute McuMultiplicationFactor (McuPII0_Parameter) detailed description (continued)

Property	Value
Default	96
Invalid	Range <=96 >=32

4.7.18.9.2.5 McuProgressiveClockSwitching (McuPII0_Parameter)

Enable the Progressive Clock Switching.

Note: Implementation specific Parameter .

Table 4-436. Attribute McuProgressiveClockSwitching (McuPII0_Parameter) detailed description

Property	Value
Label	Progressive Clock Switching
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.9.2.6 McuEnableFailInterrupt (McuPII0_Parameter)

Enable or Disable the Interrupt on System clock Fail.

Note: Implementation specific Parameter .

Table 4-437. Attribute McuEnableFailInterrupt (McuPII0_Parameter) detailed description

Property	Value
Label	Fail Interrupt
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.9.3 Form McuFmPll0_Parameter

This container contains the additional configuration parameters of the MCU PLL Configuration in FM mode. Note: Implementation specific Container.

Is included by form : [Form McuPll_0](#)

Figure 4-114. Tresos Plugin snapshot for McuFmPll0_Parameter form.

4.7.18.9.3.1 McuModulationFrequency (McuFmPll0_Parameter)

Frequency for the triangular wave modulation (Fmod). The units is in Hz.

Note: Implementation specific Parameter. The maximum configurable value for this parameter has to be configured based on the formula - Modulation frequency = ((Crystal Frequency / Input division factor) / (4*Modulation period)). It is the responsibility of the User to take of this value

Table 4-438. Attribute McuModulationFrequency (McuFmPll0_Parameter) detailed description

Property	Value
Label	Modulation Frequency
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	4000

4.7.18.9.3.2 McuPercentualModulationDepth (McuFmPll0_Parameter)

The modulation Depth value in percentage. It must be in [0.25..4]% deviation from center frequency.

Note: Implementation specific Parameter .

Table 4-439. Attribute McuPercentualModulationDepth (McuFmPII0_Parameter) detailed description

Property	Value
Label	Modulation Depth (%)
Type	FLOAT
Origin	Custom
Symbolic Name	false
Default	0.25

4.7.18.9.3.3 McuSpreadSelection (McuFmPII0_Parameter)

Select the Spread value.

Note: Implementation specific Parameter .

Table 4-440. Attribute McuSpreadSelection (McuFmPII0_Parameter) detailed description

Property	Value
Label	Spread Selection
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	Down_Spread
Range	Down_Spread Center_Spread

4.7.18.10 Form McuEMIOSSettings_A

This container enables and selects the source clocks for Motor Control and Sine Wave Generator.

Is included by form : [Form McuClockSettingConfig](#)

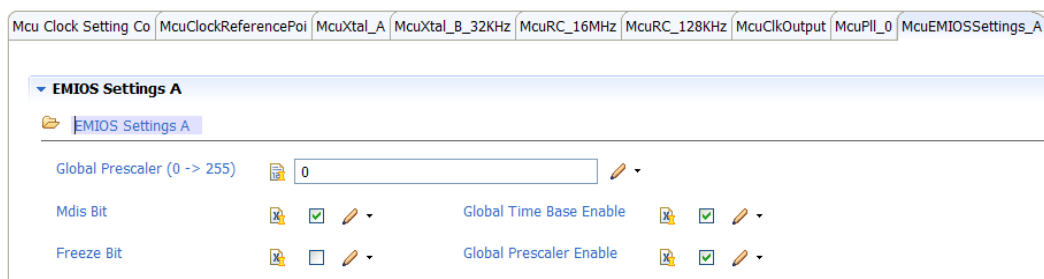


Figure 4-115. Tresos Plugin snapshot for McuEMIOSSettings_A form.

4.7.18.10.1 GlobalPrescaler (McuEMIOSSettings_A)

Global Prescaler Bits (GPRES[0:7] in EMIOS_MCR_A register). The bits select the clock divider value for the global prescaler. The allowed values are 0 up to 255 (divide ratio 1 up to 256).

Note:implementation specific parameter.

Table 4-441. Attribute GlobalPrescaler (McuEMIOSSettings_A) detailed description

Property	Value
Label	Global Prescaler
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range <=255 >=0

4.7.18.10.2 MdisBit (McuEMIOSSettings_A)

Check this to put the eMIOS200 in low power mode. The MDIS bit is used to stop the clock of the block, except the access to registers EMIOSMCR, EMIOSOUDIS and EMIOSUCDIS.

Note:implementation specific parameter.

Table 4-442. Attribute MdisBit (McuEMIOSSettings_A) detailed description

Property	Value
Label	Mdis Bit
Type	BOOLEAN

Table continues on the next page...

Table 4-442. Attribute MdisBit (McuEMIOSSettings_A) detailed description (continued)

Property	Value
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.10.3 GlobalTimeBaseEnable (McuEMIOSSettings_A)

Global Time Base Enable Bit (GTBE in EMIOS_MCR register).The GTBE bit is used to export a global time base enable from the module and provide a method to start time bases of several blocks simultaneously.

Table 4-443. Attribute GlobalTimeBaseEnable (McuEMIOSSettings_A) detailed description

Property	Value
Label	Global Time Base Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.10.4 FreezeBit (McuEMIOSSettings_A)

Freeze Bit (FRZ in EMIOS_MCR_A register).Enable the eMIOS200 to freeze the registers of the unified channels when debug mode is requested at MCU level. Each unified channel must have FREN bit set in order to enter freeze mode.

Table 4-444. Attribute FreezeBit (McuEMIOSSettings_A) detailed description

Property	Value
Label	Freeze Bit
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.10.5 GlobalPrescalerEnable (McuEMIOSSettings_A)

Select the clock source clock.

Note:implementation specific parameter.

Table 4-445. Attribute GlobalPrescalerEnable (McuEMIOSSettings_A) detailed description

Property	Value
Label	Global Prescaler Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.11 Form McuEMIOSSettings_B

This container enables and selects the source clocks for Motor Control and Sine Wave Generator.

EMIOS B is not present for 256K derivatives. These configurations will not be effective for 256K derivatives.

Is included by form : [Form McuClockSettingConfig](#)

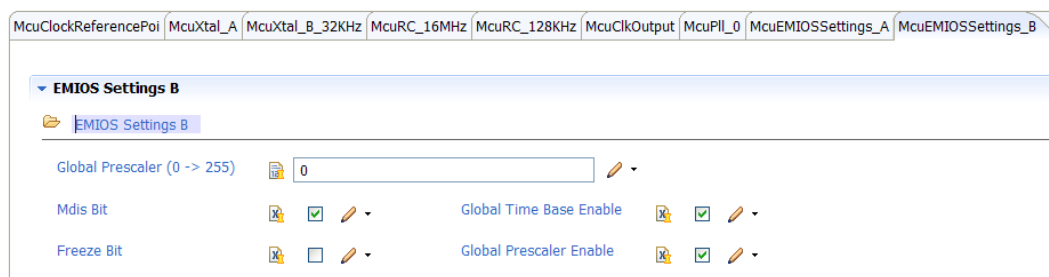


Figure 4-116. Tresos Plugin snapshot for McuEMIOSSettings_B form.

4.7.18.11.1 GlobalPrescaler (McuEMIOSSettings_B)

Global Prescaler Bits (GPRES[0:7] in EMIOS_MCR_A register). The bits select the clock divider value for the global prescaler. The allowed values are 0 up to 255 (divide ratio 1 up to 256).

Note:implementation specific parameter.

Table 4-446. Attribute GlobalPrescaler (McuEMIOSSettings_B) detailed description

Property	Value
Label	Global Prescaler
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range <=255 >=0

4.7.18.11.2 MdisBit (McuEMIOSSettings_B)

Check this to put the eMIOS200 in low power mode. The MDIS bit is used to stop the clock of the block, except the access to registers EMIOSMCR, EMIOSOUDIS and EMIOSUCDIS.

Note:implementation specific parameter.

Table 4-447. Attribute MdisBit (McuEMIOSSettings_B) detailed description

Property	Value
Label	Mdis Bit
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.11.3 GlobalTimeBaseEnable (McuEMIOSSettings_B)

Global Time Base Enable Bit (GTBE in EMIOS_MCR register).The GTBE bit is used to export a global time base enable from the module and provide a method to start time bases of several blocks simultaneously.

Table 4-448. Attribute GlobalTimeBaseEnable (McuEMIOSSettings_B) detailed description

Property	Value
Label	Global Time Base Enable
Type	BOOLEAN
Origin	Custom

Table continues on the next page...

Table 4-448. Attribute GlobalTimeBaseEnable (McuEMIOSSettings_B) detailed description (continued)

Property	Value
Symbolic Name	false
Default	true

4.7.18.11.4 FreezeBit (McuEMIOSSettings_B)

Freeze Bit (FRZ in EMIOS_MCR_A register). Enable the eMIOS200 to freeze the registers of the unified channels when debug mode is requested at MCU level. Each unified channel must have FREN bit set in order to enter freeze mode.

Table 4-449. Attribute FreezeBit (McuEMIOSSettings_B) detailed description

Property	Value
Label	Freeze Bit
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.11.5 GlobalPrescalerEnable (McuEMIOSSettings_B)

Select the clock source clock.

Note:implementation specific parameter.

Table 4-450. Attribute GlobalPrescalerEnable (McuEMIOSSettings_B) detailed description

Property	Value
Label	Global Prescaler Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.12 Form McuPeripheralClkSet_0

This container enables and selects the source clocks for peripherals set 0.

Is included by form : [Form McuClockSettingConfig](#)

Figure 4-117. Tresos Plugin snapshot for McuPeripheralClkSet_0 form.

4.7.18.12.1 McuPeripheralEnableClk (McuPeripheralClkSet_0)

Enable or Disable the clock for peripherals set 0.

Note:implementation specific parameter.

Table 4-451. Attribute McuPeripheralEnableClk (McuPeripheralClkSet_0) detailed description

Property	Value
Label	Peripheral Clk Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.12.2 McuPeripheralDivisorFactor (McuPeripheralClkSet_0)

Select the Divisor Factor for peripherals set 0.

Note:implementation specific parameter.

Table 4-452. Attribute McuPeripheralDivisorFactor (McuPeripheralClkSet_0) detailed description

Property	Value
Label	Peripheral Divisor Factor
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	1
Invalid	Range ≤ 16 ≥ 1

4.7.18.13 Form McuPeripheralClkSet_1

This container enables and selects the source clocks for peripherals set 1.

Is included by form : [Form McuClockSettingConfig](#)

Figure 4-118. Tressos Plugin snapshot for McuPeripheralClkSet_1 form.

4.7.18.13.1 McuPeripheralEnableClk (McuPeripheralClkSet_1)

Enable or Disable the clock for peripherals set 1.

Note:implementation specific parameter.

Table 4-453. Attribute McuPeripheralEnableClk (McuPeripheralClkSet_1) detailed description

Property	Value
Label	Peripheral Clk Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.13.2 McuPeripheralDivisorFactor (McuPeripheralClkSet_1)

Select the Divisor Factor for peripherals set 1.

Note:implementation specific parameter.

Table 4-454. Attribute McuPeripheralDivisorFactor (McuPeripheralClkSet_1) detailed description

Property	Value
Label	Peripheral Divisor Factor

Table continues on the next page...

Table 4-454. Attribute McuPeripheralDivisorFactor (McuPeripheralClkSet_1) detailed description (continued)

Property	Value
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	1
Invalid	Range <=16 >=1

4.7.18.14 Form McuPeripheralClkSet_2

This container enables and selects the source clock for peripherals set 2.

Is included by form : [Form McuClockSettingConfig](#)

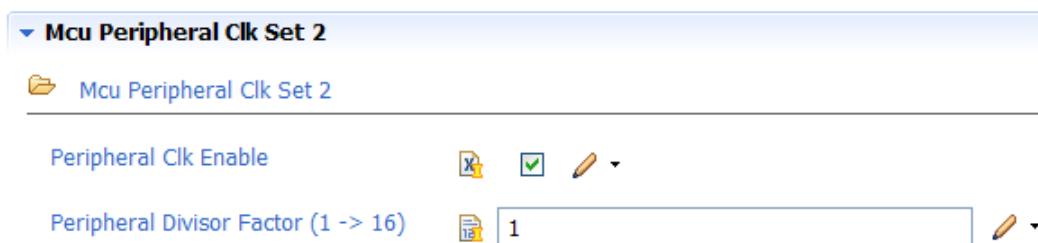


Figure 4-119. TRESOS Plugin snapshot for McuPeripheralClkSet_2 form.

4.7.18.14.1 McuPeripheralEnableClk (McuPeripheralClkSet_2)

Enable or Disable the clock for peripherals set 2.

Note:implementation specific parameter.

Table 4-455. Attribute McuPeripheralEnableClk (McuPeripheralClkSet_2) detailed description

Property	Value
Label	Peripheral Clk Enable
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.18.14.2 McuPeripheralDivisorFactor (McuPeripheralClkSet_2)

Select the Divisor Factor for peripherals set 2.

Note:implementation specific parameter.

Table 4-456. Attribute McuPeripheralDivisorFactor (McuPeripheralClkSet_2) detailed description

Property	Value
Label	Peripheral Divisor Factor
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	1
Invalid	Range ≤ 16 ≥ 1

4.7.18.15 Form McuClkMonitor

This container contains the SPECIFIC configuration (parameters) of the **Monitor 0** .

Is included by form : [Form McuClockSettingConfig](#)

Figure 4-120. Tresos Plugin snapshot for McuClkMonitor form.

4.7.18.15.1 McuFmPllMonitor (McuClkMonitor)

Enable/Disable the monitor on PLL.

Table 4-457. Attribute McuFmPIIMonitor (McuClkMonitor) detailed description

Property	Value
Label	PII Monitor
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.18.15.2 McuRCOSCSelection (McuClkMonitor)

This parameter specifies the RC oscillator selection

Table 4-458. Attribute McuRCOSCSelection (McuClkMonitor) detailed description

Property	Value
Label	Mcu RCOSC Selection
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	CK_IRCfast

4.7.18.15.3 McuRCDivisorFactor (McuClkMonitor)

This value should build the following value $x=16000000/(2^{RC})$. So, the monitor detect if the freq of external quartz is smaller than x..

Table 4-459. Attribute McuRCDivisorFactor (McuClkMonitor) detailed description

Property	Value
Label	RC Divisor Factor
Type	FLOAT
Origin	Custom
Symbolic Name	false
Default	3
Invalid	Range ≤ 3 ≥ 0

4.7.18.15.4 McuHighFrequencyRef (McuClkMonitor)

The value of the Max Frequency. So, the monitor detect if the PLL_0 frequency is greater than McuHighFrequencyRef frequency value.

Table 4-460. Attribute McuHighFrequencyRef (McuClkMonitor) detailed description

Property	Value
Label	HIGH Frequency Reference
Type	FLOAT
Origin	Custom
Default	90000000

4.7.18.15.5 McuLowFrequencyRef (McuClkMonitor)

The value of the Min Frequency. So, the monitor detect if the PLL_0 frequency value is smaller than McuLowFrequencyRef value.

Table 4-461. Attribute McuLowFrequencyRef (McuClkMonitor) detailed description

Property	Value
Label	LOW Frequency Reference
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	23000000

4.7.18.16 Form McuFlashPFCR0

This container contains the SPECIFIC configuration (parameters) of the Mcu Flash PFCR0 register.

Note: implementation specific parameter.

Is included by form : [Form McuClockSettingConfig](#)

Mcu Flash PFCR0

Mcu Flash PFCR0

Address Pipelining Control (0 -> 32)

0

Write Wait State Control (0 -> 32)

0

Read Wait State Control (0 -> 32)

0

Figure 4-121. Tresos Plugin snapshot for McuFlashPFCR0 form.

4.7.18.16.1 AddressPipeliningControl (McuFlashPFCR0)

Table 4-462. Attribute AddressPipeliningControl (McuFlashPFCR0) detailed description

Property	Value
Label	Address Pipelining Control
Type	INTEGER
Origin	Custom
Symbolic Name	false

4.7.18.16.2 WriteWaitStateControl (McuFlashPFCR0)

Bank0+2 Write Wait State Control. This field is used to control the number of wait-states to be added to the flash array access time for writes. This field must be set to a value appropriate to the operating frequency of the PFLASH. The required settings are documented in the SoC specification. Higher operating frequencies require non-zero settings for this field for proper flash operation. This field is set to an appropriate value by hardware reset. This field is set to 0b00010 by hardware reset.

Table 4-463. Detailed description.

00000	No additional wait-states are added
00001	1 additional wait-state is added
00010	2 additional wait-state is added
....	
11111	31 additional wait-state is added

Note: implementation specific parameter.

Table 4-464. Attribute WriteWaitStateControl (McuFlashPFCR0) detailed description

Property	Value
Label	Write Wait State Control
Type	INTEGER
Origin	Custom
Symbolic Name	false

4.7.18.16.3 ReadWaitStateControl (McuFlashPFCR0)

Bank0+2 Read Wait State Control. This field is used to control the number of wait-states to be added to the flash array access time for reads. This field must be set to a value corresponding to the operating frequency of the PFLASH and the actual read access time of the PFLASH. The required settings are documented in the SoC specification. Higher operating frequencies require non-zero settings for this field for proper flash operation. Shown below are the maximum operating frequencies for legal APC and RWSC settings based on estimated low-cost flash access times at 150C. The integrator is bly encouraged to verify these settings based on actual silicon results.

Table 4-465. Detailed description.

00000	No additional wait-states are added
00001	1 additional wait-state is added
00010	2 additional wait-state is added
....	
11111	31 additional wait-state is added

Higher operating frequencies require non-zero settings for this field for proper flash operation. This field must be set to a value appropriate to the operating frequency of the PFLASH and the actual read access time of the PFLASH. Shown below are the maximum operating frequencies for legal RWSC settings based on estimated low-cost flash access times at 150C.

Table 4-466. Detailed description.

Greater than 0 MHz and Less than 20 MHz	0
Greater than or equal to 20 MHz and Less than 40 MHz	1
Greater than or equal to 40 MHz and Less than 60 MHz	2
Greater than or equal to 60 MHz and Less than 80 MHz	3
Greater than or equal to 80 MHz and Less than 100 MHz	4
Greater than or equal to 100 MHz and Less than 120 MHz	5

Warning: APC must be equal to RWSC, else ECC error reporting is disabled.

Note: implementation specific parameter.

Table 4-467. Attribute ReadWaitStateControl (McuFlashPFCR0) detailed description

Property	Value
Label	Read Wait State Control
Type	INTEGER
Origin	Custom
Symbolic Name	false

4.7.18.17 Form McuFlashPFCR1

This container contains the SPECIFIC configuration (parameters) of the Mcu Flash PFCR0 register.

Note: implementation specific parameter.

Is included by form : [Form McuClockSettingConfig](#)

Figure 4-122. Tresos Plugin snapshot for McuFlashPFCR1 form.

4.7.18.17.1 AddressPipeliningControl (McuFlashPFCR1)

Table 4-468. Attribute AddressPipeliningControl (McuFlashPFCR1) detailed description

Property	Value
Label	Address Pipelining Control
Type	INTEGER
Origin	Custom
Symbolic Name	false

4.7.18.17.2 WriteWaitStateControl (McuFlashPFCR1)

Bank0+2 Write Wait State Control. This field is used to control the number of wait-states to be added to the flash array access time for writes. This field must be set to a value appropriate to the operating frequency of the PFLASH. The required settings are documented in the SoC specification. Higher operating frequencies require non-zero settings for this field for proper flash operation. This field is set to an appropriate value by hardware reset. This field is set to 0b00010 by hardware reset.

Table 4-469. Detailed description.

00000	No additional wait-states are added
00001	1 additional wait-state is added
00010	2 additional wait-state is added
....	
11111	31 additional wait-state is added

Note: implementation specific parameter.

Table 4-470. Attribute WriteWaitStateControl (McuFlashPFCR1) detailed description

Property	Value
Label	Write Wait State Control
Type	INTEGER
Origin	Custom
Symbolic Name	false

4.7.18.17.3 ReadWaitStateControl (McuFlashPFCR1)

Bank0+2 Read Wait State Control. This field is used to control the number of wait-states to be added to the flash array access time for reads. This field must be set to a value corresponding to the operating frequency of the PFLASH and the actual read access time of the PFLASH. The required settings are documented in the SoC specification. Higher operating frequencies require non-zero settings for this field for proper flash operation. Shown below are the maximum operating frequencies for legal APC and RWSC settings based on estimated low-cost flash access times at 150C. The integrator is bly encouraged to verify these settings based on actual silicon results.

Table 4-471. Detailed description.

00000	No additional wait-states are added
00001	1 additional wait-state is added

Table continues on the next page...

Table 4-471. Detailed description. (continued)

00010	2 additional wait-state is added
....	
11111	31 additional wait-state is added

Higher operating frequencies require non-zero settings for this field for proper flash operation. This field must be set to a value appropriate to the operating frequency of the PFLASH. Shown below are the maximum operating frequencies for legal APC settings based on estimated low-cost flash access times at 150C.

Table 4-472. Detailed description.

Greater than 0 MHz and Less than 20 MHz	0
Greater than or equal to 20 MHz and Less than 40 MHz	1
Greater than or equal to 40 MHz and Less than 60 MHz	2
Greater than or equal to 60 MHz and Less than 80 MHz	3
Greater than or equal to 80 MHz and Less than 100 MHz	4
Greater than or equal to 100 MHz and Less than 120 MHz	5

Note: implementation specific parameter.

Table 4-473. Attribute ReadWaitStateControl (McuFlashPFCR1) detailed description

Property	Value
Label	Read Wait State Control
Type	INTEGER
Origin	Custom
Symbolic Name	false

4.7.18.18 Form McuClockReferencePoint

This container defines a reference point in the Mcu Clock tree. It defines the frequency which then can be used by other modules as an input value. Lower multiplicity is 1, as even in the simplest case (only one frequency is used), there is one frequency to be defined.

Is included by form : [Form McuClockSettingConfig](#)

Name McuClockReferencePoint_0

General

Mcu Clock Reference Point Frequency (1000 -> 64000000) 8000000.0

Mcu Clock Frequency Select SystemClockFrequency

Figure 4-123. Tresos Plugin snapshot for McuClockReferencePoint form.

4.7.18.18.1 McuClockReferencePointFrequency (McuClockReferencePoint)

This is the frequency for the specific instance of the McuClockReferencePoint container. It shall be given in Hz. McuClockReferencePointFrequency has to be less than or equal to 64 MHz (or 48MHz for Bolero 256K).

Table 4-474. Attribute McuClockReferencePointFrequency (McuClockReferencePoint) detailed description

Property	Value
Label	Mcu Clock Reference Point Frequency
Type	FLOAT
Origin	AUTOSAR_ECUC
Symbolic Name	false

4.7.18.18.2 McuClockFrequencySelect (McuClockReferencePoint)

Select the source of your reference clock frequency.

Note: Implementation specific Parameter.

Table 4-475. Attribute McuClockFrequencySelect (McuClockReferencePoint) detailed description

Property	Value
Label	Mcu Clock Frequency Select
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	SystemClockFrequency

4.7.19 Form McuModeSettingConf

This container contains the configuration (parameters) for the Mode setting of the MCU. Please see MCU035 for more information on the MCU mode settings.

Is included by form : [Form McuModuleConfiguration](#)

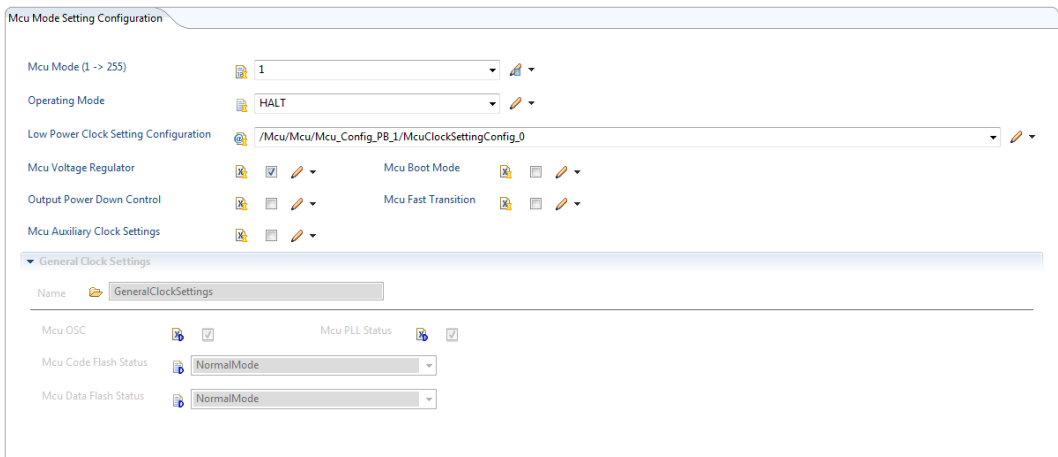


Figure 4-124. TRESOS Plugin snapshot for McuModeSettingConf form.

4.7.19.1 McuMode (McuModeSettingConf)

This parameter shall represent the number of MCU modes available for the MCU.

Table 4-476. Attribute McuMode (McuModeSettingConf) detailed description

Property	Value
Label	Mcu Mode
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	true
Invalid	Range <=255 >=1

4.7.19.2 McuOperatingMode (McuModeSettingConf)

This parameter selects the Low Power Mode used for the current configuration channel.

Note: This parameter is not used by the current Implementation

Table 4-477. Attribute McuOperatingMode (McuModeSettingConf) detailed description

Property	Value
Label	Operating Mode
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	HALT
Range	RUN3 RUN2 RUN1 RUN0 DRUN SAFE HALT STOP STANDBY

4.7.19.3 McuVoltageRegulator (McuModeSettingConf)

Main Voltage Regulator in ON or OFF state.

Note: implementation specific parameter.

Table 4-478. Attribute McuVoltageRegulator (McuModeSettingConf) detailed description

Property	Value
Label	Mcu Voltage Regulator
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.19.4 McuBootMode (McuModeSettingConf)

Specifies if the system will boot from backup RAM or flash out of STANDBY exit.

Table 4-479. Detailed description.

0	Boot from flash on STANDBY exit
1	Boot from backup RAM on STANDBY exit

Note: implementation specific parameter.

Table 4-480. Attribute McuBootMode (McuModeSettingConf) detailed description

Property	Value
Label	Mcu Boot Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.19.5 McuOutputPowerDownControl (McuModeSettingConf)

Output power-down status: This bit controls the output power-down of I/Os.

This parameter is configurable only when McuStartRunMode is configured as SAFE/TEST/ STOP0 mode.

Disabled - No automatic safe gating of I/Os used and pads power sequence driver is enabled.

Enabled - In SAFE/TEST modes, outputs of pads are forced to high impedance state and pads power sequence driver is disabled. The inputs are level unchanged. In STOP0 mode, only the pad power sequence driver is disabled, but the state of the output remains functional.

Note: Implementation specific Parameter .

Table 4-481. Attribute McuOutputPowerDownControl (McuModeSettingConf) detailed description

Property	Value
Label	Output Power Down Control
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.19.6 McuFastTransition (McuModeSettingConf)

TRUE: Perform transition to low power mode without modifying clock settings in the target low power mode (in this case LowPowerClockSettingConfig is ignored).

Note: implementation specific parameter.

Table 4-482. Attribute McuFastTransition (McuModeSettingConf) detailed description

Property	Value
Label	Mcu Fast Transition
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.19.7 McuAuxiliaryClockSettings (McuModeSettingConf)

TRUE: Clock settings applied after an exit from this low power mode are configured by the driver.

FALSE: Global Flash Settings including latency timings are not configured by the driver.

Note: implementation specific parameter.

Table 4-483. Attribute McuAuxiliaryClockSettings (McuModeSettingConf) detailed description

Property	Value
Label	Mcu Auxiliary Clock Settings
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.19.8 LowPowerClockSettingConfig (McuModeSettingConf)

This parameter is used to select the Clock setting for Low power mode of the MCU. When a clock setting is configured, the parameters in the specified McuClockSettingConfig container will be used to configure the system clock before entering the low power mode. The parameters that will not be used from the

McuClockSettingConfig container are GeneralClockSettings/McuStartRunMode. The parameters in the container McuPLL will be used only when the McuOperatingMode parameter is configured as 'HALT'.

Note: Implementation specific Parameter.

Table 4-484. Attribute LowPowerClockSettingConfig (McuModeSettingConf) detailed description

Property	Value
Label	Low Power Clock Setting Configuration
Type	REFERENCE
Origin	Custom

4.7.19.9 Form GeneralClockSettings

This container contains the clock settings used after an exit from a low power mode.

Note: Implementation specific Container.

Is included by form : [Form McuModeSettingConf](#)

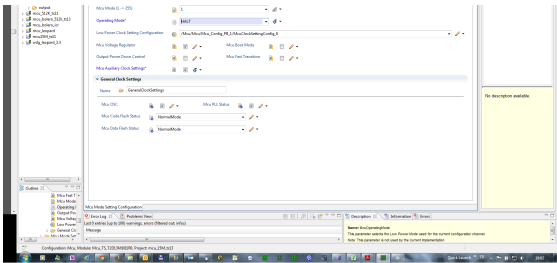


Figure 4-125. Tresos Plugin snapshot for GeneralClockSettings form.

4.7.19.9.1 McuOSC (GeneralClockSettings)

External Oscillator in ON or OFF state.

Note: Implementation specific Parameter.

Table 4-485. Attribute McuOSC (GeneralClockSettings) detailed description

Property	Value
Label	Mcu OSC

Table continues on the next page...

Table 4-485. Attribute McuOSC (GeneralClockSettings) detailed description (continued)

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false

4.7.19.9.2 McuPLLStatus (GeneralClockSettings)

Put the PLL0 in ON or OFF state.

Note: Implementation specific Parameter.

Table 4-486. Attribute McuPLLStatus (GeneralClockSettings) detailed description

Property	Value
Label	Mcu PLL Status
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true
Enable	true

4.7.19.9.3 McuCodeFlashStatus (GeneralClockSettings)

Select the Code Flash Status in new operative Mode.

Note: Implementation specific Parameter.

Table 4-487. Attribute McuCodeFlashStatus (GeneralClockSettings) detailed description

Property	Value
Label	Mcu Code Flash Status
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	NormalMode
Range	PowerDown LowPower NormalMode

4.7.19.9.4 McuDataFlashStatus (GeneralClockSettings)

Select the Data Flash Status in new operative Mode.

Note: Implementation specific Parameter.

Table 4-488. Attribute McuDataFlashStatus (GeneralClockSettings) detailed description

Property	Value
Label	Mcu Data Flash Status
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	NormalMode
Range	PowerDown LowPower NormalMode

4.7.20 Form McuRamSectorSettingConf

This container contains the configuration (parameters) for the RAM Sector setting. Please see MCU030 for more information on RAM sec-tor settings.

Current AutoSAR specification does not allow any unitialized RAM sector. In case this is needed anyway, please create a dummy RAM section and make sure Mcu_InitRamSection is not called in the code.

Note: Implementation specific Parameter.

Is included by form : [Form McuModuleConfiguration](#)

Property	Value
Mcu Ram Default Value (0 -> 255)	0xba
Mcu RAM Section Base Address (1073741824 -> 1342242816)	0x40006ba0
Mcu RAM Section Size (0 -> 131072)	0x40
Mcu RAM Section Base Address Linker Symbol	
Mcu RAM Section Size Linker Symbol	

Figure 4-126. TRESOS Plugin snapshot for McuRamSectorSettingConf form.

4.7.20.1 McuRamDefaultValue (McuRamSectorSettingConf)

This parameter shall represent the Data pre-setting to be initialized

Table 4-489. Attribute McuRamDefaultValue (McuRamSectorSettingConf) detailed description

Property	Value
Label	Mcu Ram Default Value
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	186
Invalid	Range <=255 >=0

4.7.20.2 McuRamSectionBaseAddress (McuRamSectorSettingConf)

This parameter represents the RAM section base address. The address must be aligned to 4 bytes.

Table 4-490. Attribute McuRamSectionBaseAddress (McuRamSectorSettingConf) detailed description

Property	Value
Label	Mcu RAM Section Base Address
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	1073741824
Invalid	Range <=1342242816 >=1073741824

4.7.20.3 McuRamSectionSize (McuRamSectorSettingConf)

This parameter represents the RAM section size in bytes. The size must be multiple of 4.

Table for size of different derivatives

Table 4-491. Attribute McuRamSectionSize (McuRamSectorSettingConf) detailed description

Property	Value
Label	Mcu RAM Section Size
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	1024
Invalid	Range <=131072 >=0

4.7.20.4 McuRamSectionBaseAddrLinkerSym (McuRamSectorSettingConf)

This parameter represents the RAM section base address. The address must be aligned to 4 bytes. If this parameter is empty, then the integer values from McuRamSectionBaseAddress will be used.

Note: Implementation specific Parameter.

Table 4-492. Attribute McuRamSectionBaseAddrLinkerSym (McuRamSectorSettingConf) detailed description

Property	Value
Label	Mcu RAM Section Base Address Linker Symbol
Type	STRING
Origin	Custom
Symbolic Name	false
Default	

4.7.20.5 McuRamSectionSizeLinkerSym (McuRamSectorSettingConf)

This parameter represents the RAM section size in bytes. The size must be multiple of 4. If this parameter is empty, then the integer values from McuRamSectionSize will be used.

Note: Implementation specific Parameter.

Table 4-493. Attribute McuRamSectionSizeLinkerSym (McuRamSectorSettingConf) detailed description

Property	Value
Label	Mcu RAM Section Size Linker Symbol
Type	STRING
Origin	Custom
Symbolic Name	false
Default	

4.7.21 Form McuRunConfig

This field is for Peripheral Run Configurations.

Is included by form : [Form McuModuleConfiguration](#)

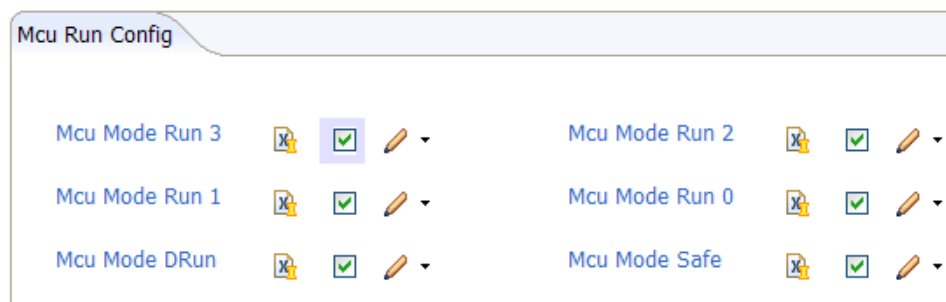


Figure 4-127. Tresos Plugin snapshot for McuRunConfig form.

4.7.21.1 McuModeRun3 (McuRunConfig)

Check if you want to enable the peripheral in RUN3 mode

Note:: Implementation specific Parameter.

Table 4-494. Attribute McuModeRun3 (McuRunConfig) detailed description

Property	Value
Label	Mcu Mode Run 3
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.21.2 McuModeRun2 (McuRunConfig)

Check if you want to enable the peripheral in RUN2 mode.

Note:: Implementation specific Parameter.

Table 4-495. Attribute McuModeRun2 (McuRunConfig) detailed description

Property	Value
Label	Mcu Mode Run 2
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.21.3 McuModeRun1 (McuRunConfig)

Check if you want to enable the peripheral in RUN1 mode.

Note:: Implementation specific Parameter.

Table 4-496. Attribute McuModeRun1 (McuRunConfig) detailed description

Property	Value
Label	Mcu Mode Run 1
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.21.4 McuModeRun0 (McuRunConfig)

Check if you want to enable the peripheral in RUN0 mode.

Note:: Implementation specific Parameter.

Table 4-497. Attribute McuModeRun0 (McuRunConfig) detailed description

Property	Value
Label	Mcu Mode Run 0
Type	BOOLEAN
Origin	Custom

Table continues on the next page...

Table 4-497. Attribute McuModeRun0 (McuRunConfig) detailed description (continued)

Property	Value
Symbolic Name	false
Default	true

4.7.21.5 McuModeDRun (McuRunConfig)

Check if you want to enable the peripheral in DRUN mode.

Note:: Implementation specific Parameter.

Table 4-498. Attribute McuModeDRun (McuRunConfig) detailed description

Property	Value
Label	Mcu Mode DRun
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.21.6 McuModeSafe (McuRunConfig)

Check if you want to enable the peripheral in SAFE mode.

Note:: Implementation specific Parameter.

Table 4-499. Attribute McuModeSafe (McuRunConfig) detailed description

Property	Value
Label	Mcu Mode Safe
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.7.22 Form McuLowPower

This field contains a possible configuration for Low-Power Peripheral Configuration Registers.

Is included by form : [Form McuModuleConfiguration](#)

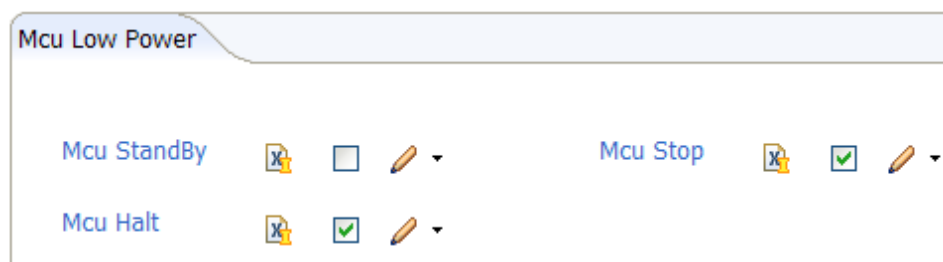


Figure 4-128. Tresos Plugin snapshot for McuLowPower form.

4.7.22.1 McuSTBY (McuLowPower)

Check if you want to enable the peripheral in StandBy mode.

Table 4-500. Attribute McuSTBY (McuLowPower) detailed description

Property	Value
Label	Mcu StandBy
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.22.2 McuStop (McuLowPower)

Check if you want to enable the peripheral in STOP mode.

Table 4-501. Attribute McuStop (McuLowPower) detailed description

Property	Value
Label	Mcu Stop
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.7.22.3 McuHalt (McuLowPower)

Check if you want to enable the peripheral in HALT mode.

Note:: Implementation specific Parameter.

Table 4-502. Attribute McuHalt (McuLowPower) detailed description

Property	Value
Label	Mcu Halt
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

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